- **Question:** Which anticholinergic has the most antisialigogue effect? (Valley p. 59)
- **Answer:** Scopolamine > Glycopyrolate > Atropine
- **FYI**:

	Atropine	Scopolamine	Glycopyrrolate
Sedation	+	+++	0
Antisialagogue	+	+++	++
Increased HR	+++	+	++
Relax Smooth Muscle	++	+	++
Mydriasis; Cycloplegia	+	+++	0
Prevent Motion Sickness	+	+++	0
Gastric H^+ secretion	+	+	+
Altered fetal HR	0	?	0

Symbols: 0, none; +, mild; ++, moderate; +++, marked; ? controversial

- **Drag/Drop Question**: MAC of all of the volatile agents

Agent	MAC	Blood:Gas (Speed of Uptake)	Oil:Gas (Lipid Solubility; Potency)	Vapor Pressure
N20	104%	0.46	1.4	--
Desflurane	6.6%	0.42	19	669
Sevoflurane	2.0%	0.69	47	170
Enflurane	1.71%	1.91	98.5	172
Isoflurane	1.2%	1.46	91	240
Halothane	0.75%	2.54	224	244

- **Question:** Which volatile agent increases CSF absorption?
- **Answer:** Isoflurane
- **FYI:**
  - Decreases CSF reabsorption : Halothane & Enflurane
  - No Change in CSF reabsorption: Desflurane
  - Volatile anesthetics affect both formation and absorption of CSF. Halothane impedes absorption of CSF but only minimally retards formation. Isoflurane, on the other hand, facilitates absorption and is therefore the only volatile agent with favorable effects on CSF dynamics (M&M).

- **Question:** IV induction doses for pediatric patient:
- **Answer:**
  - Ketamine: 1-2 mg/kg IV; 6-10 mg/kg IM
  - Propofol: 2-4 mg/kg
  - Thiopental (Pentothal): 2-4 mg/kg
  - Methohexital (Brevital): 1-2 mg/kg
  - Etomidate (Amidate): 0.3 – 0.6 mg/kg
- **FYI: Table 44–3. Pediatric Drug Dosages (M&M)**

  - Midazolam was also a choice, but is a SEDATIVE, not an induction agent (0.5 – 1 mg/kg)
  - Ketamine choice was and IM dose and you would "induce" with an IM dose
  - Methahexital was a PR dose

- **Question:** A patient with a mediastinal mass is induced and intubated and BP falls dramatically to 55/40. What should you do?
- **Answer:**
  - The mass may compress the pulmonary artery and heart as well and the IVC.
  - If symptoms worsen during induction of GA in the supine position, the sitting, leaning forward, or even prone position is advised.
  - With hypotension in the pt with a mediastinal mass, changing the position may correct the problem.

- **Fill In the Blank Question:** regarding the oculocardiac reflex (OCR) -- had to type in the Roman Numeral V (5) for Trigeminal Nerve.
- **FYI (M&M)**
  - trigeminal afferent (V1) and a vagal efferent pathway
  - oculocardiac reflex is most common in pediatric patients undergoing strabismus surgery

- **Calculate BMI**
  - Was given kg & height in cm
  - Weight in kg
  - Height in m2
  - 2.54 cm = 1 in.
  - 39.4 Inches = 1 m
  - 100 cm = 1 m
  - m → cm (move decimal 2 points)

$$BMI = \frac{\text{mass in (kg)}}{\text{height in (m)}^2}$$

$$BMI = \frac{\text{kg}}{(\text{m})^2}$$

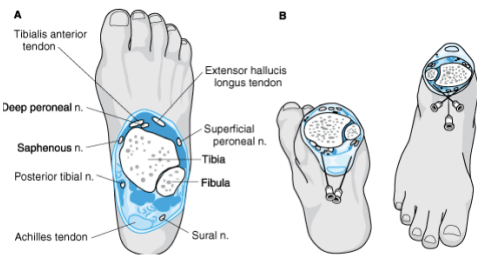
- **FYI**
  - Overweight — BMI 24 kg/m2
  - Obesity — BMI 30 kg/m2
  - Extreme obesity (morbid obesity) — BMI of 40 kg/m2
- FYI:
  - To answer this question, you must first determine the patients body mass index (BMI) which is the weight in kilograms divided by the height in meters squared. To obtain the weight in kilograms, divide the weight in pounds by 2.2. The answer is 100 kilograms. Next, multiply the height in inches (69) by 0.0254 to convert it to meters. Take this result (1.75m) and square it to arrive at 3.0 meters squared. Divide the weight in kilograms (100) by 3.0 meters squared to arrive at a BMI of 33.3.
    - Ideal body weight falls in a BMI range between 20 and 25,
    - mild obesity is 26 to 29,
    - moderate obesity is 30 to 35,
    - severe obesity is 35 to 39, and
    - morbid or extreme obesity is a BMI of 40 or above.
  - With obese patients, calculate TV, IVF, meds on IBW.

- **Fill in the Blank Question:** What is the abnormal response to painful stimuli?
- **Answer:** Hyperesthesia

- **FYI**: (M&M) Chapter 18. Pain Management

Allodynia	Perception of an ordinarily nonnoxious stimulus as pain
Analgesia	Absence of pain perception
Anesthesia	Absence of all sensation
Anesthesia dolorosa	Pain in an area that lacks sensation
Dysesthesia	Unpleasant or abnormal sensation with or without a stimulus
Hypalgesia (hypoalgesia)	Diminished response to noxious stimulation (eg, pinprick)
Hyperalgesia	Increased response to noxious stimulation
Hyperesthesia	Increased response to mild stimulation
Hyperpathia	Presence of hyperesthesia, allodynia, and hyperalgesia usually associated with overreaction, and persistence of the sensation after the stimulus
Hypesthesia (hypoesthesia)	Reduced cutaneous sensation (eg, light touch, pressure, or temperature)
Neuralgia	Pain in the distribution of a nerve or a group of nerves
Paresthesia	Abnormal sensation perceived without an apparent stimulus
Radiculopathy	Functional abnormality of one or more nerve roots

- **Question:** How many nerves do you block with ankle block?
- **Answer:** 5
  - 1. Sural
  - 2. Saphenus
  - 3. Superficial Peroneal
  - 4. Deep Peroneal
  - 5. Posterior Tibial

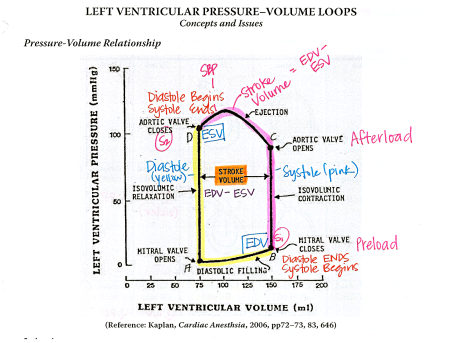


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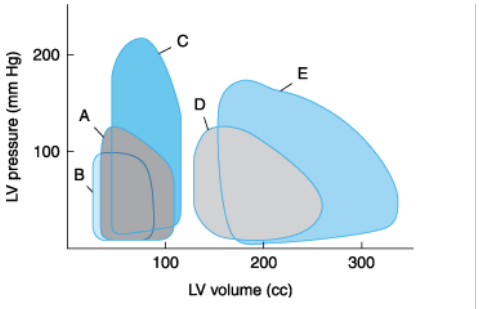
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- **Question:** given a Pressure-Volume Loop, calculate ejection fraction:

$$EF = \frac{EDV}{EDV - ESV}$$



(Reference: Kaplan, *Cardiac Anesthesia*, 2006, pp72–73, 83, 646)

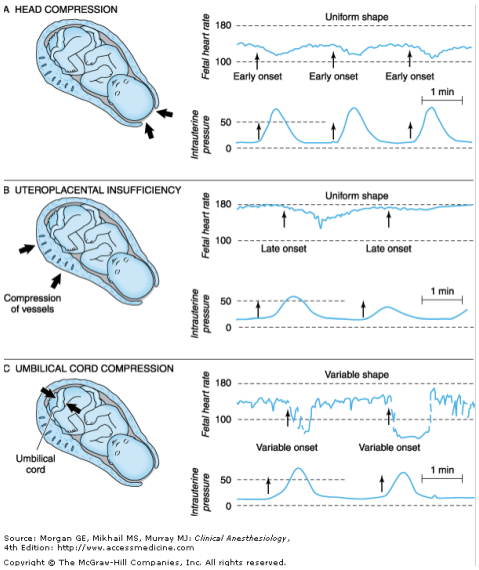


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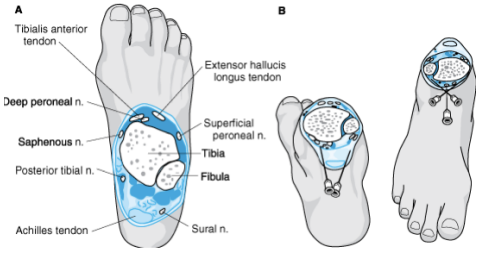
Pressure–volume loops in patients with valvular heart disease. A, normal; B, mitral stenosis; C, aortic stenosis; D, mitral regurgitation (chronic); E, aortic regurgitation (chronic). LV, left ventricular.

- **Question:** Given a choice of FHR patterns
- **Question** will ask…
  - Which one is head compression? Early Deceleration
  - Which one is cord compression? Variable Deceleration
- FYI: Chapter 43. Obstetric Anesthesia (M&M)
  - EARLY (TYPE I) DECELERATIONS
    - Early deceleration (usually 10–40 beats/min) (Figure 43–5A) is thought to be a **vagal response** to **compression of the fetal head** or stretching of the neck during uterine contractions.
    - The heart rate forms a smooth mirror image of the contraction.
    - Early decelerations are generally not associated with fetal distress and occur during descent of the head.
  - LATE (TYPE II) DECELERATIONS
    - Late decelerations (Figure 43–5B) are associated with fetal compromise and are characterized by a decrease in heart rate at or following the peak of uterine contractions.
    - Late decelerations may be as few as 5 beats/min and are thought to be due to the effect of a decrease in arterial oxygen tension on chemoreceptors or the sinoatrial node.
    - Late decelerations with normal variability may be observed following acute insults (maternal hypotension or hypoxemia) and are usually reversible with treatment.
    - Late decelerations with decreased variability are associated with prolonged asphyxia and may be an indication for fetal scalp sampling
    - Complete abolition of variability in this setting is an ominous sign signifying severe decompensation and the need for immediate delivery.
  - VARIABLE (TYPE III) DECELERATIONS
    - The most common type of decelerations are of the variable type (Figure 43–5C).
    - These decelerations are variable in onset, duration, and magnitude (often > 30 beats/min).
    - They are typically abrupt in onset and are thought to be related to umbilical cord compression and acute intermittent decreases in umbilical blood flow.
    - RULE OF 60: Variable decelerations are typically associated with fetal asphyxia when they are greater than 60 beats/min, last more than 60 secs, or occur in a pattern that persists for more than 30 min.



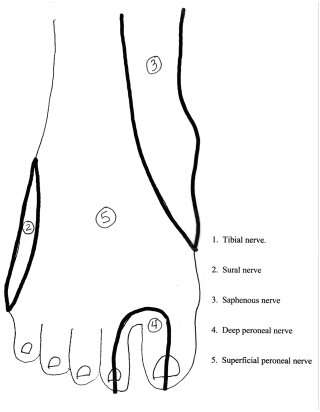
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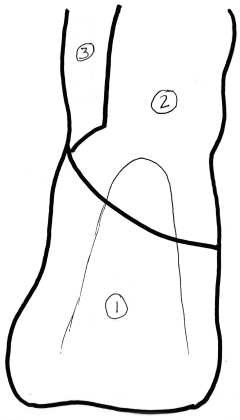
- **Question:** Picture of foot & ankle, needle pointing to nerve
- **Answer:** Sural nerve



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Chapter 17. Peripheral Nerve Blocks (M&M)





- **Question:** Why do you decrease the dose for epidural in the parturient? Identify 2 factors that decrease local anesthetic dosing by 30 – 50% in the obstetric patient. [MM p. 472]
  - [1] Engorgement of epidural veins, producing a decrease in the size of the epidural space and decreased CSF volume, which facilitates spread of the agent.
  - [2] Increased peripheral sensitivity to local anesthetics.
- A reduction in dosage of 25-50% is required in pregnancy. Barash states that the dose should be reduced 33-40%. Reducing the dosage by onethird may be a good rule of thumb (MM).

- **Question:** What is the best choice of diuretic for Cushing's syndrome?
- **Answer:** Potassium-sparing diuretics
  - Spironolactone (Aldactone)
  - Triamterene (Dyrenium)
  - Amiloride (Midamor)
- NOTES: Patients with Cushing's syndrome tend to be volume overloaded, and have hypokalemic metabolic alkalosis from the activity of glucocorticoids. This should be corrected perioperatively with supplemental K+ and Aldactone.

PHEOCHROMOCYTOMA (Chapter 36. Anesthesia for Patients with Endocrine Disease, M&M)

- What drug to have one hand after drainage of pheochromocytoma? Neosynephrine? Nipride?
  - after tumor resection → hypotension → tx with neosynephrine
- FYI (MM)
  - In the patient undergoing surgery for a pheochromocytoma, which two antihypertensives could be selected for use during the case? **Nitroprusside or phentolamine.**
  - The patient undergoing surgery for a pheochromocytoma, what is the best way to prevent hypotension after removal of the tumor?
    - Restore IV deficit, if necessary, give alpha agonist, like phenylephrine
- Alpha block before Beta block
- AVOID anesthetic drugs or techniques that
  - stimulate the sympathetic nervous system (eg, ephedrine, ketamine, hypoventilation),
  - potentiate the arrhythmic effects of catecholamines (eg, halothane),
  - inhibit the parasympathetic nervous system (eg, pancuronium), or
  - release histamine (eg, atracurium, morphine sulfate) may precipitate hypertension
- Preoperative assessment should focus on the adequacy of adrenergic blockade and volume replacement
- Intraoperative hypertension can be effectively treated with phentolamine, nitroprusside, or nicardipine.
  - Nitroprusside is favored by some anesthesiologists because of a more rapid onset of action, a shorter duration of action, and increased familiarity with the drug.
  - Phentolamine specifically blocks adrenergic receptors and prevents the effects of excessive circulating catecholamines.
  - Nicardipine is being used more frequently preoperatively and intraoperatively.
- After tumor ligation and resection, the primary problem frequently becomes **hypotension** from
  - hypovolemia,
  - persistent adrenergic blockade, and
  - prior tolerance to the high levels of endogenous catecholamines that have abruptly ended.
- Fluid resuscitation should include consideration of surgical bleeding and third-space fluid loss. Assessment of intravascular volume includes urinary output, central venous pressure, arterial blood pressure, and, if available, pulmonary capillary occlusion pressure.
- Infusions of adrenergic agonists, such as phenylephrine or norepinephrine, may occasionally prove necessary.

- **Question:** What is most common complication in PACU > 3 years old?
- **Answer:**
  - Hypoxia and Airway Obstruction were not choices
  - Picked **N/V**, not laryngeal edema.

- Some disorder with the eye (**amblyopia**) -- > look for **MH**
- "lazy eye," amblyopia is reduced vision in a healthy eye.

- **Question:** What causes a substantial increase in CVP (> 20) during CABG?
- **Answer:** Obstruction to venous drainage.
  - Examples:
    - Improper positioning of venous cannula in the patient
    - A kink in the venous return tubing
    - Intrained air in the venous return tubing – creating an air-lock
    - Unintended partial clamping of the venous return tubing
- **FYI (MM, 374)**
  - Central venous pressure (CVP) increases substantially (to, for example 25 mm Hg) during cardiopulmonary bypass. Why?
    - An increased central venous pressure during cardiopulmonary bypass occurs if there is an obstruction to venous drainage. Causes of obstruction to venous return include: improper positioning of the venous cannula in the patient; a kink in the venous return tubing; entrained air in the venous return tubing creating an "air lock"; and unintended partial clamping of the venous return tubing.
- **FYI (M&M)**
  - Malpositioning of the venous cannulas can interfere with venous return or impede venous drainage from the head and neck (superior vena cava syndrome). Upon initiation of CPB, the former is manifested as poor venous return to the reservoir, whereas the latter produces edema of the head and neck. Under these circumstances, central venous pressure increases only if the tip of the catheter is high in the vena cava.

- What disease most effect geriatric lungs?
- * Emphysema??????

- An elderly pt becomes hypotensive and tachycardia when going from supine to upright position… why?
- **Autonomic dysfunction**. Baroreceptor reflexes are impaired (MM 490).

- What are 2 important changes in the ANS that take place with aging? (MM 490).
  - A decrease in response to beta receptor stimulation, &
  - An increase in SNS activity

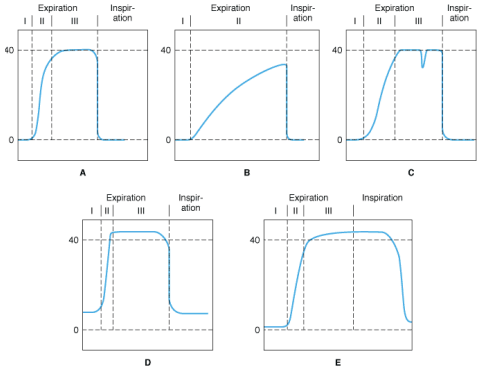
- OBESITY
  - Obese: BMI > 28 - 30
  - Morbid Obesity: BMI > 35 – 40
  - Calculate drug doses on IBW, not actual body weight

- Dose of succinylcholine? Increased or Decrease dose?
  - Base the dose of Succinylcholine on IBW

- Which drugs do you base on ideal body weight?
  - Remifentanil,
  - Succinylcholine,
  - Roccuronium, &
  - Thiopental

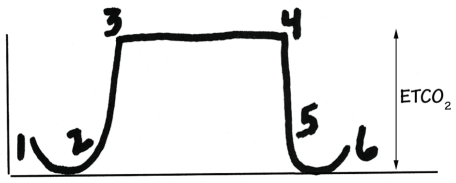
- FYI: Theoretically, larger fat stores provide an increased volume of distribution for lipid-soluble drugs (eg, benzodiazepines, opioids). Thus, a larger loading dose would be required to produce the same plasma concentration. This is the rationale for basing some drug doses on actual body weight in obese patients. By the same reasoning, maintenance doses should be administered less frequently because clearance would be expected to be slower with a larger volume of distribution. In contrast, water-soluble drugs (eg, NMBAs) have a much more limited volume of distribution, which should not be influenced by fat stores. The dosing of these drugs should therefore be based on ideal body weight to avoid overdosing. In reality, however, clinical practice does not always validate these expectations.

CAPNOGRAPHY



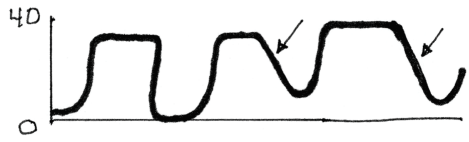
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A: A normal capnograph demonstrating the three phases of expiration: phase I—dead space; phase II—mixture of dead space and alveolar gas; phase III—alveolar gas plateau.
B: Capnograph of a patient with severe chronic obstructive pulmonary disease. No plateau is reached before the next inspiration. The gradient between end-tidal CO2 and arterial CO2 is increased.
C: Depression during phase III indicates spontaneous respiratory effort. D: Failure of the inspired CO2 to return to zero may represent an incompetent expiratory valve or exhausted CO2 absorbent.
E: The persistence of exhaled gas during part of the inspiratory cycle signals the presence of an incompetent inspiratory valve.



1 = baseline
2 = begin expiration
2–3 = combination of dead space and alveolar gases
3–4 = alveolar plateau, also called alveolar gas plateau
4 = actual alveolar ETCO2: patient begins to inspire CO2-free gases
4–5 = inspiratory limb, inspiration to baseline of 0 (zero), unless there is rebreathing
6 = end of inspiration

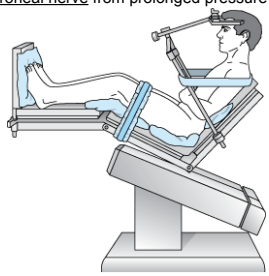
- **Question:** Inspiratory Valve Failure Capnograph



- **Question:** Which position is V/Q mismatch the greatest?
- **Answer: L**ateral decubitus?

- Lateral Position (MM 315):
  - awake & spontaneously breathing dependent lung →
    - better ventilated & perfused than non-dependent
  - anesthetized & spontaneouly breathing →
    - ventilation is better in the non-dependent lung
    - dependent is better perfused
  - anesthetized & mechanically ventilated →
    - V/Q mismatch is greater
- Lateral position compromises pulmonary function the MOST
- See Valley Review page 419

- **Question:** What nerve injury will be seen with sitting/beach-chair position? (Sciatic was not an option … picked common peroneal)
  - The chair position prevents injury to the sciatic nerve
  - Pillows under knees to protect sciatic nerve, but can <u>cause peroneal nerve</u> from prolonged pressure in popliteal fossa



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- Hypoperfusion of the spinal cord with posterior fossa crani → <u>extreme flexion of the neck</u>???
- NOTE: Placement of the head and neck at the limits of motion may produce anatomic changes that compress the ipsilateral and/or contralateral vertebral arteries.
- **FYI (M&M)**
  - Careful positioning helps avoid injuries. Pressure points such as the elbows, ischial spines, heels, and forehead must be protected. Excessive neck flexion has been associated with swelling of the upper airway (due to venous obstruction) and, rarely, quadriplegia (due to compression of the cervical spinal cord). Preexisting cervical spinal stenosis probably predisposes patients to the latter injury.

- What would increase the GFR?
- **<u>Afferent arteriole</u>** DILATES more than the efferent arteriole.
- GFR will decrease if the afferent arteriole constricts more than the efferent.
- FYI (M&M)
  - Glomerular filtration pressure (about 60 mm Hg) is normally about 60% of mean arterial pressure and is opposed by both plasma oncotic pressure (about 25 mm Hg) and renal interstitial pressure (about 10 mm Hg). Both afferent and efferent arteriolar tones are important determinants of filtration pressure: Filtration pressure is directly proportional to efferent arteriolar tone, but inversely proportional to afferent arteriolar tone. Approximately 20% of plasma is normally filtered as blood passes through the glomerulus.

- **Multiple/Multiple question**: What drugs would you give because of the side effects of a pain pump?
- "Albuterol, narcan, labetalol?

- **FYI: Chapter 18. Pain Management**
  - Excess glucocorticoid activity can produce <u>hypertension</u>, <u>hyperglycemia</u>, increased susceptibility to <u>infection</u>, <u>peptic ulcers</u>, <u>osteoporosis</u>, aseptic necrosis of the femoral head, proximal myopathy, cataracts, and, rarely, psychosis. Patients can also develop the physical features characteristic of Cushing's syndrome (see Chapter 36). Excess mineralocorticoid activity causes sodium retention and hypokalemia, and can precipitate congestive heart failure.
  - opioids are nausea, vomiting, itching, urinary retention and ileus
  - respiratory depression – narcan 0.04 mg increments; doxapram
  - pruritus – narcan
  - itching – diphenhydramine
  - n/v – reglan, scopolamine, droperidol, ondansetron

- What is the most common reason for a liver transplant in children? (MM 454)
  - Cholecistic liver disease secondary to biliary atresia **<u>biliary atresia</u>**

- A routine antenatal ultrasound reveals **<u>polyhydramnios</u>** in the pregnant patient. What is polyhydramnios and what are the implications for the fetus? (MM 485)
- Polyhydramnios → presence of excessive amniotic fluid around the unborn infant.
- Occurs when the infant fails to swallow and absorb amniotic fluid in normal amounts.
- The most common causes are GI obstruction & neurologic disorders.
- **<u>ESOPHAGEAL ATRESIA</u>**, as seen in TEF, is of greatest concert for the infant.
- If polyhydramnios is present, check for TEF by passing a nasogastric tube shortly after delivery.

- **Question:** What is **<u>double-circuit ventilator</u>**? (MM 260)
  - Double-circuit ventilators (bellows in a box, bag in a bottle), in which one circuit contains patient gas, and the other circuit contains drive gas, are used most commonly on modern anesthesia workstations.
  - The bellows is housed in a pressure chamber, and the inside of the bellows is connected to the breathing system.
  - The **<u>DC separates breathing system gas from driving gas</u>** → they are pneumatically driven

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- OBESITY

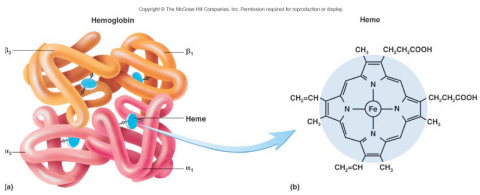
Increased	Decreased
Diaphragm Elevated Work of Breathing CO Blood Volume (50 mL/kg) Cholesterol Triglycerides PHTN	Chest Wall Compliance Expiratory Reserve Volume Inspiratory Capacity Vital Capacity FRC PaO2 (arterial hypoxemia)

- **Question:** Autoregulation with liver is dependent on what ?
- **Answer:** **Blood flow**

- **Question:** What test is increased with liver dysfunction? [MM p. 106, #22]
- **Answer:** 5NT > GST > ALT > AST

- Conn's Syndrome (Chapter 36. Anesthesia for Patients with Endocrine Disease)
  - Intrinsic hypersecretion of aldosterone by the adrenal cortex
  - Primary hyperaldosteronism → excess in mineralocorticoid
  - FYI: Clinical manifestations of mineralocorticoid excess include an elevation in blood pressure, hypervolemia, hypokalemia, muscle weakness, and metabolic alkalosis. Prolonged hypokalemia may lead to a renal concentrating defect and polyuria. Alkalosis will lower ionized calcium levels and can cause tetany. Serum sodium is often normal.

- **Question:** RBC 50% occupied by hemoglobin. How many?
- **Answer:** 2
- **FYI:** each hemoglobin can carry 4 O2's



- HGB COMBINES WITH O_2 AT THE ALVEOLAR-CAPILLARY LEVEL AND RELEASES THE O_2 AT CAPILLARY-TISSUE LEVEL.
- NOTE: The iron in hemoglobin combines loosely and reversibly with O_2. The iron remains in the Fe^{++} (ferrous) form. It is not oxidized to Fe^{+++} (ferric) by O_2. Oxidation to ferric is irreversible, so the hemoglobin/iron could not release the O_2 to cells.
- EACH Fe^{++} CAN COMBINE WITH ONE OXYGEN MOLECULE (O_2). SINCE A HGB MOLECULE CONTAINS FOUR Fe^{++}, EACH HGB MOLECULE CAN COMBINE WITH 4 MOLECULES OF OXYGEN (O_2)/8 ATOMS OF OXYGEN.

- **Question:** If you fail to follow the standards of practice, what could you be charged for?
- **Answer:** Malpractice [MM p. 532]

- **Question:** What risk management tool is best used for anesthesia outcomes?
- **Answer:** Indicator wasn't an option? [MM p. 530]

- **Question:** How do narcotics work?
- Prostagladins, thromboxanes, enkephalin, leukotrienes???

FYI

- Opioids bind to specific receptors located throughout the central nervous system and other tissues. Four major types of opioid receptor have been identified (Table 8–4): mu (μ , with subtypes μ -1 and μ -2), kappa (k), delta (δ), and sigma (σ).

- **Question:** Where does spinal cord terminate in adult?
- **Answer:** **conus medularis** MM p. 56

- Dantrolene directly interferes with muscle contraction by binding to the **Ryr1 receptor** calcium channel and inhibiting calcium ion release from the SR.
- RYR = ryanodine receptor

EPIDURAL STEROID INJECTIONS

- Effective for relief of pain associated with ACUTE RADICUOLOPATHY (nerve root compression)
- Provides relief when 3 things are present:
  - [1] Nerve root EDEMA
  - [2] Nerve root INFLAMMATION
  - [3] Increased levels of PLA2 expression
- Most effective when given within 2 weeks of onset of pain → don't provide long-term pain relief > 3 months
- Affect UNMYELINATED C fibers
- 3 Approaches:
  - [1] Interlaminar: 5-10 mL
  - [2] Intervertebral (Tranforaminal): 2-3 mL *Requires XRay *
  - [3] Sacral Hiatus (Caudal): 25 mL
- Advantages of Local Anesthetic & Steroid Mixure:
  - Helpful if pt has muscle spasm & LA provides immediate pain relief until anti-inflammatory effects take place (12-48 hours)
- Disadvantages:
  - Risk of intrathecal, subdural, and intravascular complications (decreased BP, arrhythmias & seizures)
- Most commonly used drugs:
  - Methylprednisone acetate (Depo-Medrol): 40-120 mg
  - Triamcinolone diacetate (Aristocort): 40-80 mg
  - Both can be injected alone, mixed with saline dilutent, or mixed with LA.

- Multiple/Multiple Question about Mu 2 receptors → sedation, bradycardia, spinal (not supraspinal), diuresis?

Mu-1	Mu-2	Kappa	Delta
Spinal Analgesia	**Spinal Analgesia**	Spinal analgesia (kappa-1)	Spinal analgesia (delta-1 & delta-2
Supraspinal analgesia		Supraspinal analgesia (kappa-3)	Supraspinal analgesia (delta-1 delta-2)
	Respiratory Depression		Respiratory Depression
Decreased heart rate			
Euphoria		**Dysphoria**, sedation	
Low abuse potential	**Physical Dependence (Addiction)**	Low abuse potential	Physical Dependence
Urinary Retention		Diuresis	
Pruritus	Constipation (marked)		Constipation (minimal)

- Increased dose to elderly → atropine/glycopyrolate
- Why do you give more atropine/glycopyrolate?
  - Increased parasympathetic (vagal tone)

- Denervation of carotid baroreceptors? → Tachycardia?
- MM p. 388 # 14

- What is the most common post-op complication after a carotid endarterectomy (CEA)? [MM p. 388]
  - HTN (?)
- Anywhere from 10 – 66% of pts experience post-op hypertension following CEA.
- The exact cause of post-op HTN following CEA is unclear, but all proposed mechanisms involve the carotid sinus baroreceptors.
- Post-op HTN may be due to surgical denervation of the ipsilateral carotid baroreceptor, chemical denervation of the baroreceptor or carotid sinus dysfunction from surgical trauma.
- 1st & 2nd most common cause of morbidity & mortality with CEA?
  - [1] MI
  - [2] Stroke

- Pt moved from supine to sitting position. What causes decreased BP?
- Impaired baroreceptor reflex response [MM p. 324, #17]

- Which condition is associated with large tongue? [MM p. 72, #32]
- **Sarcoidosis** -- hyperplastic lymphoid tissue & LARGE TONGUE (difficult airway).
- Down's syndrome, Pierre Robin syndrome, acromegaly, and hypothyroidism (MM 447)
- Cretinism (MM 87) – congenital hypothyroidism
- Treacher-Collins syndrome is not associated with macroglossia
- (large tongue) or mental retardation (MM 448).

- FIBEROPTIC INTUBATION
- What would you not do a fiberoptic bronchoscopy?
  - Epiglottitis?????????
  - Epiglottitis is a contraindication to fiberoptic nasopharyngoscopy by inexperienced personnel, as it may result in laryngospasm and subsequent airway compromise
- What can you put down the suction port on FO scope?
  - lidocaine ??? supplemental O2 ??? defog ???
- Advantages of FOI for difficult airway → spontaneous ventilation
  - A flexible FOB allows indirect visualization of the larynx in such cases or for any situation in which awake intubation is planned
- When would you not do FOI → severe respiratory distress
- FYI
  - a guide wire will pass easily through the suction port
  - may be connected to high flow oxygen
  - spray lidocaine
  - suctioning of secretions, insufflation of oxygen, or instillation of local anesthetic.

- You have 0.0625% Bupivacaine. You give a bolus dose of 4 mL. How much did you give?
- 0.625mg/mL X 4 mL
- = 2.5 mg

- **Ritodrine (Yutopar)**
  - Used to stop premature labor by relaxing uterine smooth muscle
  - Adrenergic beta-2 agonist
  - S/E: hyperglycemia, hypokalemia, tachycardia, and possibly **pulmonary edema**
  - Do not administer atropine to a pt receiving ritodrine b/c it will exacerbate the tachycardia and produce pulmonary edema\
- FYI (MM 470)
  - Side-effects of ritodrine, a beta-2 adrenergic agonist, include hyperglycemia, hypokalemia, tachycardia and possibly pulmonary edema.

- B2 question – opposite, not tachycardia???? [Valley p. 41]

Organ	Receptors	Response
Pupil of eye	Alpha-1	Mydriasis (pupil dilates)
Heart		
SA node	**Beta-1**	**↑ heart rate**
AV node	**Beta-1**	**↑ conduction speed**
Muscle fibers	**Beta-1**	**↑ contractility**
Arterial blood vessels		
most systemic vessels	**Alpha-1**	**Vasoconstriction (↑SVR)**
Skeletal muscle vessels	**Beta-2**	**Vasodilation (↓ SVR)**
Veins	**Alpha-1**	**Vasoconstriction (↑ preload)**
Lung		
Bronchiolar muscle	**Beta-2**	**Bronchodilation**
Secretory glands	**Beta-2**	**↑ secretions**
Spleen	Alpha-1	Release RBCs
Kidney	**Beta-1**	↑ Renin release
	Alpha-1	↓ Renin release
Liver	**Beta-2**	**Glycogenolysis** ↑ glucose
	Beta-2	**Gluconeogenesis** ↑ glucose
Gallbladder	Beta-2	Relaxation
Stomach & Intestines	Alpha-1	Relaxation
Urinary bladder wall	Beta-2	Relaxation
Pancreas	Beta-2	↑ insulin secretion
	Alpha-2	↓ insulin secretion
	Alpha-1	↑ glucagon secretion
Adipose tissue	Beta-1	Lipolysis
Uterus	**Beta-2**	**Relaxation**
	Alpha-1	Contraction
Sodium-potassium pump	**Beta-2**	**Stimulates Na^+–K^+ pump (↓ plasma K^+)**

- What drug would give for a bronchoscopy, with an upper airway obstruction?
- Glycopyrolate—to decrease secretion
- Vasoconstrictor – to decrease bleeding

LMA Sizes
Valley p. 521

Patient	LMA Size	Maximum Cuff Volume	Largest ETT (mm)	FOB size (mm)
Neonates: (< 5 kg)	1	4 mL	3.5	2.7
Infants: 5 – 10 kg	1 ½	7 mL	4.0	3
10 – 20 kg	2	10 mL	4.5	3.5
20 – 30 kg	2 ½	14 mL	5.0	4
30 – 50 kg	**3**	**20 mL**	**6.0 Cuffed**	**5**
Adult	4	30 mL	6.0 Cuffed	5
Adult	5	40 mL	7.0 Cuffed	5
Large Adult	6	50 mL	7.0 Cuffe	5

- Decreased FRC, what would decrease???
  - FRC = RV + ERV

- Math questions
- Calculate how much 1:1,000 EPI to 20 mL solution

- 1 gram
- 1,000 mL

- 1,000,000mcg
- 1,000 mL

- 1000 mcg
- mL

- 0 .1?????

- Delayed wake up in elderly with Versed (midazolam)

- What are the 9 parameters that the anesthesia machine must monitor in order to comply with ASTM F1850-00?
  1. Continuous breathing system pressure
  2. Exhaled TV
  3. Ventilatory CO2 concentration
  4. Anesthetic vapor concentration
  5. Inspired O2 concentration
  6. Arterial O2 concentration
  7. O2 supply pressure
  8. Arterial BP
  9. Continuous EKG

- FDA checklist for anesthesia machine? First step → Turn ventilator ON (duh!!!)

- Drag & Drop: damage to what cranial nerve?
  - Droopy tongue

Number	Name	Function(s)	Sensory/Motor
I	Olfactory	Smells (One Nose Smells)	Sensory
II	Optic	Sees (Two Eyes See)	Sensory
III	**Occulomotor**	**Moves eyes; ADduction of eye (medial rectus); pupil size**	Motor
IV	Trochlear	Moves eyes	Motor
V	**Trigeminal**	**Chews (muscles of mastication); Sensory from face**	Both
VI	**Abducens**	**Moves eyes; Abduction of eye (lateral rectus)**	Motor
VII	**Facial**	**Facial muscles**, taste (from anterior 2/3 of tongue)	Both
VIII	Acoustic	Balance (vestibular); audition (cochlear)	Sensory
IX	Glossopharyngeal (Hering's Nerve)	Taste (from posterior 1/3 of tongue); Carotid body and Carotid sinus afferents	Both
X	Vagus	Decreased heart rate; motor control of larynx and pharynx	Both
XI	Accessory	Shoulder and head movements	Motor
XII	**Hypoglossal**	**Moves tongue**	Motor

- Chronic abdominal pain associated with chronic pancreatitis: what block?
  - Celiac plexus block?

- Autoregulation of liver →
  - splanchnic blood flow/mesenteric blood flow

- Cross clamp → severe HYPOTENSION after release? [MM p. 389/390]
- Metabolic acidosis → increased PVR after the cross clamp is released
- Following unclamping of the abdominal aortic cross-clamp, there is a "washout" of local tissue mediators (e.g., prostaglandins) and metabolic products (especially lactate), a decrease in systemic vascular resistance (SVR), and volume shifts leading to a central hypovolemia and ultimately decreased venous return to the heart. Decreased venous return leads to decreased cardiac output and potentially significant hypotension.

- What is the best indicator of renal reserve/function?
  - Creatinine clearance

- **Poiseuille's Law** states that laminar flow is proportional to what 2 factors?
  - Flow is DIRECTLY proportional to => the pressure drop along the tube & the r^4 of the tube
  - Flow is INVERSELY proportional to => length of the tube & viscocity
  - If the radius of the tube doubled, how much does laminar flow increase? $r^4 = (2)^4 = 16x$
  - Tripled? $r^4 = (3)^4 = 81x$

fyi

- Laminar (streamlined) flow is described mathematically by Hagen-Poiseuille's law:
  - a. F is directly proportional to the fourth power of the radius (r).
    - Doubling the radius increases flow 16 times. (2r)4 = 2.2.2.2r4 = 16r4.
    - Tripling the radius increases flow 81 times. Halving the radius decreases flow to 1/16th of the original value.
    - Decreasing the radius by 1/3 decreases flow to 1/81 of its original value.
    - Changing radius has the most dramatic effect on flow. Flow through a large bore needle is faster than through a small one.
  - b. F is directly proportional to the hydrostatic pressure gradient (ΔP).
    - ΔP = inflow pressure minus outflow pressure.
    - Raising the IV pole increases the hydrostatic pressure in the fluid delivered to the patient by gravity; flow increases proportionately.
  - c. F is inversely proportional to fluid viscosity (η). Know that the property of a fluid that determines flow when flow is laminar ("streamLined") is viscosity.
    - Polycythemic patients have high blood viscosity, so flow through blood vessels is reduced, other factors remaining constant.
    - The anemic patient has decreased blood viscosity, and hence greater flows through tissues will result.
  - d. F is inversely proportional to the length (l) of the tube. Reducing the needle length increases flow of fluids from an IV bag hanging on a pole.

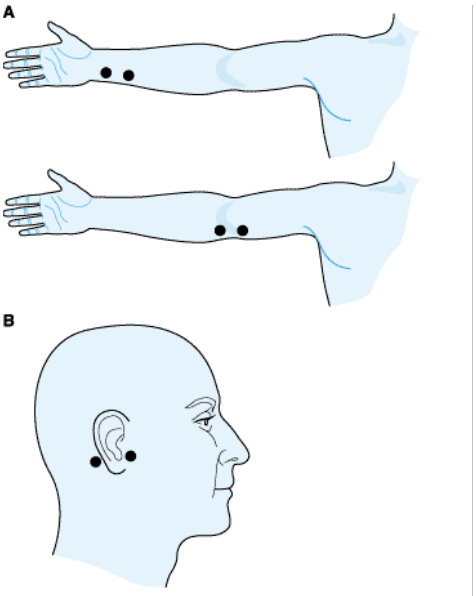
- Mitral Stenosis
  - Mitral oriface is reduced from normal (4-6 cm^2) to an opening of < 2 cm^2
  - (M&M Chapter 20. Anesthesia for Patients with Cardiovascular Disease)

- Drag & Drop: put in order which most to least effects on beta
  - Isoproterenol > Epinephrine > NE (no beta effect)

- VAE: what position do you place patient in?
  - LLD, and
  - Reverse Trendelenburg

- Regional block, sciatic block most beneficial for what surgery?
  - Tibial

- What muscle innervated by ulnar?
  - Adductor pollicis [MM. p. 357, #17-18]



Source: Morgan GE, Mikhail MS, Murray MJ: *Clinical Anesthesiology*, 4th Edition: http://www.accessmedicine.com

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A: Stimulation of the ulnar nerve causes contraction of the adductor pollicis muscle. B: Stimulation of the facial nerve leads to orbicularis oculi contraction. The orbicularis oculi recovers from neuromuscular blockade before the adductor pollicis.

- Lap. Sx, → pt with left shoulder pain → MI????

- Hotspot question: What nerves would be damaged with PDA repair? ??? SLN? RLN?

- Drag & Drop Question: pKa of Mepivacaine & Levobupivacaine & Benzacaine

Drug	pKa
Chloroprocaine	9.1
Procaine	8.9
Tetracaine	8.6
Bupivacaine	8.1
Ropivacaine	8.1
Levobupivacaine	**8.09**
Lidocaine	7.7
Etidocaine	7.7
Mepivacaine	**7.6**
Benzacaine	**3.5**

- What do all non-rebreathing systems have in common?
  - ?? they all have check valves ??

- Drag & Drop [Valley p. 200]
- Sensory to glottis (vocal cords) and UP → Internal Branch of Superior Laryngeal Nerve
- Sensory to subglottis (everything below vocal cords) → Recurrent Laryngeal Nerve (sensory)
- Motor Innervation to everything except cricothyroid → Recurrent Laryngeal Nerve (motor)
- Motor innervation to cricothyroid →external branch of Superior Laryngeal Nerve

- NOTE: IS ME
  - Internal SLN – sensory
  - External SLN – motor (cricothyroid only)

- Drag & Drop: Which one has shortest/intermediate/longest half-life?
  - Oxycodone (Shortest) 3-4 hours
  - Levorphanol (Intermediate) 11-16 hours
  - Methadone (Longest) 24-36 hours

- Drag & Drop:
  - Bleomycin -- Lungs
  - Doxoyrubicin (Adriamycin) -- Heart
  - Cisplatin – Kidneys
  - **Cyclophosphamide**, Steptozocin, Methotrexate – Liver

- What drug uses cAMP as a 2nd messenger?
  - EPI

- Drag & Drop: Identify drug with classification
  - Histamine-2 Antagonist: (Raise gastric pH and decrease gastric volume)
    - Cimetidine (Tagamet)
    - Ranitidine (Zantac)
    - Famotidine (Pepcid)
  - Antacid:
    - Metoclopramide (Reglan)
    - Sodium Citrate (Bicitra)

- Drag & Drop: Match words with definitions of … [M&M p. 361]
  - **Allodynia** → perception of an ordinarily nonnoxious stimulus as pain
  - **Hyperalgesia** → increased response to noxious stimuli
  - **Radiculopathy** →functional abnormality of one or more nerve roots
  - **Central sensitization** → nociceptive neurons in the dorsal horns of the spinal cord become sensitized by peripheral tissue damage or inflammation. This type of sensitization has been suggested as a possible causal mechanism for chronic pain conditions.

- (Valley Review p. 296)
- Volume of Distribution Question: 75 L and 3 (something)/mL. How much Fentanyl would you give?
  - $V_d = \frac{\text{Dose}}{\text{Concentration}}$
  - Volume of distribution is the theoretical volume necessary to account for the total amount of drug in the body if it were present throughout the body in the same concentration as in the plasma
  - Vd = Q / Cpt=0
  - Q = Amount of Drug
  - Cpt = Plasma Concentration 3 mg/mL
  - Vd = volume of distribution = 75L
- *** Answer is 25 mcg of Fentanyl ***

NOTE: My answer!!!
- Q = Vd x Cpt=0
- Q = (75 L) (3 mg/L)
- Q = 225 mg??????

- ?? Med with ESRD ??
  - ?? Morphine ??

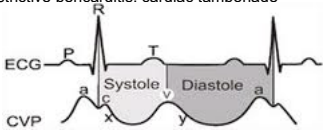
- What electrolyte disturbances will you see with over-production of cortisol (Cushing's Syndrome)?
  - INCREASED glucose
  - DECREASED K+

- Pick 3 about spinal narcotics ???
  - * ??? prostaglandins, thromboxanes, enkephalins???

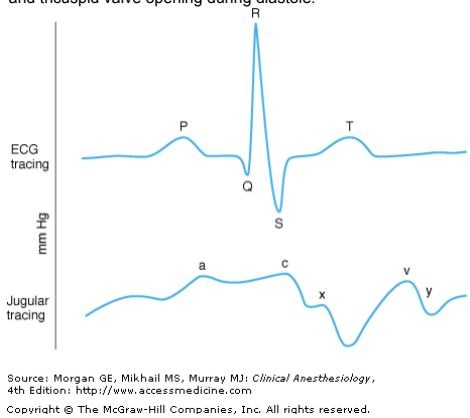
- What is the most sensitive indicator for a carcinoid tumor? [MM p. 128]
  - **Serotonin**
- What is a carcinoid syndrome?
  - a complex of s/s caused by vasoactive substances released from enterochromaffin tumors, or carcinoid tumors
- What is the primary substance released by a carcinoid tumor?
  - Serotonin (5-HT) is the primary substance.
  - Then kallikrein & histamine
- The diagnosis of carcinoid syndrome is confirmed by detection of the serotonin metabolite (5-HIAA) in the urine
- Are substances released from carcinoid tumors (serotonin, histamine, bradykinin, prostaglandins, kallikrein) considered hormones? … **YES**
- What is the key to anesthetic management of pts with carcinoid syndrome?
  - Avoid anesthetic techniques that could activate the tumor including: hypotension, catecholamine administration, histamine-releasing drugs (morphine, atracurium, curare, mivacurium, and trimethaphan)

- Fiber-Optic Intubations:
- What would help you visualize glottis?
  - Neck flexion
- Contraindication to FOI?
  - Hypoxia (Severe respiratory distress)

- Which would be most beneficial for LA toxicity?
  - Hypoxemia ?? (Hyperventilation was NOT a choice)
- MM p. 386 (# 368): After giving O2, hypERventilate the patient. HypERventilation of the lungs produces hypOcapnia and constriction of cerebral vessels to reduce delivery of additional local anesthetics to the brain; ALSO, respiratory alkalosis and hypOkalemia associated with hypERventilation result in hypERpolarization of the nerve membranes, which opposes the effects of local anesthetic. HypOcapnia increases the seizure threshold of local anesthetics.

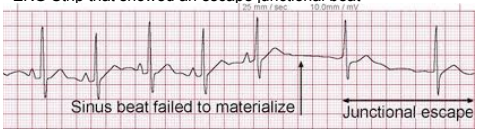
- CVP Waveform:
  - **a wave:** d/t atrial contraction, shortly after P wave from atrial depolarization.
    - absent in atrial fibrillation, and exaggerated in junctional rhythms
    - Large A wave d/t: Tricuspid or Pulmonic Stenosis, PHTN, or decreased RV compliance
  - **C Wave:** due to tricuspid valve elevation (toward atrium) during early ventricular contraction
  - **V Wave:** reflects venous return against a closed tricuspid valve; systolic filling of the atrium
    - Large V wave d/t: tricuspid regurgitation, RV failure, constrictive pericarditis, cardiac tamponade
- 

M&M (Chapter 6: Patient Monitoring)
- a waves from atrial contraction are absent in atrial fibrillation and are exaggerated in junctional rhythms (cannon waves); c waves are due to tricuspid valve elevation during early ventricular contraction; v waves reflect venous return against a closed tricuspid valve; and the x and y descents are probably caused by the downward displacement of the tricuspid valve during systole and tricuspid valve opening during diastole.



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- EKG Strip that showed an escape junctional beat



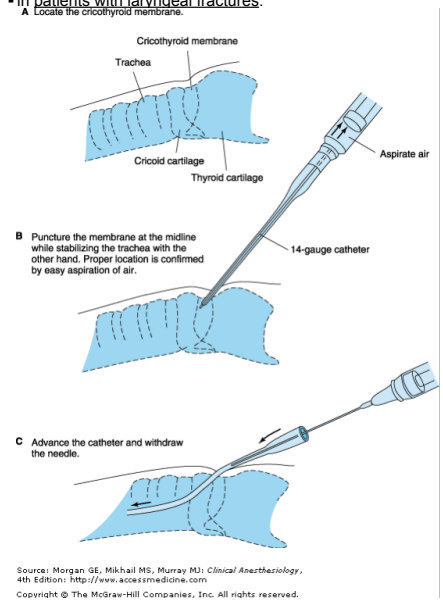
- What is in EMLA cream? (MM 170)
  - 2.5 % Lidocaine & 2.5 % Prilocaine
  - Applied under an occlusive bandage for 45 – 60 minutes
  - Don't use orally (duh!)
  - Can be a total of 5% or 10%
- FYI (MM 170)
  - A combination of 2.5% prilocaine and 2.5% lidocaine cream (5% total concentration) is called a eutectic mixture of local anesthetics (EMLA), because the melting point of the combined drugs is lower than prilocaine or lidocaine alone. ("Eutectic" means equal proportions of solid crystals of prilocaine and lidocaine, and this mixture remains a liquid at room temperature.)
  - EMLA cream is not recommended for: (1) use on mucous membranes, (2) use on skin wounds, (3) use on patients being treated with the antidysrhythmic agent mexiletine, and (4) use in patients with known history of allergy to amide local anesthetics.

- Patient with bowel prep and NGT suctioning for surgery prep …
  - HYPOchloremic metabolic alkalosis

- What is the MAC of Sevo for a geriatric patient?
  - 1.45

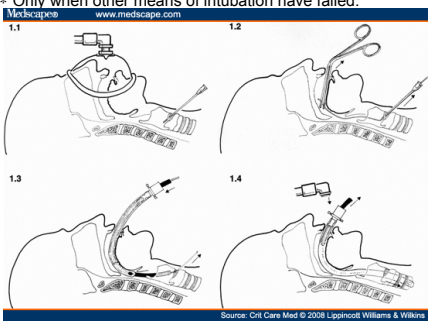
- FYI about MAC (MM): 1These minimum alveolar concentration (MAC) values are for 30- to 55-year-old human subjects and are expressed as a percentage of 1 atmosphere. High altitude requires a higher inspired concentration of anesthetic to achieve the same partial pressure.

- Major complications of **cricothyroidotomy**: [MM p. 335]
  - pneumothorax, subQ emphysema, mediastinal emphysema, bleeding, esophageal puncture, aspiration, respiratory acidosis
- Where is the needle inserted during an emergency tracheostomy?
  - The 12-14 gauge extra catheter needle should be inserted through the cricothyroid membrane
- Where is the cricothyroid membrane located? Is it palpable?
  - It is a palpable transverse indentation between the thyroid and cricoid cartilages.
- What pressure is needed to ventilate through a cricothyroidotomy? (TTJV)
  - 50 psi
- Contraindications?
  - Cricothyroidotomy, cricothyrotomy, coniotomy, and minitracheostomy are synonyms for establishing an air passage through the cricothyroid membrane.
  - Although cricothyrotomy is the procedure of choice in an emergency situation, it may also apply to an elective situation when there is limited access to the trachea.
  - Cricothyrotomy is **CONTRAINDICATED
    - in neonates & children younger than 6 years of age, and
    - in patients with laryngeal fractures.



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- How is a retrograde intubation performed? [MM p. 328]
  - A large bore needle is placed through the cricothyroid membrane, a flexible J wire is directed cephalad through the needle and out the mouth or nares, the endotracheal tube is placed over the wire and directed through the glottic opening
- When should retrograde intubation be performed?
  - Only when other means of intubation have failed.



- What triggers the Hering-Breuer reflex? (MM p. 41)
  - Lung inflation triggers the Hering-Breuer reflex.
  - When the HB reflex is triggered by lung inflation, inspiration is inhibited (apnea).
  - The HB reflex plays a minor role in normal ventilation in the adult. It is mainly a protective mechanism that is probably not activated until the tidal volume increases to greater than 1.5 L.
- The vagus nerve carries sensory (afferent) impulses of the Hering-Breuer reflex.

- What is the most common cause of heparin resistance? How can this resistance be treated? [MM p. 220]
  - **Antithrombin III deficiency** is the most common cause of heparin resistance.
  - Adequate anticoagulation can be achieved following infusion of two units of FFP.

- What do you give to reverse Coumadin immediately? (MM p. 122)
  - FFP (vtamin K will also reverse Coumadin, but FFP works immediately)
- List 4 indications to administer FFP:
  - Isolated coagulation factor deficiencies
  - Reversal of Coumadin therapy
  - Correction of coagulopathies associated with liver disease.
  - After massive transfusions with continued bleeding even after platelet transfusions.
- FFP contains what clotting factors?
  - All of them except platelets.
- What are the most abundant factors in FFP?
  - Factor V & VIII. (5 & 8)

- FYI (MM): The antidote is vitamin K, fresh whole blood, or fresh frozen plasma. Vitamin K (1-5 mg) takes 3-6 hours to begin working and 4 to 24 hours to return prothrombin time (PT) to the normal range. Fresh frozen plasma and banked blood, which contain all the vitamin K-dependent factors, takes less time.

- Know pacemaker settings: VVI setting? [MM p. 431]
  - First Letter: chamber paced
    - (A) Atrium; (V) Ventricle; (O) None; (D) Dual – both atrium & ventricle are paced
  - Second Letter: chamber sensed
    - (A) Atrium; (V) Ventricle; (O) None; (D) Dual – both atrium & ventricle are sensed
  - Third Letter: area to respond to sensing ☒
    - (T) Triggered; (I) Inhibited
- Example of VVI: In the VVI mode, the ventricle (V) is sensed, the ventricle (V) is paced, and, when the ventricle (V) is sensed, the ventricle (V) is inhibited (I).

- Sentinel Event vs Critical Event: [MM p. 532]
  - **Sentinel Event**:
    - An unexpected occurrence or variation involving serious physical or psychological injury, or risk thereof.
    - Sentinel means sends an alarm or signal that requires immediate attention.
    - FYI (wickipedia)
      - A Sentinel Event is defined by The Joint Commission (TJC) as any unanticipated event in a healthcare setting resulting in death or serious physical or psychological injury to a patient or patients, not related to the natural course of the patient's illness. Sentinel events specifically include loss of a limb or gross motor function, and any event for which a recurrence would carry a risk of a serious adverse outcome.
      - Besides "unexpected occurrences involving death or serious physical or psychological injury, or the risk thereof", sentinel events also include the following, even if the outcome was not death or major permanent loss of function:
        - Infant abduction, or discharge to the wrong family.
        - Unexpected death of a full term infant.
        - Severe neonatal jaundice (bilirubin over 30 milligrams/deciliter).
        - Surgery on the wrong individual or wrong body part.
        - Surgical instrument or object left in a patient after surgery or another procedure.
        - Rape in a continuous care setting.
        - Suicide in a continuous care setting, or within 72 hours of discharge.
        - Hemolytic transfusion reaction due to blood group incompatibilities. [2]
        - Radiation therapy to the wrong body region or 25% above the planned dose.
- **Critical Event**
  - Event that causes or has the potential to cause injury if not noticed and corrected in a timely manner.
  - Ventilator disconnect is an example of critical event.
- In continuous quality improvement programs, events are identified and analyzed as opportunities to improve processes. An event that could potentially produce in harm in a patient, such as a circuit disconnect, whether the event was detected before harm was committed or not is referred to as a critical incident. A sentinel event is a single, isolated event such as a patient death or surgery on the wrong patient or body part, that indicates a potentially systemic problem.
- As part of the patient safety goals of the JCAHO guidelines, any sentinel event such as wrongful patient death, surgery on the wrong patient or wrong body part, or injury from medication errors should undergo root causal analysis to determine the point of error so that it can be corrected in the future.
- The most common critical incident in anesthesia is due to breathing circuit disconnections which most commonly occur at the y-piece.

- During abdominal laparoscopy, the surgeon creates a pneumoperitoneum. What hemodynamic changes are seen in the otherwise healthy patient? What additional hemodynamic changes are seen in an ASA III-IV patient? [MM p. 389]
  - With creation of a pneumoperitoneum, intra-abdominal pressures reach 10-15 mmHg, compressing the vasculature and often shifting the diaphragm cephalad.
  - Hemodynamic consequences of these changes in the otherwise healthy patient are INCREASED SVR, INCREASED MAP, AND DECREASED CARDIAC INDEX. Cardiac output and heart rate are generally unchanged in the healthy patient.
  - In addition to these hemodynamic changes, ASA III – IV patients exhibit DECREASED CO, INCREASED PAOP, and INCREASED LV STROKE WORK INDEX upon creation of the pneumoperitoneum.
- FYI (M&M)
  - The hallmark of laparoscopy is the creation of a pneumoperitoneum with pressurized CO2. The resulting increase in intraabdominal pressure displaces the diaphragm cephalad, causing a decrease in lung compliance and an increase in peak inspiratory pressure. Atelectasis, diminished FRC, ventilation/perfusion mismatch, and pulmonary shunting contribute to a decrease in arterial oxygenation. These changes should be exaggerated in this obese patient with a long history of tobacco use.
  - The high solubility of CO2 increases systemic absorption by the vasculature of the peritoneum. This, combined with smaller tidal volumes because of poor lung compliance, leads to increased arterial CO2 levels and decreased pH.

- List 5 pulmonary changes associated with pneumoperitoneum [MM p. 389]
  - 1: Increased PIP
  - 2: Decreased Vital Capacity
  - 3: Decreased FRC
  - 4. Increased intrapleural pressure
  - 5: Decreased respiratory system compliance

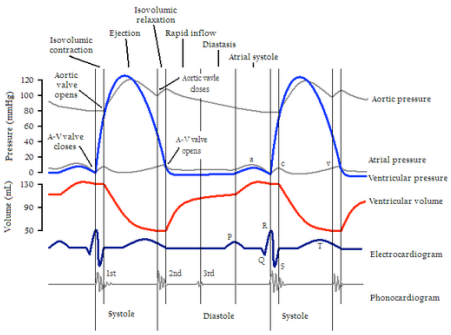
- Complications with a stellate ganglion block? [MM p. 349]
  - Horner's syndrome is caused by blockade of the stellate ganglion which is located at the lateral border of the vertebral body of C7. Because of this autonomic location, the axillary approach would be least likely to cause stellate ganglion blockade.

- S/S of Horner's syndrome: **ipsilateral**
  - Ptosis
  - Miosis
  - Facial and arm flushing (d/t vasodilation),
  - Increased skin temperature
  - Anhydrosis (lack of sweating on the face)
  - Nasal congestion.

- What nerves may be damaged during face mask ventilation?[MM p. 318]
  - The buccal branch of the FACIAL NERVE (CN VII)
  - The supraorbital branch of the TRIGEMINAL NERVE (CN V)
- FYI M&M: Mask ventilation for long periods may result in pressure injury to branches of the trigeminal or facial nerves.

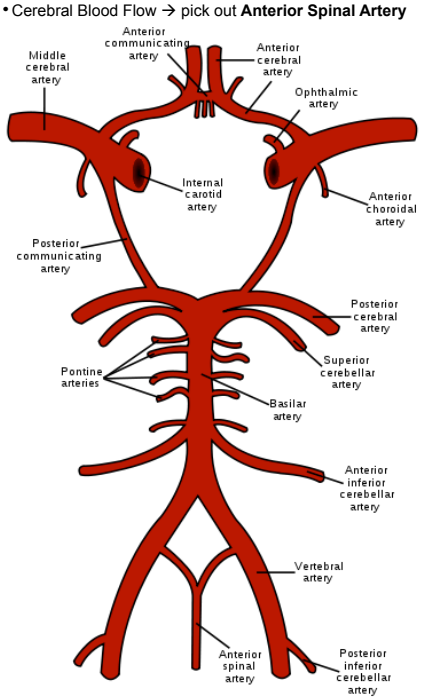
- How long after sulfur hexafluoride bubble is placed in the eye do you need to avoid N20? [MM p. 395]
  - The bubble of sulfur hexafluoride will increase in size (3 times within 30 minutes if 70% N20 is used) and IOP couldincrease. Thus, avoid N20 for **TEN DAYS** after treatment of ophthalmic pt with sulfur hexafluoride.
  - Turn N20 OFF at least 15 minutes before air bubble injection

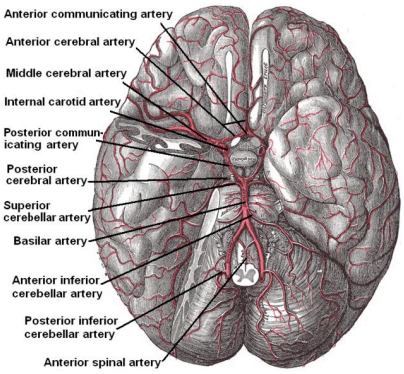
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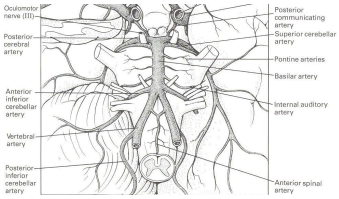


- Wigger's Cycle: Where does Aortic Valve open?

- Cerebral Blood Flow → pick out **Anterior Spinal Artery**







- Pick three with bladder perforation [MM p. 392]
  - Extraperitoneal: periumbilical, inguinal, suprapubic pain; lower abdominal distension, and pain
  - Intraperitoneal: abdominal rigidity, distension, pain; referred shoulder pain; hiccups; SOB, tachycardia, hypotension or HTN, diaphoresis; vomiting
- Most signs are extraperitoneal and the awake pt complains of: N/V, diaphoresis, retropubic or lower abdominal pain.
- Sudden nausea and vomiting and abdominal pain in the awake patient.

- The FDA Safe Medical Device Act of 1990:
  - Requires hospitals to "report instances in which medical devices cause or contribute to death, serious illness, or serious injury."
  - The act also requires any medical personnel whom become aware of the problem to remove the equipment from patient contact and report the problem to their supervisor(s).
  - The incident report must be made by the hospital risk manager to the FDA within 10 days of the incident (MM 518)

Anatomic Site	Leads	Coronary Artery Supply
Posterior, Inferior	II, III, aVF	Right (RCA)
Septum, Anterior	**V2 – V5**	**Left Anterior Descending (LAD)**
Lateral Wall	I, aVL, V4-V6	Left Circumflex

- Best lead to detect ischemia????
  - V5???
- FYI (MM 138)
  - What ECG leads best detect ischemia in the myocardium resulting from occlusion of the right coronary artery? Where is the myocardium supplied by the right coronary artery located?
    - Leads 11,111, and aVF monitor myocardium supplied by the right coronary artery. The inferior myocardium is supplied by the right coronary artery.
  - What ECG leads best detect myocardial ischemia in myocardium resulting from occlusion of the left circumflex artery? Where is the myocardium supplied by the left circumflex artery located?
    - Leads I, aVL, V5-V6 monitor the lateral myocardium, which is supplied by the left circumflex artery.
  - What ECG leads best detect ischemia in the myocardium resulting from occlusion of the left coronary artery? Where is the myocardium supplied by the left coronary artery located?
    - Leads I, aVL, V1-V4 monitor myocardium supplied by the left coronary artery. The anterior myocardium is supplied by the left coronary artery.
  - What ECG leads best detect ischemia in myocardium resulting from occlusion of the left anterior descending branch of the left coronary artery? Where is the myocardium supplied by the left descending branch of the left coronary artery located?
    - Leads V1-V4 monitor the myocardium supplied by the left descending branch of the left coronary artery; the anteroseptal myocardium is supplied by the left anterior descending branch of the left coronary artery.

	Formula	Normal Range
Cardiac Output	Heart Rate X Stroke Volume	
Cardiac Index:	Cardiac Output / BSA	2.5 – 4 L/min/m2
Ideal Body Weight (Males)	50 kg + 2.3 kg for each inch over 5 feet	
Ideal Body Weight (Females)	45.5 kg + 2.3 kg for each inch over 5 feet	
SVR	80 X (MAP – CVP) / CO	1200 – 1500 dynes/sec/cm-5
PVR	80 X (PAP – LAP) / CO	
Cerebral Perfusion Pressure	MAP – ICP	
Coronary Perfusion Pressure	DBP – PCWP (same as LVEDP)	60 -160 mmHg
MAP	(SBP + 2DBP) / 3	
Ejection Fraction	SV / EDV = EDV - ESV / EDV	(0.6 – 0.8) or 60 - 80%
BMI		
BSA		
Stroke Volume	EDV – ESV	60 – 90 mL (in a 70kg patient)
Stroke Index	**Stroke Volume** / BSA in m2	40 – 60 mL/beat/m2
Volume of Distribution	Q / Cp Q = quantity of drug injected (amount) Cp = (plasma conc. of drug that would be achieved)	$V_d = \frac{\text{Dose}}{\text{Concentration}}$

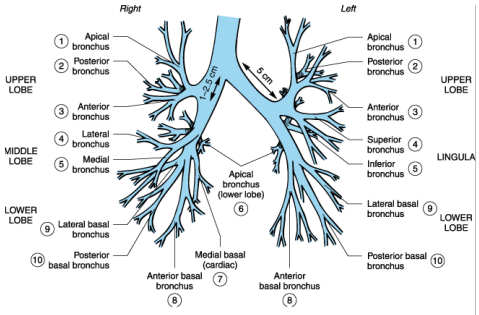
- Be able to calculate Cerebral Perfusion Pressure if the only information you were given were the BP & ICP.
- CPP = MAP – ICP

LMA Sizes

Valley p. 521

Patient	LMA Size	**Maximum Cuff Volume**	**Largest ETT (mm)**	**FOB size (mm)**
Neonates: < 5 kg	1	4 mL	3.5	2.7
Infants: 5 – 10 kg	1 ½	7 mL	4.0	3
10 – 20 kg	2	10 mL	4.5	3.5
20 – 30 kg	2 ½	14 mL	5.0	4
30 – 50 kg	**3**	**20 mL**	**6.0 Cuffed**	**5**
Adult	4	30 mL	6.0 Cuffed	5
Adult	5	40 mL	7.0 Cuffed	5
Large Adult	6	50 mL	7.0 Cuffed	5

- What are 4 absolute indications for one-lung ventilation? [MM p. 377]
  - [1] Presence of blood or infectious secretions of one-lung (contamination of both is life-threatening)
  - [2] Presence of brochopleural fistula or bronchocutaneous fistula
  - [3] Presence of giant cysts or unilateral bullae (which may rupture during PPV)
  - [4] Performing brochopulmonary lavage is an absolute indication for OLV (it is mandatory to avoid spillage of fluid)



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- What thyromental (chin to thyroid notch) distance in cm indicated a probably difficult airway?
  - TMD ≤ 6

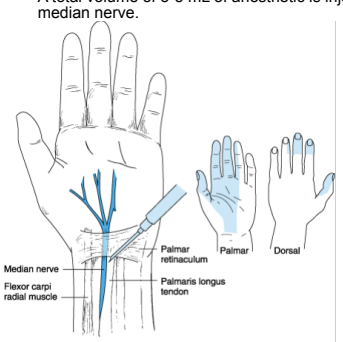
- What cranial nerve innervates the POSTERIOR 1/3 of the tongue & carries the sensation of taste?
  - Glossopharyngeal Nerve (CN IX)

- What cranial nerve innervates the ANTERIOR 2/3 of the tongue & carries the sensation of taste?
  - Facial Nerve (CN VII)

- What increases IOP?
  - succinylcholine, coughing, straining, gagging, bucking, increased BP increased PaCO2, increased CVP
- IOP is determined by the rate of aqueous humor formation and rate of outflow

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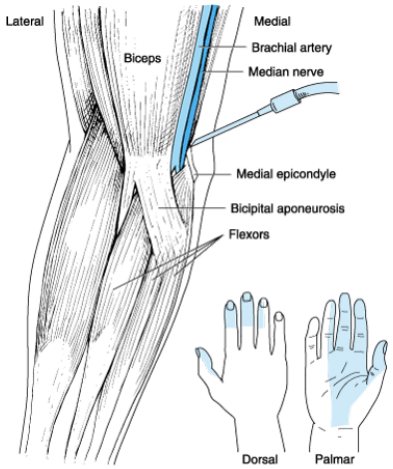
- Describe the landmarks and relative needle location in order to perform a median nerve block at the wrist. [MM p. 356]
  - In order to perform a median nerve block at the wrist, a 22-gauge needle is directed just medial to the ulnar artery pulse, or, if the ulnar pulse is not palpable, just medial to the flexor carpi radialis.
  - A total volume of 3-5 mL of anesthetic is injected to block the median nerve.



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Median nerve block at the wrist.

Asking the patient to flex the wrist against resistance can identify the palmaris longus tendon. It is marked at the proximal flexion crease. A 22- to 25-gauge, B-bevel needle is inserted just medial and deep to the palmaris longus, and 3–5 mL of anesthetic is injected.



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Median nerve block in the antecubital space.

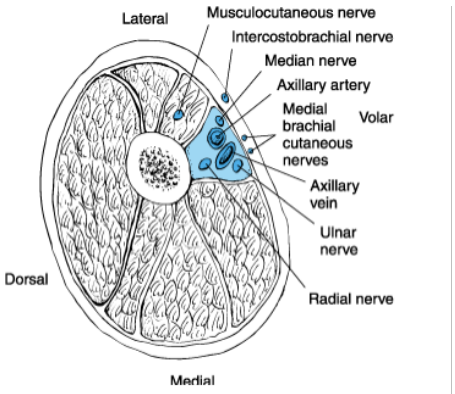
The brachial artery can be identified in the antecubital crease just medial to the biceps insertion. A 22-gauge, 1.5-in B-bevel needle is inserted just medial to the artery and directed toward the medial epicondyle until a paresthesia, motor-evoked response (wrist flexion), or periosteum is encountered. If periosteum is encountered, the needle is withdrawn 0.5–1 cm. Then 3–5 mL of anesthetic is injected.

- The National Practitioner Data Bank (NPDB) requires input (information) from what 5 sources?
  - [1] Medical Malpractice Payments
  - [2] License actions by medical boards
  - [3] Professional review or clinical privilege actions taken by hospitals and other health care entities
  - [4] Actions taken by DEA
  - [5] Medicare/Medicaid exclusions

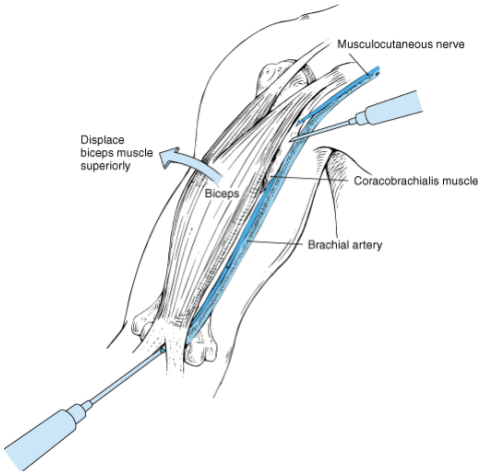
- What nerves are damaged with direct laryngoscopy? [MM p. 429]
  - Superior Laryngeal
  - Recurrent Laryngeal
  - Vagus, and/or
  - Hypoglossal
- Bilateral hypoglossal nerve damage may be found in patients with amyotrophic lateral sclerosis or myasthenia gravis.

- KNOW THIS DIAGRAM & position of nerves in relation to the artery
  - RUMM ☒ PISS [MM p. 348]
    - The median nerve lies anterior (superior) to the axillary artery, the ulnar nerve is medial and slightly posterior (inferior) to the artery, and the radial nerve is posterior and slightly lateral to the artery.

  - ?? If digging inferior/posterior… what nerve? – Ulnar n.??
  - **Radial n.**: posterior and slightly lateral to the artery
  - **Ulnar n.**: medial/slightly posterior inferior
  - **Median n.**: anterior (superior) to axillary nerve.
  - **Musculocutaneous n.**: most superior in the plexus, not the bundle; lies in the coracobrachialis muscle.



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Musculocutaneous block, showing injection into the coracobrachialis.

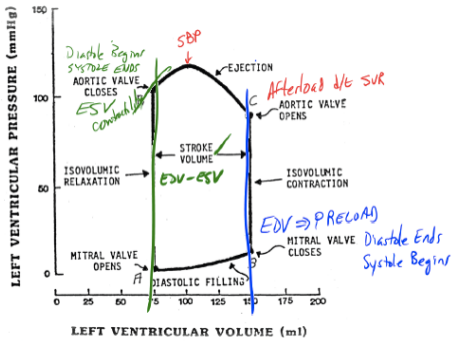
- Why are opioids prolonged in the elderly?
  - Decreased clearance
  - Increased volume of distribution for lipid soluble drugs, which decreases their plasma concentration and rate of elimination
  - Decreased dose requirement for barbs, opioids, and benzos
  - Increase body lipid storage sites prolongs the action of lipid soluble drugs
  - Decreased renal and hepatic blood flow
  - Prolonged action of the drugs dependent on kidneys & liver for elimination
  - Decreased volume of distribution for water soluble drugs (high initial plasma concentration)

- Afib



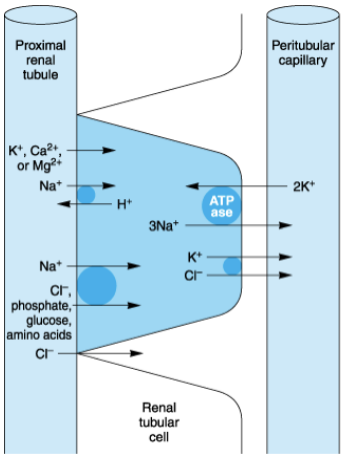
ECG of atrial fibrillation (top) and normal sinus rhythm (bottom). The purple arrow indicates a P wave, which is lost in atrial fibrillation.

- Pressure-Volume Loop
  - Where is isovolumeteric contraction?
    - between EDV (preload), the MV closure (preload) AND the beginning of systole with the AV closure
  - Where is isovolumetric relaxation?
    - from AV closure to MV closure



- Pick 3 with renal transplant:
  - Diuretic (Mannitol)
  - Steroids
  - Immunosuppresive therapy

- Where is Na more readily absorbed in the Kidneys?
  - Proximal Tubule
- FYI: Approximately two-thirds of all reabsorption and secretory processes in renal tubules takes place inproximal tubules. (MM 103)

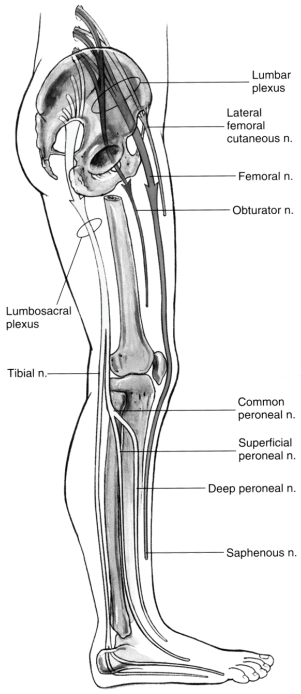


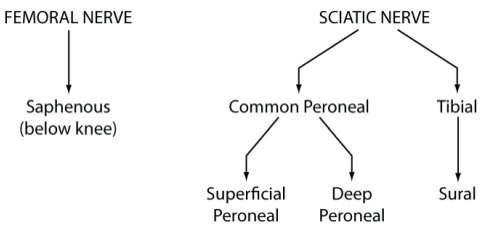
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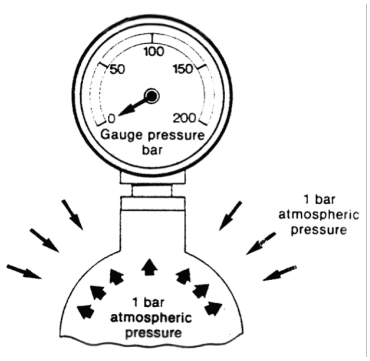
Reabsorption of solutes in proximal tubules. Note that Na+–K+-ATPase supplies the energy for reabsorption of most solutes by maintaining a low intracellular concentration of sodium.

- You want to block the superficial peroneal nerve, you would need to block the branches of what nerve? **Sciatic n.**

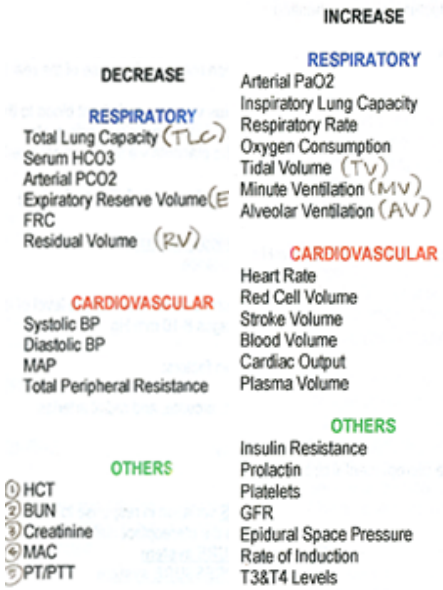




- Pressure of O2 in cylinder… at 1 atm ????
  - At 20 degrees C, a full E-cylinder of O2, has a pressure of 2200 psi and contains 660L
- FYI (Valley Review): Bourdon Gauges
  - Bourdon gauges measure high pressures (e.g., cylinder pressures); gauge pressures measure the pressure relative to atmospheric pressure, not the absolute pressure.
  - When the gauge pressure reads zero, the pressure in the cylinder is the same as atmospheric.



- Pick 3 that INCREASE in the parturient



- Cardiac Changes:
  - Greatest decrease is SVR by 20%
  - Greatest increase is CO by 50%

- Respiratory parameters that DON'T CHANGE in the pregnant patient
  - Deadspace
  - Lung compliance
  - Arterial blood pH
  - FEV1
  - VC
  - Diffusing capacity
  - Closing volume, and
  - TLC

- Lateral decubitus in the awake patient, what are the V/Q mismatches?
  - No Change
  - In the spontaneously breathing individual in the LD position, ventilation & perfusion are BEST in the DEPENDENT lung.
  - *

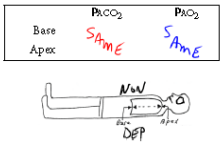
Ventilation and Perfusion - Lateral Decubitus (Summary)

	Unanesthetized		Anesthetized	
	V	Q	V	Q
Non-Dependent Lung	↓	↓	↑	↓
Dependent Lung	↑	↑	↓	↑

Know and understand this table.

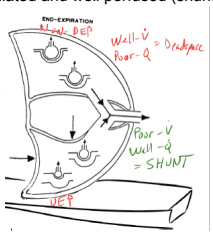
Know: When the patient in the lateral decubitus position is anesthetized and paralyzed, a ventilation:perfusion (V/Q) mismatch develops. Ventilation of the non-dependent lung increases (the dependent lung is difficult to inflate), but the distribution of blood flow does not change, compared to the awake, spontaneously breathing subject.

- FYI Valley Review
  - The awake, spontaneously breathing patient is in the supine (or prone or lateral decubitus) position. Indicate in the table below how PAO2 in the base compares with PAO2 in the apex; also indicate how PACO2 in the base compares with PACO2 in the apex.



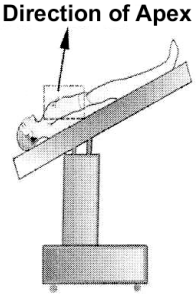
  - Lateral Decubitus -- Unanesthetized
    - In the spontaneously breathing individual in the lateral decubitus position, the distributions of ventilation and perfusion to dependent and nondependent lung are similar to those found in the upright position.
  - Lateral Decubitus -- Anesthetized and Paralyzed

- When the patient in the lateral decubitus position is anesthetized and paralyzed, the dependent lung is "compressed" by the weight of the abdominal contents (see arrow). With positive pressure ventilation, inspiratory gases are preferentially distributed to the nondependent lung. A clinically significant mismatch in ventilation and perfusion develops. The nondependent lung is well ventilated but poorly perfused (deadspacing), and the dependent lung is poorly ventilated and well perfused (shunting).



- Complications of Steep-Trendelenburg
  - Hypotension, Blood loss/hypovolemia, VAE, ocular complications (retinal detachment, cerebral edema), Venous thrombosis, **ETT migration (endobronchial)**, atelectasis, neuropathy, arthralgia, finger injuries, regurgitation.

Direction of Apex



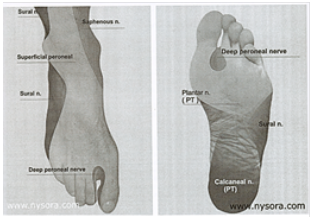
- What are contraindications to FOI? Five contraindications for fiberoptic bronchoscopy are (MM 332):
  - **hypoxia**
  - heavy airway secretions not relieved with suction or antisialagogues
  - bleeding from the upper or lower airway not relieved with suction
  - local anesthetic allergy (for awake attempts)
  - inability to cooperate (for awake attempts)

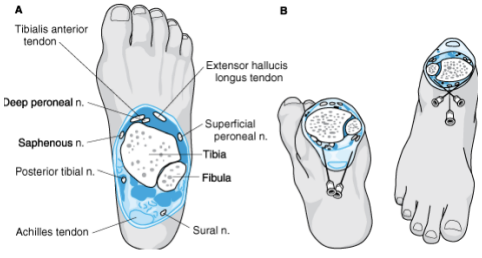
- What is a common complication in the PACU for pediatric patient > 3 years old?
  - IV access, Increased HR, **Hypoxia** ?? → oropharyngeal obstruction

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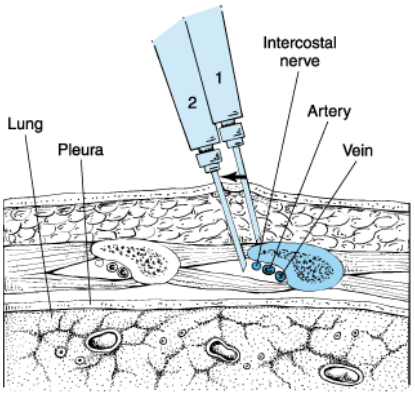
- Geriatric with decreased sensitivity to atropine?
  - Elderly require an INCREASED dose of atropine d/t increased vagal tone.

- Ankle Block
  - Ankle block consists of anesthetizing five nerves.
    - 1. Posterior tibial: L4-5, S1-3
    - 2. Sural: branch of tibial
    - 3. Saphenous: branch of femoral
    - 4. Deep peroneal: L4-5, S1-2
    - 5. Superficial peroneal: L4-5, S1-2
- Which nerve of the ankle is the most difficult to block?
  - Posterior Tibia
- Which nerves of the ankle are most superficial? "S"
  - Sural
  - Saphenous
  - Superficial peroneal (the most superficial)
- NO EPI





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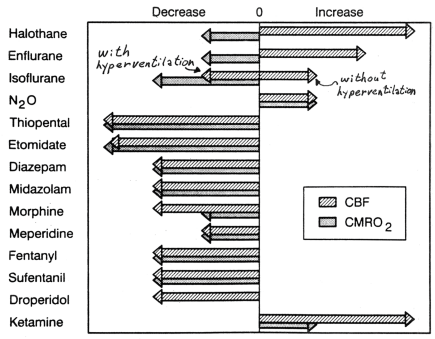


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- Anesthetic goals for aneurysm clipping?
  - (1) avoid aneurysm rupture (avoid increases in arterial blood pressure)
  - (2) maintain cerebral perfusion pressure and transmural pressure (transmural pressure = MAP- ICP), and
  - (3) provide a "slack" brain.
    - Slack brain: hyperventilate, mannitol, barbs, and CSF drainage.
    - Reduce intracranial pressure slowly to "relax" the brain. Diuretics (Lasix, mannitol) and head-up position may be helpful in this regard (MM 385).

- Lead II is the best to detect arrhythmias → presence of larger p waves.

- What inhalation agent decreases CMO2, but increases CBF the most?
  - Halothane



CBF = Cerebral Blood Flow
CMRO2 = Cerebral Metabolic Rate of Oxygen (Oxygen Consumption)

FYI: Know: (1) With volatile agents, cerebral metabolism decreases and cerebral blood flow increases; during hyperventilation, cerebral blood flow may be decreased during isoflurane anesthesia; (2) with the exception of ketamine, intravenous anesthetics decrease both cerebral metabolism and cerebral blood flow; (3) ketamine increases both cerebral blood flow and cerebral metabolism; (4) nitrous oxide by itself increases both cerebral blood flow and cerebral metabolism.

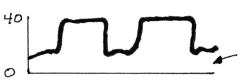
- What decreases EEG the most?
  - Hypothermia, **propofol**, thiopental, something else ????
  - Desflurane & Isoflurane produce dose dependent depression of EEG

- What drug do you NOT give to a patient with muscle contractures??????
  - Dantrolene, **Muscle Relaxants**, Propofol, Thiopenthal
  - Don't give Sux

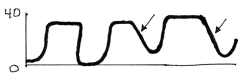
- What would you see with patients receiving glycine??????
  - Increased glucose, hyperthermia, hypOkalemia, pulmonary edema, resp. failure
  - Transient blindness with TURP
  - (MM 391): Hyperglycinemia leads to transient blindness, ammonia toxicity, nausea and vomiting, headache, ECG changes, myocardial depression, and loss of light and accommodation reflexes.

- What is the most likely reason a patient who undergoes a non-opthalmic procedure develops blindness post-operatively?
  - Optic nerve and retinal ischemia

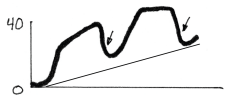
- Inspiratory valve failure



- Incompetent Expiratory Valve
- Incompetent Inspiratory Valve
- Exhausted soda lime canister due to storage of machine

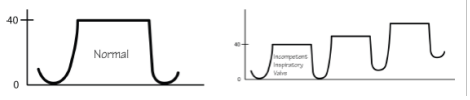


- Incompetent Inspiratory Valve
  - not returning to zero axis
  - inspiratory limb prolonged



- Incompetent Inspiratory Valve
  - note the "take-off" angle

4. If the $ETCO_2$ waveform is on the x-axis and then begins to "take-off" and leave the x-axis, there is an incompetent inspiratory valve.



- Consider a ventilator in pressure control mode: what parameter fluctuates with each cycle? What patient parameters determine this fluctuation?
  - In pressure control mode, the ventilator is set so that the inspiratory pressure is greater than the positive end-expiratory pressure. In this mode, tidal volume fluctuates (varies) with alterations in patient pulmonary compliance, pulmonary resistance, and with patient-ventilator asynchrony.

- Calculate the partial pressure of CO2 in expired gas if ETCO2 is 5%?

$$PPgas = \frac{\%\ \text{concentration of gas}}{100} \times \text{atm}$$

$$PPgas = \frac{5}{100} \times 760\ \text{mmHg}$$

$$PPgas = 0.05 \times 760\ \text{mmHg}$$

$$PPgas = 38\ \text{mmHg}$$

Dalton's law of partial pressure

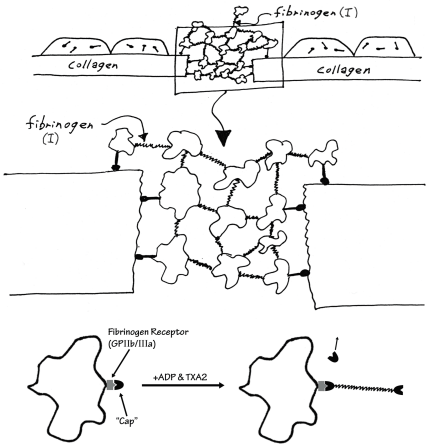
- Describe the diameter index safety system (DISS), and state its purpose.
  - The pipeline inlet fittings are gas specific DISS threaded body fittings. The DISS provides threaded non-interchangeable connections for medical gas lines, and this minimizes the risk of misconnection.
  - Ex: pipeline inlet fittings

- Standard of Care:
  - The standard ofcare is the conduct and skill of a prudent practitioner that can be expected by a reasonable patient.
  - Standard of care is also defined as reasonable care by a professional. (Nagelhout).
  - Standard of care is a very important medicolegal concept because a bad medical result due to failure to meet the standard of care is malpractice.

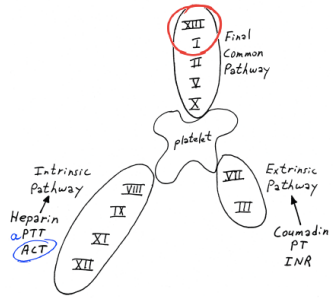
- What anesthesia agents inhibit HPV?
  - HPV is inhibited by a high concentration of a volatile agent, possibly at 1.0-1.5 MAC, or higher.
  - Nipride, NTG, Hydralazine
- Opioids do NOT decrease HPV

- Compare hydrophilic and lipophilic opioids with regard to early depression of ventilation after intrathecal or epidural placement. Early ventilatory depression, due to intravascular absorption of opioid,occurs with lipophilic opioids when placed either in the intrathecal or epidural space, but ventilator depression is most pronounced after epidural placement. "Most reports of clinically important depression of ventilation involve epidural depression of ventilation by fentanyl or sufentanil." "Clinically significant early depression of ventilation after intrathecal injection of morphine is unlikely." Although unlikely, early depression of ventilation may be seen with epidural morphine. Early depression of ventilation is not observed with intrathecal morphine. Lipophilic opioids will produce early depression of ventilation following either intrathecal or epidural placement.
- Compare hydrophilic and lipophilic opioids with regard to late (delayed) depression of ventilation after intrathecal or epidural placement. "All reports of late (delayed) depression of ventilation involve morphine." Morphine is hydrophilic.

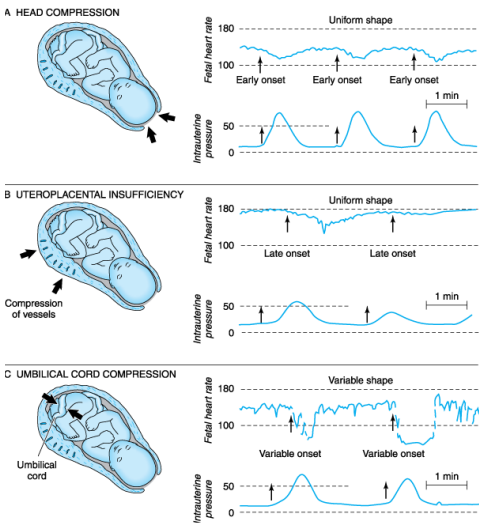
- Know how Plavix works in the coagulation pathway.
- Plavix (clopidogrel) is a an anti-ADP agent (life-time of the platelets).
  - Irreversible inhibitor of the ADP receptor on platelet membranes
- FYI (valley review): Mechanism of Platelet Aggregation
  - Thromboxane A2 and ADP uncover fibrinogen receptors. Fibrinogen attaches to its receptors, thereby linking platelets to each other. At this point, the clot is still water-soluble and friable.



- Intrinsic pathway includes what factors?
  - The intrinsic pathway is initiated by damage inside the blood vessel.
  - 12, 11, 9, 8
  - "If you cannot buy the intrinsic pathway for $12, you can get it for $11.98."
  - Heparin : aPTT, ACT
- Extrinsic pathway includes what factors?
  - The extrinsic pathway is initiated in response to damage occurring outside the blood vessel.
  - 3, 7
  - Coumadin: **PT**, INR
  - "For 37 cents you can purchase the extrinsic pathway."
- Final common pathway factors include what factors?
  - 10, 5, 2, 1, 13
- The clotting cascade can be activated in response to tissue trauma (extrinsic pathway), which triggers the release of tissue factor (factor III, also known as thromboplastin), or damage to the vascular endothelium or blood (intrinsic pathway).



- Uteroplacental insufficiency causes?
- EARLY DECELERATIONS (TYPE I)
  - Etiology: head compression or stretching of neck during uterine contractions (vagal stimulation), decreasing fetal heart rate to 10–14 bpm
  - Action: mirror the patterns of the uterine contraction and usually not associated with fetal distress; occurs during descent of the fetal head
  - occur with each uterine contraction
  - start and end with the contraction
  - gradually decrease in rate and then end in a return to baseline
  - are uniform in appearance
  - are associated with a mild decrease in fetal heart rate (FHR) (20 bpm or less)
- LATE DECELERATIONS (TYPE II)
  - Etiology: **uteroplacental insufficiency** and fetal compromise with a decrease in heart rate. Occurs at or following the peak of uterine contraction. Decrease in heart rate due to a decrease in arterial oxygen tension at chemoreceptors or the sinoatrial (SA) node
  - Action: begin near onset of uterine contraction
  - occur with each uterine contraction
  - start between 10 and 30 seconds after the uterine contraction
  - gradually decrease in rate and end in a return to baseline
  - are uniform in appearance
  - vary in depth according to the strength of the uterine contraction
  - may or may not be accompanied by beat-to-beat variability
  - are classified as severe if the FHR decreases by more than 45 bpm
- VARIABLE DECELERATIONS
  - Etiology: cord compression usually seen in the second stage of labor
  - Action: nonuniform shape with variable waveform
  - most common
  - variable onset, duration, and magnitiude (>30 bpm)
  - abrupt onset due to umbilical cord compression and acute intermittent decreases in umbilical blood flow
  - typically associated with fetal asphyxia when:
    - (1) greater than 60 bpm fetal heart rate
    - (2) duration greater than 60 seconds or,
    - (3) occur in a pattern that persists for more than 30 minutes
  - vary in appearance, duration, depth, and shape
  - demonstrate abrupt onset and recovery
  - maintains beat-to-beat variablilty with the deceleration
  - are classified as severe if the FHR decreases by 60 bpm, if the FHR decreases to less than 60 bpm, or if the decelerations last 60 seconds or longer

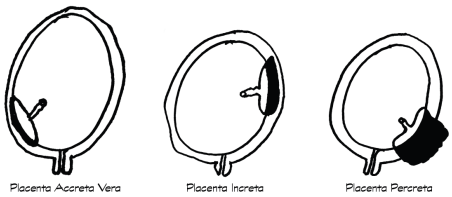


Source: Morgan GE, Mikhail MS, Murray MJ: *Clinical Anesthesiology*, 4th Edition: http://www.accessmedicine.com

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A: Early (type I) decelerations. B: Late (type II) decelerations. C: Variable (type III) decelerations.

- Abnormal Placental Implantation



- **Placenta accreta vera**: adherence to the myometrium without invasion of or passage through uterine muscle.
- **Placeta increta**: invades and is confined to the myometrium
- **Placenta percreta**: invades and may penetrate the myometrium, the uterine serosa, or other pelvic structures.
- Remember, if mom has had a previous c-section or had uterine trauma , she is at risk for developing placenta accreta. Also, the more C-sections the greater the incidence of accreta.

- Signs/Symptoms of aortocaval compression:
  - Signs and symptoms: (1) Diaphoresis, (2) nausea, (3) vomiting, (4) changes in cerebration, (5) dizziness, (6) vertigo, (7) tachycardia, and (8) apprehension
  - Treatment: Left uterine displacement (right hip up, LUD). This maneuver is most readily accomplished by placing a wedge (> 15°) under the right hip.

- Central venous pressure (CVP) increases substantially (to, for example 25 mm Hg) during cardiopulmonary bypass. Why?
  - An increased central venous pressure during cardiopulmonary bypass occurs if there is an obstruction to venous drainage. Causes of obstruction to venous return include: improper positioning of the venous cannula in the patient; a kink in the venous return tubing; entrained air in the venous return tubing creating an "air lock"; and unintended partial clamping of the venous return tubing.

- Atrial Natriuretic Peptide (ANP)
  - Released into circulation as a hormone in response to atrial stretching
  - Binds to G-protein coupled receptors and signal the kidneys to excrete sodium & water.
  - Elevated ANP & BNP levels in the CHF patient may be a predictor of mortality.
- FYI MM: BNP is B-type natriureticpeptide, also known as brain natriuretic peptide. BNP is produced in the atria of the heart (it is called brain natriuretic peptide because it was first discovered in the brain) and functions similarly to ANP-atrial natriuretic peptide. ANP & BNP are released into the circulation (hormones) in response to atrial stretching. Natriuretic peptides bind to G-protein coupled receptors and signal the kidney to excrete sodium and water. Elevated ANP & BNP levels in CHF may be a predictor of mortality.

- Poiseuille's Law (MM 246)
  - Flow is directly proportional to:
    - (1) the pressure drop along the tube;
    - (2) the fourth power of the radius (r4) of the tube.
  - Flow is inversely proportional to:
    - (3) the length of the tube; and
    - (4) the viscosity of the fluid.
- If the radius of a tube decreases to one-third of its original size, flow will decrease to 1181 of its original value. Flow is proportional to the radius to the fourth power (r4). 1/3 x 1/3 x 1/3 x 1/3 = 1/81. Poiseuille's law applies. Note also that if radius is halved, flow is reduced to 1/16th of its original value. If radius is doubled, flow increases 16-fold. If radius is tripled, flow increases 81-fold.
- According to Poiseuille's law, when radius is doubled, resistance is decreased to 1/16th of its original value and flow is increased 16-fold; when radius is halved, resistance is increased 16-fold and flow is decreased to 1/16th of its original value; when radius is tripled, resistance is decreased to 1/81 of its original value and flow is increased 81-fold; when radius is decreased to 1/3rd of its original value, resistance is increased 81-fold and flow is decreased to 1/81 of its original value.

$$F = \frac{\pi r^4 \Delta P}{8\eta L}$$

$$R = \frac{\Delta P}{F}$$

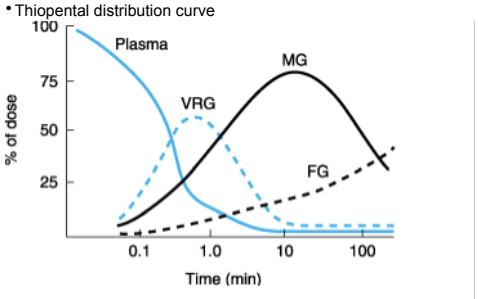
$$R = \frac{8\eta L}{\pi r^4}$$

- What inhalation agents are most metabolized by the liver?

  Halothane > **Sevo** > Enflorane > Iso > Des > N2O

  - Halothane (15-20%)
  - Sevoflorane (5%)
  - Enflurane (2.4%)
  - Isoflurane (0.2%)
  - Desflurane (0.02)
  - N20 (0.004)

- Thiopental distribution curve



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Distribution of thiopental from plasma to the vessel-rich group (VRG), to the muscle group (MG), and finally to the fat group (FG).

-

- The duration of action of highly lipid-soluble barbiturates (thiopental, thiamylal, and methohexital) is determined by redistribution, not metabolism or elimination. For example, although thiopental is highly protein bound (80%), its great lipid solubility and high nonionized fraction (60%) account for maximal brain uptake within 30 s. If the central compartment is contracted (eg, hypovolemic shock), if the serum albumin is low (eg, severe liver disease), or if the nonionized fraction is increased (eg, acidosis), higher brain and heart concentrations will be achieved for a given dose. Subsequent redistribution to the peripheral compartment— specifically, the muscle group—lowers plasma and brain concentration to 10% of peak levels within 20–30 min (Figure 8–4). This pharmacokinetic profile correlates with clinical experience—patients typically lose consciousness within 30 s and awaken within 20 min. Induction doses of thiopental depend on body weight and age. Lower induction doses in elderly patients is a reflection of higher peak plasma levels because of slower redistribution. In contrast to the rapid initial distribution half-life of a few minutes, the elimination halflife of thiopental ranges from 3–12 h. Thiamylal and methohexital have similar distribution patterns, whereas less lipid-soluble barbiturates have much longer distribution half-lives and durations of action. Repetitive administration of barbiturates saturates the peripheral compartments, so that redistribution cannot occur and the duration of action becomes more dependent on elimination.

- What drives K into the cell within minutes?
  - HCO3 (3-5 minutes) – fastest shift of K+ into the cell
  - Hyperventilate (10-15 min)
  - Insulin-glucose (~ 30 min)
  - Beta Adrenergic Agonists
  - Dialysis (Days)
  - Kayexalate (Hours)

  - IV Calcium (Fastest) – BUT doesn't correct the problem, rather it protects the heart.

- FYI Valley Review: Therapies and Mechanisms for Treating Hyperkalemia (fastest to slowest)
  - 1. Administer HCO3
    - When HCO3 is administered, H+ concentration in plasma decreases (metabolic alkalosis). H+ shifts out of cells to buffer the alkalosis, and, in exchange, K+ shifts into cells.
    - can decrease plasma [K+] within 15 min (M&M)
  - 2. Hyperventilate
    - With hyperventilation, H+ concentration in plasma decreases (respiratory alkalosis). H+ shifts out of cells to buffer the alkalosis, and, in exchange, K+ shifts into cells.
    - Know: for each 10 mmHg decrease in PaCO2, serum [K+] decreases 0.5 mEq/L.
  - 3. Give insulin-glucose
    - Insulin, by stimulating the sodium-potassium pump, drives K + into cells. Insulin also opens glucose channels. Glucose is administered along with the insulin to prevent hypoglycemia.
  - 4. Give Ca++
    - (5–10 mL of 10% calcium gluconate or 3–5 mL of 10% calcium chloride) partially antagonizes the cardiac effects of hyperkalemia and is useful in patients with marked hyperkalemia. Its effects are rapid but unfortunately short lived. Care must be exercised in patients taking digoxin, as calcium potentiates digoxin toxicity.
  - 5. Administer a β2-adrenergic receptor agonist (to stimulate the sodium-potassium pump)
  - 6. Dialyze patient
  - 7. Give Kayexalate

- What is the best choice of diuretic for Cushing's syndrome?
  - Potassium-sparing diuretics
    - Spironolactone (Aldactone)
    - Triamterene (Dyrenium)
    - Amiloride (Midamor)
- Pts with Cushing's tend to be volume overloaded, and have hypokalemic metabolic alkalosis from the activity of glucocorticoids. This should be correct perioperatively with supplemental K and Aldactone

- FYI: MM 93
- Cushing's syndrome (hyperadrenocorticism) is a cluster of clinical abnormalities due to excessive circulating levels of cortisol…but secretion of androgens is also sometimes significant.
- In two-thirds of all cases, Cushing's disease is caused by over-production of ACTH by the anterior pituitary.
- signs and symptoms of Cushing's disease (hypertension, hypoka lemia, and hyperglycemia) are related to excessive release of glucocorti coids (namely, cortisol) from the adrenal cortex. Excessive levels of cortisol promote increased sodium reabsorption and sodium retention. Accompanying the sodium retention is water retention, which leads to hypervolemia and hypertension. Cortisol has slight mineralocorticoid (aldosterone-like) actions. Excessive levels of cortisol promote potassium secretion and excretion, and hence the development of hypokalemia. Excessive levels of cortisol stimulate gluconeogenesis and block entry of glucose into skeletal muscle cells and adipose tissue, so hyperglycemia develops.

- A patient with a cervical injury has a urine osmolality of 100 mOsm/kg. What is the cause?
  - Diabetes Insipidus

- FYI: MM 83
- The untreated patient with diabetes insipidus has (1) hypernatremia (hyperosmolality) and (2) a dilute urine (osmolality < 200 m0sm/kg). The hallmark of diabetes insipidus is polyuria (large volumes of dilute urine), confirmed by the kidney's inability to concentrate urine, despite water deprivation.

- Murmur at the right & left sternal border 2nd ICS?
  - AS: 2nd Right ICS
  - AR: Left Sternal Border (2nd LEFT ICS ?????)
- fyi: MM p. 1
  - Aortic stenosis is recognized by its characteristic systolic murmur, best heard in the second right intercostal space (over the aortic arch) with transmission into the neck.
  - Aortic regurgitation is recognized by its diastolic murmur, best heard along the left sternal border.

- Pyloric Stenosis → projectile vomiting → **hypochloremic metabolic alkalosis**

- What drugs use cAMP as a 2nd messenger?
  - NE, Epi, Albuterol, glucagon, beta agonist, histamine, serotonin
  - PDI increases intracellular cAMP

- What two drugs have synergistic effects?
  - EPI & Cocaine → tachycardia / HTN ???
  - Cocaine inhibits re-uptake of NE

- What are two causes of ST segment ELEVATION?
  - Transmural ischemia (Prinzmetal's angina) or Transmural infarction
  - Transmural has a Q wave
- FYI MM 139
  - What are two causes of ST segment depression? Subendocardial ischemia or subendocardial infarction may lead to ST segment depression. Recall also that digitalis may cause downsloping of the ST segment.

- What is a problem associated with Scoliosis (a restrictive disease)
  - Reduced lung volumes
  - Decreased chest wall or lung compliance
  - Decreased PaO2, and increased PaCO2 (secondary to V/Q mismatch)
  - **PHTN (secondary to increased PVR from chronic hypoxia)**

- Hypothyroidism → possible difficult airway
- FYI MM 447
  - Name four conditions in which the patient presents with a large tongue. What is the concern if the patient has a large tongue?
  - Down's syndrome, Pierre Robin syndrome, acromegaly, and hypothyroidism. Patients with large tongues are prone to respiratory obstruction.

- Anesthetic concerns with Trisomy 21 (Down's Syndrome)
  - Difficult Airway: (1) Intubation may be difficult owing to the large tongue, short neck, small mouth, and subglottic stenosis (select endotracheal tube one size smaller than anticipated)
  - (2) neckflexion during laryngoscopy and intubation may result in atlanto-occipital (cervical spine) dislocation because of congenitally weak ligaments (considergetting cervical-spinefilms)
  - (3) congenital heart disease is present in 40% of patients (anesthesia compatible with the patient's congenital heart defects should be delivered).

- Patients with myasthenia gravis have greatly increased sensitivity to nondepolarizing agents. These patients may be either very sensitive or resistant to succinylcholine. If possible, muscle relaxants should be avoided in the patient with myasthenia gravis.

- What are contraindications of lithotripsy?
  - Inability to position the pt with the lung and intestines away from the wave focus
  - Urinary obstruction below the stone, Untreated infection, a bleeding diathesis, pregnancy
  - **The presence of a near-by AAA, or orthopedic prosthetic device (relative contraindications)**

- What has both Factor 8 and Factor 5?
  - FFP has all except platelets

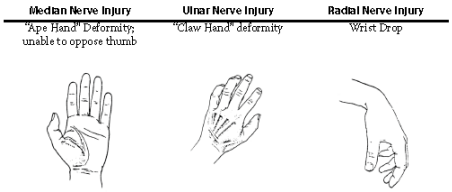
- TURP Syndrome (MM 390)

Cardio	Hemotologic/Renal	CNS
Respiratory Distress	Hemolysis	N/V
Cyanosis	ARF	Confusion
HTN	HypOnatremia	Twitching
HypOtension	HypOmolarity	Visual disturbances
Widened QRS/Increased ST	HypERglylcemia	Seizures
Dysrhythmias	HypERammonemia	Paralysis
Bradycardia	Coma	Shock

- Complications of TURP
  - Central pontine myelinolysis, MI, fibrinolysis, bladder perforation, hyperglycemia
  - Hyperglycinemia leads to transient blindness, ammonia toxicity, N/V, ha, EKG changes, myocardial depression, and loss of light and accommodation reflexes.

- Nerve Injuries

Median Nerve	"Ape Hand" deformity; unable to oppose thumb
Ulnar Nerve	**"Claw Hand" deformity**
Radial Nerve	Wrist drop, inability to abduct thumb
Axillary	Inability to abduct arm
Musculocutaneous	Inability to flex forearm



- Dosages for caudal block
  - 0.5–1 mL/kg of Bupivacaine 0.125% - 0.25%

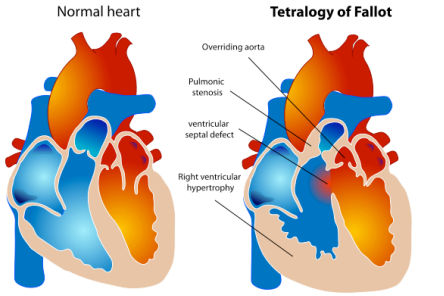
- Patient with acute trauma of elbow… consider?
  - Regional block may disguise compartment syndrome???

- Avogadro's Number and Hypothesis. Avogadro described the relationship between the amount of gas and volume of gas.
  - a. Avogadro's number: the number of molecules in one mole of a substance is 6.022 x 1023.
  - b. Avogadro's hypothesis: one mole of a gas at standard temperature (0°C) and standard pressure (1 atm) occupies a volume of 22.4 liters.
  - c. Problem: Two moles of gaseous N2O occupy what volume under standard conditions? Under standard conditions, two moles of gaseous N2O occupy 22.4 x 2 = 44.8 liters.
  - d. Problem: One-half mole of O2 occupies what volume under standard conditions? 0.5 x 22.4 = 11.2 liters.

Action	Muscle	pneumonic
ABducts the vocal cords	POSTERIOR cricoarytenoids	"Pab"
ADducts the vocal cords	LATERAL cricoarytenoids	"Lad"
Tenses the vocal cords	Cricothyroids	"CT scan"
Relaxes the vocal cords	Thyroarytenoids	"RT"

- Goals for managing patient with Tetralogy of Fallot (TOF)
  - Maintain intravascular volume & SVR
  - Increases in PVR, such as might occur from acidosis or excessive airway pressures, should be avoided.
  - The right-to-left shunting tends to slow the uptake of inhalation anesthetics; in contrast, it may accelerate the onset of intravenous agents.
    Ketamine (intramuscular or intravenous) is a commonly used induction agent because it maintains or increases SVR and therefore does not aggravate the right-to-left shunting. Patients with milder degrees of shunting generally tolerate inhalation induction with halothane.
  - The right-to-left shunting tends to slow the uptake of inhalation anesthetics (see Chapter 7); in contrast, it may accelerate the onset of intravenous agents. Oxygenation often improves following induction of anesthesia.
  - Muscle relaxants that release histamine should be avoided.
  - Hypercyanotic spells may be treated with intravenous fluid and phenylephrine (5 g/kg).
  - Propranolol (0.1 g/kg) may also be effective in relieving infundibular spasm. Sodium bicarbonate, to correct the resulting metabolic acidosis, may also be helpful when the hypoxemia is severe and prolonged.

- FYI: Tetralogy of Fallot
  - This tetralogy classically includes (1) right ventricular obstruction, (2) right ventricular hypertrophy, and a (3) VSD with an overriding aorta.
  - The combination of right ventricular obstruction and a VSD results in ejection of unoxygenated right ventricular blood as well as oxygenated left ventricular blood into the aorta.
  - The right-to-left shunting across the VSD has both fixed and variable components. The fixed component is determined by the severity of the right ventricular obstruction, whereas the variable component depends on SVR and PVR.
  - Neonates with severe right ventricular obstruction may deteriorate quickly as pulmonary blood flow decreases when a PDA starts to close. Intravenous prostaglandin E1 (0.05–0.2 g/kg/min) is used to prevent ductal closure in such instances.

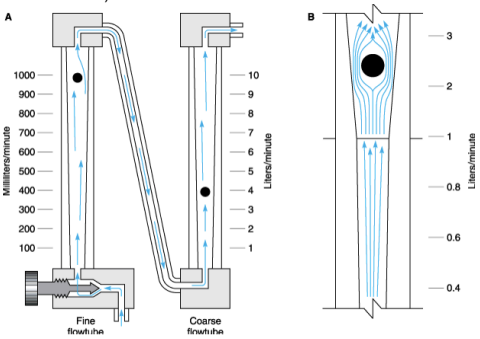


- FRC = RV + ERV
  - Calculate FRC: TV is 10% of TLV. So in a 70kg patient, we can assume that TV is 700 mL making TLV 7000. FRC is 30% of TLV, so FRC is 2100mL
  - Summary:
    - TV is 10% of TLV
    - FRC is 30% of TLV

- Estimated Blood Volume
- Infants
  - Premature (< term) 95 mL/kg
  - Full-Term 85 mL/kg
- Infant (< 6 weeks) 80 mL/kg
- Toddler (6 wks – 2 years) 75 mL/kg
- Child (2 yrs – 12 yrs) 72 mL/kg
- Adult Male 70-75 mL/kg
- Adult Female 65 mL/kg

- FYI: The transfusion point can be determined preoperatively from the hematocrit and by estimating blood volume (Table 29–5). Patients with a normal hematocrit should generally be transfused only after losses greater than 10–20% of their blood volume. The exact point is based on the patient's medical condition and the surgical procedure. The amount of blood loss necessary for the hematocrit to fall to 30% can be calculated as follows:
  - **1.** Estimate blood volume from Table 29–5.
  - 2. Estimate the red blood cell volume (RBCV) at the preoperative hematocrit (RBCVpreop).
  - 3. Estimate RBCV at a hematocrit of 30% (RBCV30%), assuming normal blood volume is maintained.
  - 4. Calculate the red cell volume lost when the hematocrit is 30%; RBCVlost = RBCVpreop – RBCV30%.
  - 5. Allowable blood loss = RBCVlost x 3.

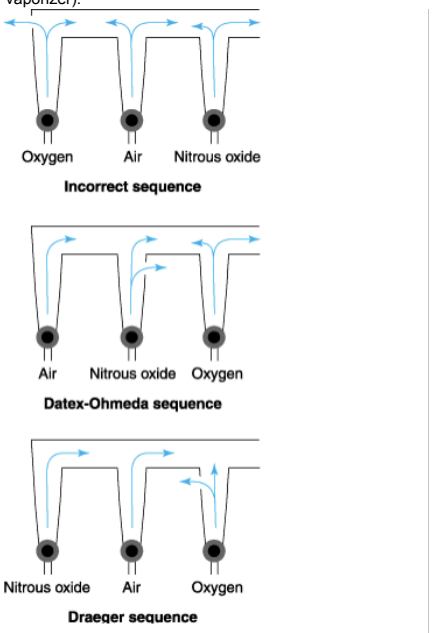
- Thorpe Tube
  - The tapered (variable orifice) tube with an indicator bobbin that is part of the constant pressure variable oriface flow meter found in the anesthesia machine
  - Located in the manifold area of the anesthesia machine and is tapered with the smallest diameter at the bottom and the largest at the top.
  - Variable orifice flow meters use rotameters (rotating indicators)



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fyi

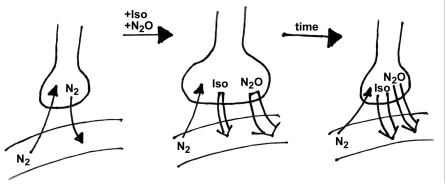
- Flowmeters on anesthesia machines are classified as either constant-pressure variable-orifice or electronic flowmeters. In constant-pressure variable-orifice flowmeters, an indicator ball, bobbin, or float is supported by the flow of gas through a tube (Thorpe tube) whose bore (orifice) is tapered. Near the bottom of the tube, where the diameter is small, a low flow of gas will create sufficient pressure under the float to raise it in the tube. As the float rises, the orifice of the tube widens, allowing more gas to pass around the float. The float will stop rising when its weight is just supported by the difference in pressure above and below it. If flow is increased, the pressure under the float increases, raising it higher in the tube until the pressure drop again just supports the float's weight. This pressure drop is constant regardless of the flow rate or the position in the tube and depends on the float weight and tube cross-sectional area.
- Flowmeters are calibrated for specific gases, as the flow rate across a constriction depends on the gas's viscosity at low laminar flows and its density at high turbulent flows. To minimize the effect of friction between them and the tube's wall, floats are designed to rotate constantly, which keeps them centered in the tube. Coating the tube's interior with a conductive substance grounds the system and reduces the effect of static electricity. Some flowmeters have two glass tubes, one for low flows and another for high flows (Figure 4–10A); the two tubes are in series and are still controlled by one valve. A dual taper design can allow a single flowmeter to read both high and low flows (Figure 4–10B). Causes of flowmeter malfunction include dirt in the flow tube, vertical tube misalignment, and sticking or concealment of a float at the top of a tube.
- Should a leak develop within or downstream from an oxygen flowmeter, a hypoxic gas mixture can be delivered to the patient (Figure 4–11). To reduce this risk, oxygen flowmeters are always positioned downstream to all other flowmeters (nearest to the vaporizer).



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- Sequence of flowmeters in a three-gas machine. A: An unsafe sequence. B: Typical Datex-Ohmeda sequence. C: Typical Draeger sequence. Note that regardless of sequence a leak in the oxygen tube or further downstream can result in delivery of a hypoxic mixture.
-

- Second Gas Effect
  - When one gas delivered at a hight inspired concentration (first gas such as N20) accelerates the blood uptake of a concurrently administered companion gas (such as VA) … this is the 2nd gas effect
  - Fick's Law of Diffusion
- fyi
  - Before administering inhalational agents, atmospheric N2 is in equilibrium with the body N2 (left figure). The middle figure shows N2O and a second gas (isoflurane in this case) turned on simultaneously. N2O diffuses into the blood from the alveoli in much greater amounts than N2 diffuses from the blood into the alveoli (N2 O is 34X more soluble in blood than is N2). Consequently, because N2 O is diffusing out of alveoli at a greater rate than N2 is diffusing into the alveoli, the alveoli "shrink" (right figure). The concentration of both the second gas (isoflurane) and N2 O remain elevated. Because concentration of the second gas (isoflurane) is sustained, the rate of diffusion of the second gas (isoflurane) remains elevated. Put another way, the uptake of a volatile agent is increased when it is administered simultaneously with N2O. This is the second gas effect, which is explained by Fick's law of diffusion.



- Ohm's Law
- 1. Calculation of systemic vascular resistance: (SVR = 80 x (MAP-CVP)/CO, where SVR is systemic vascular resistance, MAP is mean arterial pressure, CVP is central venous pressure, and CO is cardiac output.
- 2. Calculation of resistance (R) of flow through a tube: **R = (Pin – Pout)/Flow**, where Pin and Pout are inflow and outflow pressures, respectively.

- Enkephalin
  - An enkephalin is a pentapeptide involved in regulating nociception in the body. The enkephalins are termed endogenous ligands, or specifically endorphins, as they are internally derived and bind to the body's opioid receptors.
  - An endogenous peptide that binds to opioid receptors

- Know that most drugs act on Phase 4 of the action potential

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- What is the best initial test of thyroid function test?
  - TSH

CLOTTING FACTORS

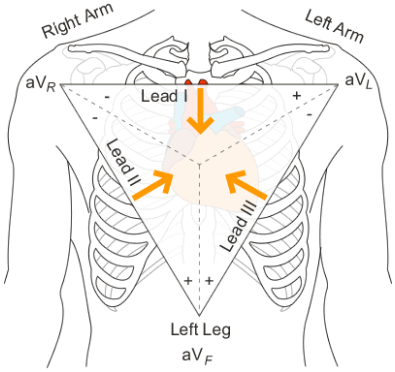
Factor	Synonym	Source	Vitamin K Dependent
I	Fibrinogen		
II	Prothrombin		YES
III	Tissue Factor Thromboplastin	Vascular wall & Extravascular cell membranes; Released from traumatized cells	
IV	Calcium	Diet	
V	Proaccelerin		
THERE IS NO FACTOR VI			
VII	Proconvertin		YES
VIII:C	Antihemophiliac Factor (AHF)		
VIII:vWF	vonWillibrand's factor	Vascular Endothelial Cells	
IX	Christmas factor		YES
X	Stuart-Prower Factor		YES
XI	Plasma thromboplastin antecedent		
XII	Hageman Factor		
XIII	Fibrin-Stabilizing Factor		
**	Protein C/Protein S		YES

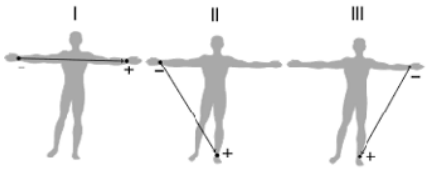
Test	Normal Value
Bleeding Time (Platelet Function)	3 – 10 minutes
Platelet Count (Platelet #)	150k – 400k cells/mL
PT (Factor 3 & 7 & Final Common Pathway)	12 – 14 seconds
aPTT (12, 11, 9, 8 & Final Common Pathway)	25 – 35 seconds
ACT (Adequacy of Heparinzation (Intrinsic Pathway)	80 – 150 seconds
Fibrinogen	> 150 mg/dL
Thrombin Time (TT): (Final Common Pathway)	< 30 seconds

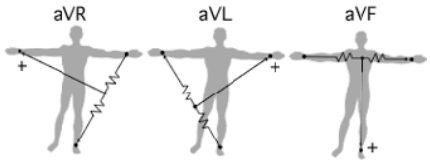
- IgG is the ONLY immunoglobulin that can pass through the placenta.
- IgG antibodies are predominantly involved in the secondary immune response.
- IgG is the only isotype that can pass through the human placenta, thereby providing protection to the fetus in utero.
- Along with IgA secreted in the breast milk, residual IgG absorbed through the placenta provides the neonate with humoral immunity before its own immune system develops.

- ??? Prone to sitting crani … decreased transfusion needs???

- ??? 3 Lead EKG, where is Limb Lead I placed????







- Lead I: left arm positive, right arm negative, left leg ground
- Lead II: right arm negative, left leg positive, left arm ground
- Lead III: left arm negative, left leg positive, right arm ground

---

- What causes hemolytic anemia?
  - Even though the number of red blood cells formed is normal, the red blood cell life span is so short, that serious anemia results.
  - Causes of hemolysis are diverse and include abnormalities of red blood cell membranes (hereditary spherocytosis), enzyme defects (**glucose-6-phosphate dehydrogenase deficiency**), and abnormal hemoglobin (sickle cell disease).
- FYI
  - Red blood cells are normally well protected against oxidizing agents. The sulfhydryl groups on hemoglobin are protected by reduced glutathione. The latter is regenerated by NADPH (reduced nicotinamide adenine dinucleotide phosphate), which itself is regenerated by glucose metabolism in the hexose monophosphate shunt. Glucose-6-phosphate dehydrogenase (G6PD) is a critical enzyme in this pathway. A defect in this pathway results in an inadequate amount of reduced glutathione, which can potentially result in the oxidation and precipitation of hemoglobin in red cells (seen as Heinz bodies) and hemolysis.
  - Most patients are typically not anemic but can develop hemolysis following oxidant stresses such as viral and bacterial infections or ingestions of some drugs (Table 29–9). Hemolysis can also be precipitated by metabolic acidosis. Hemolytic episodes can present with hemoglobinuria and hypotension. They are generally self-limited because only the older population of cells is destroyed.
  - For the patient with glucose-6-phosphate-dehydrogenase deficiency (G6PDH), avoid aspirin, penicillin, streptomycin, sulfonarnides, quinidine, doxorubicin, and ***methylene blue***, in addition to nitroprusside and prilocaine.

- Thalassemia → hemolytic anemia
  - Defect in the production of the one or more of the subunits of hemoglobin.
    - Alpha Thalassemia (defect in alpha chain)
    - Beta Thallassemia (defect in beta chain)
  - Anemia is the principle sign seen in thalassemia. Various degrees of hemolysis and ineffective RBC formation cause the anemia (MM 112)

- Glucagon
  - 1-2 mg
  - Endocrine: secreted by the islet of Langerhan's cells (alpha cells)
  - CV: Increases myocardial contractility (has a + inotropic effect) and HR → Increased CO
  - GI: relaxes GI smooth muscle (including the sphincter of Oddi), promotes glycogenolysis. **INHIBITS gastric motility**
- FYI (MM 216)
  - (1) Treatment of insulin-induced hypoglycemia, (2) relaxation of intestinal tract, including treatment of sphincter of Oddi spasm, and (3) glucagon's positive inotrope properties make it useful for treating heart failure when patient is unresponsive to beta-1 agonists.

- List and describe the three types of damages a patient may be entitled to if the anesthetist has been negligent? (MM 506)
  - (1) General damages (those such as pain and suffering that directly result from the injury);
  - (2) special damages (those actual damages that are a consequence of the injury, such as medical expenses, lost income, and funeral expenses); and
  - (3) **punitive damages** (those that are intended to punish the physician for negligence that was reckless, wanton, fraudulent, or willful).

- ??? Neonate vs. Infant … the Neonate excretes more: Na, Cl, H20, or K???
  - **Sodium**
  - GFR @ birth is 15-30% of the normal adult. Infants do not tolerate large volumes of water and salts because of low GFR and decreased tubular concentrating ability.

- When do liver enzymes become completely functional in the neonate?
  - During the first week of life the liver functions poorly and is incapable of conjugating significant quantities of bilirubin with glucuronic acid. Reduced quantities of blood clotting factors are also synthesized. The cytochrome P450 enzyme system is fully functional at one month of age.
  - Summary: Liver metabolism is decreased in the neonate until one month of age.

- syndrome of inappropriate antidiuretic hormone (SIADH) secretion (MM 84)
  - There is excessive reabsorption of water from renal tubules secondary to high levels of circulating ADH.
  - Water retention in excess of salt retention resulting in hyponatremia and hypoosmotic plasma.
  - An inappropriately increased sodium concentration in the urine (greater than 100-150 mOsm/kg) and decreased plasma osmolality (less that 270m0sm/kg) or hyponatremia are diagnostic of SIADH. The hyponatremia is due to the retention of water in excess of solutes.
  - four causes of the syndrome of inappropriate ADH secretion (SIADH)
    - Intracranial tumors, hypothyroidism, porphyria and carcinoma of the lung are associated with the syndrome.
    - Intracranial disease (tumors) is typically responsible for SIADH.
  - Treatment
    - (1) Remove the underlying cause, (2) restrict water intake, (3) antagonize the effects of ADH on the renal tubule by giving demecolcine, (4) administer hyperosmotic saline with or without diuretics.

- Diabetes Insipidus (DI) MM83
  - 2 causes of DI
    - Central diabetes insipidus results from failure to release antidiuretic hormone (ADH) from the posterior pituitary.
    - Nephrogenic diabetes insipidus occurs when the renal tubules fail to respond to antidiuretic hormone
  - What agent may be administered to distinguish central from nephrogenic diabetes insipidus? Desmopressin (DDAVP). If the urine becomes concentrated after desmopressin is administered, central diabetes insipidus is indicated.
  - Signs and symptoms of diabetes insipidus are
    - (1) polydipsia (increased thirst); (2) **high output of dilute (hypoosmotic) urine** (polyuria); (3) **hypernatremia:** (4) hypovolemia (dehydration); and (5) hypotension.
  - The untreated patient with diabetes insipidus has (1) hypernatremia (hyperosmolality) and (2) a dilute urine (osmolality < 200 m0smlkg).
  - The hallmark of diabetes insipidus is polyuria (large volumes of dilute urine), confirmed by the kidney's inability to concentrate urine, despite water deprivation.
- Causes
  - Diabetes insipidus is seen in many neurosurgical patients, particularly those undergoing pituitary procedures.
  - Diabetes insipidus may occur after transsphenoidal hypophysectomy.
  - Damage to the pituitary gland leading to diabetes insipidus following transcranial resection of a craniopharyngioma.
  - head trauma
- Tx: ADH (vasopressin) or desmopressin (DDAVP) administration; adequate fluid intake.

- Positioning
  - Best for ventilation (sitting/beechchair)

- changes in lung volumes with sitting to supine; and supine to sitting
- Upright → Supine:
  - RV, TV, & TLC, CV do not change significantly.
  - FRC & RV decreases
  - IRV increases
  - FYI (MM312)
    - Functional residual capacity (FRC) and pulmonary compliance are **most decreased** in the Trendelenburg position. When patients shift from the upright to the supine position, FRC decreases about 800 mL. Going from supine to the head down position causes an additional decrease in FRC and pulmonary compliance. Obviously, the Trendelenburg position has a greater effect on FRC and pulmonary compliance than does the supine position. The ventilatory changes associated with the lithotomy position are not much different than those associated with the supine position the lateral decubitus position, vital capacity decreases lo%, which is comparable to the decrease in vital capacity in the supine position.
    - When going to from the upright to supine position RV, TV and TLC do not change significantly. When going from the upright to supine position FRC decreases because gravity is no longer pulling the abdominal contents away from the diaphragm. Since FRC decreases and RV is unchanged, ERV must decrease (FRC = RV + ERV) and the IRV must increase (TLC = FRC + TV + IRV). Closing capacity seems to be independent of body position, therefore CV does not change (CC = RV + CV).

- Upright → Prone:
  - SV, CO, & FRC decrease

- HEALTHY patient goes from supine → prone:
  - FRC INCREASES

- Supine → Standing
  - FRC increases

- Prone breast position
  - Normal or large breast: medial & cephalad
  - Large breast: can be placed laterally, but surgeon's access may be impaired
  - FYI (MM 318)
    - If the patient is to be placed prone and has either normal-sized or large breasts, the breasts should be placed in the medial and cephalad (headward) position. The patient with breasts of normal size may experience pain or postoperative tenderness if the breasts are positioned laterally. The patient with large breasts may not experience pain if the breasts are positioned laterally, but the surgeon's access may be impaired if the arms are positioned at the side.

- Prone Blindness: What is the most likely reason a patient who undergoes a non-ophthalmic procedure develops blindness postoperatively?
  - Optic nerve and retinal ischemia.
  - The perfusion of the optic nerve and retina can be seriously compromised because of eye compression, which can occlude the central retinal artery and decrease retinal perfusion. Eye compression may occur in the down eye in the lateral or the prone position. Systemic hypotension may also contribute to postoperative blindness. The decrease in retinal perfusion pressure produces retinal ischemia.

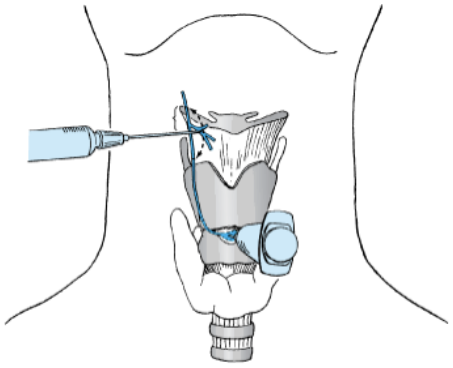
- 2-3 questions changes from sitting to supine, and vice versa.
  - CO, HR, and Peripheral Vascular Resistance

- The obese patient is turned prone, know the changes in airway volumes???
  - In the upright position, FRC is decreased and closing capacity increased. Closing capacity may be reached during tidal ventilation. This means some airways close during tidal breathing. In the supine position, FRC decreases further and tidal breathing continues to occur with some airways closed. If the patient is positioned so that the abdomen hangs free, FRC is decreased less in the prone than in either the supine or the lateral positions, so events in the morbidly obese prone patient would fall between the obese upright and obese supine positions. (MM 494)

- What is the most serious complication of the sitting position?
  - Air embolism occurs most commonly in sitting position.
  - Hypotension is the most frequent complication of the sitting position.
  - The open vasculature of the cerebral ventricles allows for entrainment of air causing pulmonary air emboli (MM 317)

- ?? Demerol works on what receptor to decrease shivering??
  - Delta ???
  - FYI: meperidine is a potent agonist at the α2 adrenoreceptors and antishivering action of meperidine possibility is tranduced by this mechanism??????

- ??Picture of a guy with a needle in his neck… what nerve?? SLN???



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Superior laryngeal nerve block and transtracheal block.

- Pick 2 regarding **thromboxane A2**:
  - vessel constriction, **platelet aggregation, decreased uterine contractions**, decreased uterine blood flow
  - FYI (M&M Chapter 43. Obstetric Anesthesia): TXA2 is a potent vasoconstrictor and promoter of platelet aggregation

---
- Best meds to give to a vascular patient?
  - Beta Blocker/ACE inhibitor OR Beta Blocker/Statins???

- What relieves a myotonic muscle contraction?
  - Local anesthetic infiltration???
  - FYI (MM 79, M&M Chapter 37. Anesthesia for Patients with Neuromuscular Disease):
    - Treatment of myotonic dystrophy is symptomatic and may include the use of phenytoin, quinine, and procainamide. These agents delay the return of membrane excitation by blocking rapid sodium (Na') influx into muscles. Quinine and procainamide should be used with caution as they may worsen cardiac conduction abnormalities (prolonged P-R interval).
    - Myotonic dystrophies are characterized by persistent contracture (myotonia) after voluntary contracture of skeletal muscle or following electrical stimulation. Electromyographicfindings are diagnostic and are characterized by prolonged discharges or repetitive action potentials. Skeletal muscle resting membrane potentials are also lowered (less polarized) in patients with MD. The inability of the skeletal muscle to relax after voluntary contraction or stimulation results from abnormal calcium metabolism-ATP-driven pumps fail to return calcium to the sarcoplasmic reticulum (SR) thus the unsequestered calcium remains available to produced sustained skeletal muscle contraction

---
- What electrolyte disturbance is found in patients with pancreatic disease?
- **Hypomagnesemia**, hyperchloremia, hypokalemia, hyponatremia
- FYI (MM 121)
  - Pancreatitis is associated with (1) dehydration, (2) hypocalcemia, (3) hyperglycemia, and (4) adult respiratory distress syndrome.
  - Patients with pancreatitis need to be evaluated for malnutrition, abnormal liver function, and early signs of alcohol withdrawal. Alcoholism is a frequent cause of pancreatitis.

- Giving FFP….. shortly after, patient decompensates… what happened?
  - ABO incompatibility

-

- What anesthetics can be applied topically? (MM 175)
  - tetracaine, lidocaine, cocaine (TLC)

- How do you treat HTN caused by cocaine?
  - Labetalol
- FYI (M&M)
  - Cocaine's cardiovascular reactions are unlike those of any other local anesthetic. Adrenergic nerve terminals normally reabsorb norepinephrine after its release. Cocaine inhibits this reuptake, thereby potentiating the effects of adrenergic stimulation. Cardiovascular responses to cocaine include hypertension and ventricular ectopy. The latter contraindicates its use in patients anesthetized with halothane. Cocaine-induced arrhythmias have been successfully treated with adrenergic and calcium channel antagonists. Cocaine produces vasoconstriction when applied topically.
  -

- Most common complication in PACU … airway obstruction

- Standing → CO/Venous return decrease, HR increased
- Supine → CO/Venous return increased, HR decreased

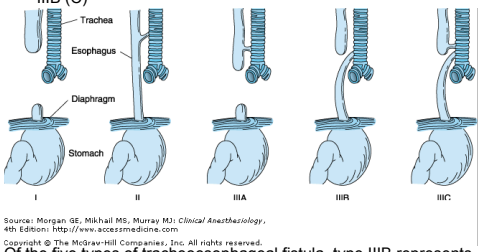
LEFT to RIGHT	**RIGHT to LEFT**
PDA	Tetrology of Fallot
VSD	Pulmonary Atresia with VSD
ASD	Patent Foramen Ovale
(Acyanotic)	(Cyanotic)

- FYI (M&M Table 20–18. Classification of Congenital Heart Disease)
- Lesions causing outflow obstruction
  - Left ventricle
  - Coarctation of the aorta
  - Aortic stenosis
  - Right ventricle
  - Pulmonic valve stenosis
- Lesions causing left-to-right shunting
  - Ventricular septal defect
  - Patent ductus arteriosus
  - Atrial septal defect
  - Endocardial cushion defect
  - Partial anomalous pulmonary venous return
- Lesions causing right-to-left shunting
  - With decreased pulmonary blood flow
    - Tetralogy of Fallot
    - Pulmonary atresia
    - Tricuspid atresia
  - With increased pulmonary blood flow
    - Transposition of the great vessels
    - Truncus arteriosus
    - Single ventricle
    - Double-outlet right ventricle
    - Total anomalous pulmonary venous return
    - Hypoplastic left heart

- What keeps the PDA open?
  - Prostaglandins
  - Low arterial oxygen tension (PaO2)
- What is responsible for closure of PDA?
  - Increased O2 tension & decreased circulating prostaglandins
- What 4 factors may cause a neonate to return to fetal circulation?
  - Hypoxemia, acidosis, pneumonia, hypothermia ☒ Increased PVR
- FYI (MM 443)
  - Persistent fetal circulation (or a return to fetal circulation) is common preterm infants and infants with metabolic derangements (asphyxia, sepsis, meconium aspiration, congenital diaphragmatic hernia). Hypoxemia, acidosis, pneumonia, and hypothermia are 4 primary precipitating factors in persistent fetal circulation. The pathologic mechanism common to all 4 factors is increased pulmonary vascular resistance (PVR) and right-to-left-shunting.
  - Normal closure of the ductus arteriosus occurs in response to increased arterial oxygen tension (PaO2), as well as to reduction in circulating prostaglandins that follow separation of the placenta. (Realize that a number of other substances such as eicosanoids and factors such as PaC02 and pH have been implicated, but that increased oxygen tension seems to be the major factor precipitating ductus arteriosus closure.

- Dose of succinylcholine in pregnant patients? (MM 464)
  - Plasma cholinesterase (pseudocholinesterase) levels will decrease by 24% before delivery and decrease further (to 33% less) by 3 days postpartum.
  - Plasma cholinesterase (pseudocholinesterase) levels will return to normal levels in 2-6 weeks postpartum.

- Tracheoesophageal Fistula (TEF)
  - Most common variation is the form that ends in a blind pouch and a lower esophagus that connects to the trachea – Type IIIB (C)



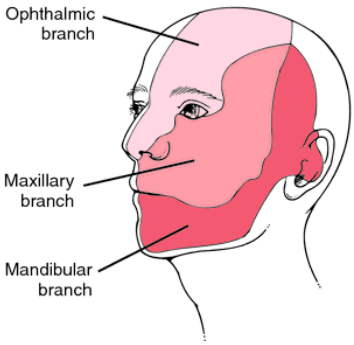
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Of the five types of tracheoesophageal fistula, type IIIB represents 90% of cases.

- **VACTERL Syndrome:**
  - Vertebral defect
  - Anal atresia
  - Cardiac Anomalies
  - Tracheoesophageal fistula
  - Esophageal atresia
  - Radial/Renal dysplasia
  - Limb anomalies

- What nerves may be damaged during face mask ventilation?
  - The buccal branch of the facial nerve (CN VII), and
  - The supraorbital branch of the trigeminal nerve (CN V)
- FYI (M&M): Mask ventilation for long periods may result in pressure injury to branches of the trigeminal or facial nerves.





- What physiological changes should be expected in the patient receiving radiocontrast media?
  - (1) Systemic hypertension resulting from increased intravascular volume secondary to increased plasma osmolality
  - (2) hypotension due to hypovolemia caused by an osmotic diuresis
  - (3) osmotic diuresis (high volume urine output)
  - (4) nausea, vomiting and urticarial, and
  - (5) chills, fever and cutaneous flushing.
  - These items cited above are described as "common" or "frequent" occurrences. There are other less common but more severe toxic reactions to contrast media.

- What is the most likely nerve damaged in the prone position?
  - Ulnar???

- problems that should be anticipated in the patient with scleroderma.
  - (1) poor gastric emptying; pretreat with antacids and H2 antagonists;
  - (2) the risk for aspiration pneumonitis during induction is increased due to a high incidence of gastroesophageal reflux;
  - (3) venous access may be difficult; and (4) skeletal muscle involvement increases the sensitivity to muscle relaxants.
- anesthetic problems related to the airway and pulmonary system
  - (1) Severe jaw limitation (limited temporomandibular joint mobility) may require nasal approach or a fiberoptic technique;
  - (2) ventilation may be difficult because of decreased compliance;
  - (3) arterial hypoxemia may be present because of decreased diffusion of O2 across the alveolar capillary
  - membrane; and
  - (4) rapid sequence induction may not be appropriate; awake intubation with head up position is safer.
- Other problems:
  - (1) Profound hypotension may occur following induction;
  - (2) regional anesthesia is preferable although there may be prolonged effects from the use of epinephrine in local anesthetics;
  - (3) blood pressure measurements may be difficult; ultrasonic or Doppler measurements may be necessary; arterial catheterization is not recommended;
  - (4) skin temperature is reduced; warm the room and use warm compresses for the extremities.
- A gastroparesis-like state is a concern in the patient with scleroderma. Progressive fibrosis of the gastrointestinal tract causes hypomotility of the lower esophagus and small intestine. Both conditions increase the risk of reflux and aspiration. With intestinal hypomotility, gastric outlet may be diminished so the likelihood of reflux and aspiration is greater.

- Pickwickian Syndrome (Obese Hypoventilation Syndrome)
  - A morbidly obese patient with classic triad of LVH, RVH, and OSA
  - Hematologic alteration: cyanosis-induced polycythemia d/t chronic hypoxia & hypercapnia

---

- Will a right-to-left intracardiac shunt theoretically slow or accelerate inhalation induction? Is the effect be clinically significant?
  - A right-to-left intracardiac shunt will theoretically slow inhalation induction, because less anesthetic is absorbed from the lung, and mixing will further dilute blood passing to the left, decreasing the arterial concentration of the blood going to the brain, especially the less soluble agents. This effect is rarely problematic.

- Will a right-to-left intracardiac shunt theoretically slow or accelerate intravenous induction?
  - An intravenous induction will be theoretically accelerated with a right-toleft intracardiac shunt.

- **Will a left-to-right intracardiac shunt theoretically slow or accelerate inhalation induction? Why is this phenomenon rarely evident clinically?**
  - **A left-to-right intracardiac shunt should accelerate the speed of inhalation induction because the rate of transfer of anesthetic agent from the lungs to the blood is increased. However, this effect is rarely clinically evident because decreased delivery of anesthetic to the target tissues negates the increased uptake of agent with a left-to-right intracardiac shunt.**

- Will a left-to-right intracardiac shunt theoretically slow or accelerate intravenousinduction? Why is this phenomenon rarely evident clinically?
  - Intravenous induction should be slowed by a left-to-right shunt; however, unless cardiac output is very poor, the effect is clinically irrelevant.

- List 10 anesthetic concerns associated with the rheumatoid arthritis patient (M&M 81)
  - (1) cervical spine involvement,
  - (2) synovitis of the temporomandibular joint,
  - (3) cricoarytenoid arthritis manifested by hoarseness,
  - (4) pericarditis,
  - (5) aortic regurgitation,
  - (6) pulmonary fibrosis,
  - (7) peripheral nerve compression,
  - (8) hepatitis,
  - (9) anemia, and
  - (10) drug-induced side-effects associated with aspirin and/or steroid therapy.

- Most critical monitor for sitting position?
  - most common complication of sitting is hypotension
  - most serious complication of sitting is venous air embolism

---

- What is the function of an auxiliary flowmeter on the gas machine? What is the advantage of an auxiliary flowmeter?
  - Auxiliary flowmeters are useful for attaching supplemental oxygen delivery devices, such as a nasal cannula, to the gas machine.
  - The auxiliary flowmeter is advantageous because the breathing circuit and gas delivery hose remain intact while supplemental oxygen is delivered to a spontaneously breathing patient.
  - Another advantage is that an oxygen source is readily available for the Ambu bag if the patient needs to be ventilated manually for any reason during the case.
- What is the primary disadvantage of an auxiliary flowmeter?
  - If pipeline supply has lost pressure or has been contaminated, the auxiliary flowmeter becomes unavailable. Another disadvantage is that the fraction of inspired inspiration cannot be varied with the auxiliary flowmeter.

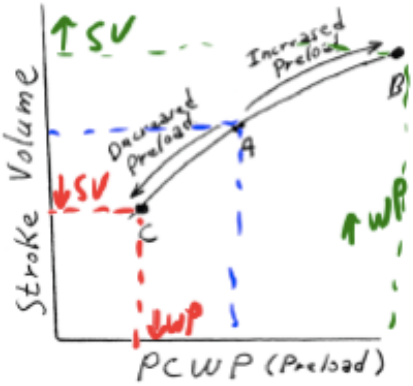
- Stimlation of posterior pituitary … Direct or Indirect?
  - ADH is synthesized in the hypothalamus, and transported INSIDE AXONS to nerve endings in the posterior pituitary.
  - DIRECT???
- FYI (M&M Chapter 26. Anesthesia for Neurosurgery)
  - The posterior pituitary secretes antidiuretic hormone (ADH, also called vasopressin) and oxytocin. These hormones are actually formed in supraoptic and paraventricular neurons, respectively, and are transported down axons that terminate in the posterior pituitary. Hypothalamic osmoreceptors and, to a lesser extent, peripheral vascular stretch receptors regulate secretion of ADH
  - Specialized neurons in the supraoptic and paraventricular nuclei of the hypothalamus are very sensitive to changes in extracellular osmolality. When ECF osmolality increases, these cells shrink and release ADH (arginine vasopressin) from the posterior pituitary. ADH markedly increases water reabsorption in renal collecting tubules (see Chapter 31), which tends to reduce plasma osmolality to normal again. Conversely, a decrease in extracellular osmolality causes osmoreceptors to swell and suppresses the release of ADH. Decreased ADH secretion allows a water diuresis, which tends to increase osmolality to normal. Peak diuresis occurs once circulating ADH is metabolized (90–120 min). With complete suppression of ADH secretion, the kidneys can excrete up to 10–20 L of water per day (see below).
  - The carotid baroreceptors and possibly atrial stretch receptors can also stimulate ADH release following a 5–10% decrease in blood volume (see below). Other nonosmotic stimuli include pain, emotional stress, and hypoxia.

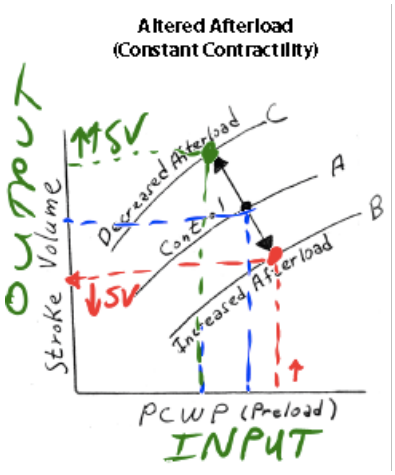
- **At what postconceptual age do you no longer worry about apnea???**
- FYI (MM 456) Would a formerly premature infant be a candidate for outpatient surgery? What are the anesthetic concerns for the formerly premature infant?
  - No, the formerly premature infant is not an appropriate candidate for outpatient surgery. Formerly premature infants (less than **46 weeks** postconceptual age), even if healthy, have an increased rate of post-anesthetic apnea and bradycardia. These formerly premature infants should have cardiorespiratory monitoring for a minimum of 24 hours postoperatively and thus are not good candidates for ambulatory day surgery.
- The infant patient is high-risk for postoperative apnea what agent may be given prophylactically to decrease the risk of apnea?
  - The infant at risk for postoperative apnea may be given caffeine prophylactically to ensure adequate serum levels exist prior to surgery and during the postoperative period. Caffeine is a respiratory and CNS stimulant and is generally preferred to theophylline because caffeine has a wider therapeutic margin and a decreased propensity for toxicity. The recommended loading dose is 10 mglkg of caffeine base, which is often obtained from 20 mg of caffeine citrate. The clinical effects of caffeine may last several days after a single dose, but do not administer caffeine and then discharge the patient, assuming that the caffeine will prevent apnea
- FYI (M&M)
  - Premature infants who are less than **50 (some authorities would say 60) weeks postconceptional age** at the time of surgery are prone to postoperative episodes of obstructive and central apnea for up to 24 h. In fact, even term infants can experience—albeit rarely—apneic spells following general anesthesia. Risk factors for postanesthetic apnea include a low gestational age at birth, anemia (< 30%), hypothermia, sepsis, and neurological abnormalities. The risk of postanesthetic apnea may be decreased by intravenous administration of caffeine (10 mg/kg) or aminophylline.
  - Nonetheless, elective or outpatient procedures should be deferred until the preterm infant reaches the age of at **least 50 weeks postconception**. A 6-month symptom-free interval has been suggested for infants with a history of apneic episodes or bronchopulmonary dysplasia. If surgery must be performed earlier, monitoring with pulse oximetry for 12–24 h postoperatively is mandatory for infants less than 50 weeks postconception; infants between 50 and 60 weeks postconception should be closely observed in the postanesthesia recovery unit for at least 2 h.

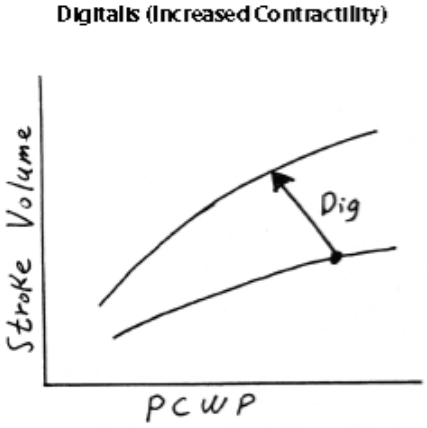
- Epidural Steroid Injections (MM p. 497-500)
- Lightening Bolts & Bull's eyes

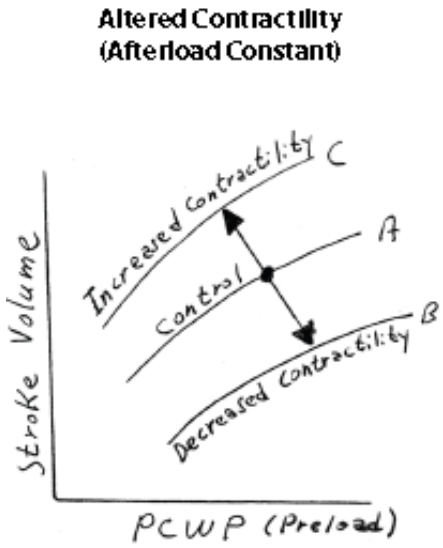
- VENTRICULAR FUNCTION CURVES

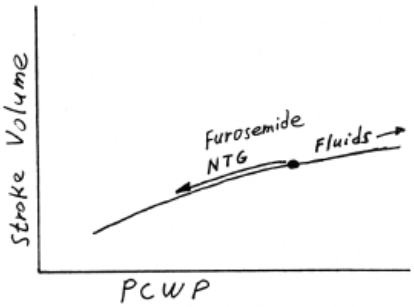
Altered Preload
(Constant Contractility and Afterload)





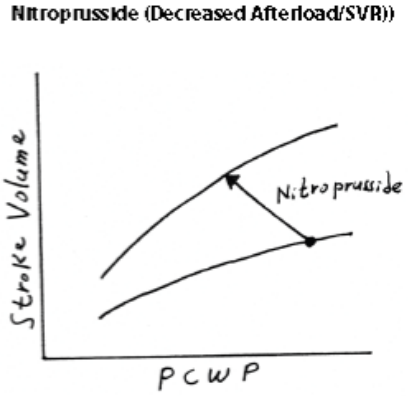




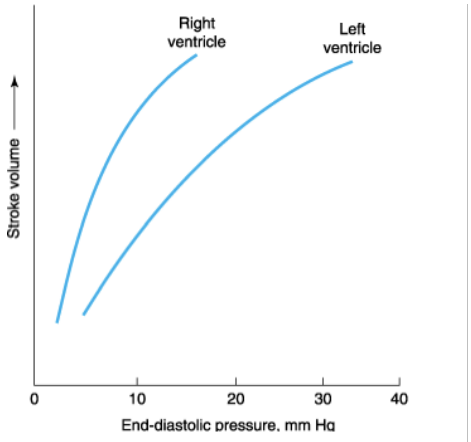


- a. A ventricular function curve is a plot of some index of ventricular function (stroke volume, stroke volume index, etc.) versus some measure of end-diastolic volume (e.g., PCWP).
- b. When there is an increase or a decrease in preload, there is a shift up or down of a single ventricular function curve, respectively.
- c. When there is an increase in contractility or a decrease in afterload, there is an upward and to-the-left shift in the ventricular function curve (reflecting an increase in stroke volume and a decrease in size of the left ventricular chamber).
- d. When there is a decrease in contractility or an increase in afterload, there is a downward and to-the right shift in the ventricular function curve (reflecting a decrease in stroke volume and an increase in the size of the left ventricular chamber).

- Drug-Induced Shifts in the Ventricular Function Curve

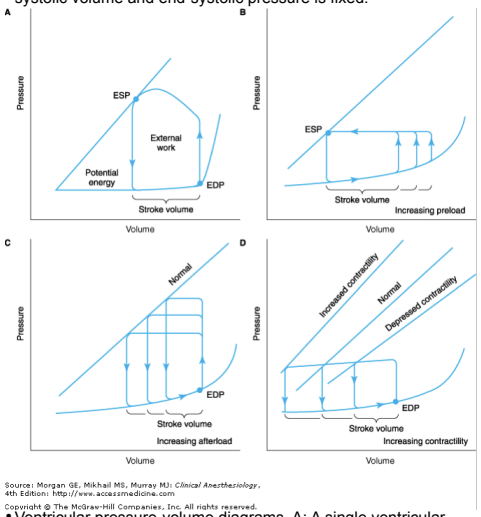


- Morgan and Mikhail
- Plotting cardiac output or stroke volume against preload is useful in evaluating pathological states and understanding drug therapy. Normal right and left ventricular function curves are shown in Figure 19–9.



Source: Morgan GE, Mikhail MS, Murray MJ: *Clinical Anesthesiology*, 4th Edition: http://www.accessmedicine.com
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Function curves for the left and right ventricles.

- Ventricular pressure–volume diagrams are even more useful because they dissociate contractility from both preload and afterload. Two points are identified on such diagrams: the end-systolic point (ESP) and the enddiastolic point (EDP) (Figure 19–10). ESP is reflective of systolic function, whereas EDP is more reflective of diastolic function. For any given contractile state, all ESPs are on the same line—ie, the relationship between end-systolic volume and end-systolic pressure is fixed.



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- Ventricular pressure-volume diagrams. A: A single ventricular contraction. Note that stroke volume represents change in volume on the x-axis (difference between end-systolic volume and end-diastolic volume). Note also that the circumscribed area represents external work performed by the ventricle. B: Increasing preload with constant contractility and afterload. C: Increasing afterload with constant preload and contractility. D: Increasing contractility with constant preload and afterload. ESP, end-systolic point; EDP, end-diastolic point.

- Best lead to monitor A-fib?
  - Lead II

- How do case-control and cohort studies differ?
  - Cohort and case-control studies are called longitudinal studies because they are involve an extended period of time defined by point when the study begins and the point when it ends.
  - A case-control study begin "at the end" with the presence or absence of an outcome and then looks backward in time to try to detect possible causes or risk factors that may explain the outcome. Therefore, a case-control study is a retrospective study that asks, "What happened?"
  - In a cohort study, the researchers select group ofpeople who have something in common, say a risk factor, and then monitor the cohort forward in time. A cohort study asks, "What will happen?" and thus is aprospective study.

- Methergine???
- Used for bleeding after delivery. 0.2mg IM (repeat q 15 min); max dose 0.8 mg??????????

- FYI (M&M)
  - Methylergonovine (Methergine) causes intense and prolonged uterine contractions. It is therefore given only after delivery (postpartum) to treat uterine atony. Moreover, because it also constricts vascular smooth muscle and can cause severe hypertension, it is usually administered only as a single 0.2 mg dose intramuscularly.
  - Carboprost tromethamine (Hemabate) is a synthetic analogue of prostaglandin F2 that stimulates uterine contractions. It is often used to treat refractory postpartum hemorrhage. As with methergine, it should be administered only intramuscularly. An initial dose of 0.25 mg may be repeated every 15–90 min to a maximum of 2 mg. Common side effects include nausea, vomiting, and diarrhea.

- What is the intrathecal dose of Duramorph?
  - 0.5 – 1mg????

- FYI (M&M)
  Table 43–2.Spinal Opioid Dosages for Labor and Delivery.

Agent	Intrathecal	Epidural
Morphine	0.25–0.5 mg	5 mg
Meperidine	10–15 mg	50–100 mg
Fentanyl	12.5–25 mcg	50–150 mcg
Sufentanil	3–10 mcg	10–20 mcg

- Which two laws involve Fick's Law?
  - Grahm's Law and Fick's Law – diffusion of N20 and other gases (MM 239)
  - Ohm's Law and Fick's Principle???? – cardiac output calculation????

FYI (Valley Review)
- **Fick's Law of Diffusion**

$$\text{Diffusion rate} = \frac{(P_1 - P_2)\cdot(\text{Area})\cdot(\text{Solubility})}{(\text{Membrane thickness})\cdot(\sqrt{\text{MW}})}$$

  - 1. Diffusion rate of a gas is directly proportional to:
    - a. partial pressure gradient (P1-P2 = ΔP)
    - b. membrane area
    - c. solubility of gas in membrane
  - 2. Diffusion rate is inversely proportional to:
    - a. membrane thickness
    - b. the square root of the molecular weight
- **Graham's Law**

$$\text{Diffusion} \propto \frac{1}{\sqrt{\text{MW}}}, \text{ where MW is molecular weight}$$

  - Graham's law states that a gas diffuses at a rate that is inversely proportional to the square root of its molecular weight.

FYI (Valley Review)

Law of Laplace
- 1. A blood vessel that is aneurysmal has greater wall tension.
- 2. With greater ventricular filling during diastole, there is greater tension in the wall at end-diastole.
- 3. In ARDS, smaller alveoli empty into larger alveoli (atelectasis).
- 4. In healthy individuals, atelectasis does not occur.

Formula: the tension in the wall of a hollow structure (blood vessel, left ventricle, or normal alveolus) is proportional to the radius. The greater the radius in a chamber or vessel, the greater the tension in the walls of the chamber or vessel.
- For cylindrically shaped structures,
  T = P x r
  (where T = wall tension, P = pressure of liquid within cylinder, and r = radius).
- For spherically-shaped structures,
  T = (P x r)/2
  (where T = tension in the wall of the sphere and P = pressure within the sphere.)

Poiseuille's Law
- 1. For IV fluid administration, flow is increased by
  - a. increasing diameter (decreasing gauge) of needle;
  - b. decreasing length of the needle;
  - c. raising the IV bag to a greater height.
- 2. The polycythemic patient has decreased flow through tissues.
- 3. The anemic patient has increased flow through tissues.

Formula

$$F = \frac{\pi r^4 \Delta P}{8\eta L}$$

$$R = \frac{\Delta P}{F}$$

$$R = \frac{8\eta L}{\pi r^4}$$

- Flow is directly proportional to:
  - (1) the pressure drop along the tube;
  - (2) the fourth power of the radius (r4) of the tube.
- Flow is inversely proportional to:
  - (3) the length of the tube; and
  - (4) the viscosity of the fluid.

Venturi Effect (Bernoulli's Principle)
- 1. Nebulizer (pneumatic and jet)
- 2. Venturi oxygen mask
- 3. Jet ventilator
- 4. Injector
- 5. Anterior leaflet in IHSS

Henry's Law
- 1. Permits calculation of the amount of O2 dissolved in blood.
- 2. Permits calculation of the amount of CO2 dissolved in blood.

Formula
- oxygen dissolved in blood = PaO2 x 0.003
- carbon dioxide dissolved in blood = PaCO2 x 0.067

Boyle's Law
- 1. Explains the release into the atmosphere of a large amount of gas from a compressed cylinder.
- 2. Squeezing the Ambu bag raises pressure and decreases volume

Formula: Pressure (P) is inversely proportional to volume (V) at constant temperature; if the volume of a gas is halved, the pressure has doubled.
- P ∝ 1/V
- PV = k
- P1V1 = P2V2

Charles' Law
- 1. The inflatable cuff of an endotracheal tube or laryngeal mask airway expands during sterilization in an autoclave.

Formula: Volume is directly proportional to the absolute temperature (K) at constant pressure; when temperature increases, volume of a gas increases. Note: K = 273 + ºC.
- V ∝ T
- V/T = k
- V1/T1 = V2/T2)

Gay-Lussac's Law
- 1. When the temperature of a closed cylinder increases, cylinder pressure increases.
- 2. When the temperature of a closed cylinder decreases, cylinder pressure decreases.
- 3. As a cylinder containing liquid N2O empties, temperature decreases (Joule-Thompson effect) as temperature decreases, the pressure of N2O within the tank decreases.
- 4. Wood's metal blows when temperature increases substantially

Formula: Pressure is directly proportional to absolute temperature (oK) if volume is constant; when the temperature of a gas in a container at constant volume increases, pressure increases.
- P ∝ P/T = k
- P1/T1 = P2/T2

Ideal Gas Law
- 1. As a cylinder of compressed gas empties, the pressure in the cylinder falls (because the number of moles, n, in the tank is decreasing).

Dalton's Law
- 1. Permits calculation of the % concentration of a gas by dividing the partial pressure of the gas by the total pressure.
- 2. Permits calculation of the partial pressure of a gas by multiplying % concentration by total pressure.

Formula: The total pressure in a mixture of gases is equal to the sum of the pressures of the individual gases (each gas is said to exert a partial pressure). The partial pressure of a gas is calculated by multiplying the percent gas (fractional concentration) times the atmospheric pressure.
- Ptot = P1 + P2 + …PN
- PPgas = %gas/100 x atm

Joule-Thompson Effect
- 1. As cylinder of compressed gas empties, the cylinder cools; condensation and ice crystals may be seen.
- 2. The cryoprobe operates on the basis of the Joule-Thompson effect.

Fick's Law of Diffusion
- 1. Concentration effect
- 2. Second gas effect
- 3. Diffusional hypoxia
- 4. Expansion of gas spaces, or increase in pressure of closed gas spaces, when N2O is turned on

Formula

$$\text{Diffusion rate} = \frac{(P_1 - P_2)\cdot(\text{Area})\cdot(\text{Solubility})}{(\text{Membrane thickness})\cdot(\sqrt{\text{MW}})}$$

- 1. Diffusion rate of a gas is directly proportional to:
  - a. partial pressure gradient (P1-P2 = ΔP)
  - b. membrane area
  - c. solubility of gas in membrane
- 2. Diffusion rate is inversely proportional to:
  - a. membrane thickness
  - b. the square root of the molecular weight

Fick Principle
- 1. Calculation of cardiac output from A–V O2 or A–V CO2 difference.

Ohm's Law
- 1. Calculation of systemic vascular resistance: (SVR = 80 x (MAP -CVP)/CO, where SVR is systemic vascular resistance, MAP is mean arterial pressure, CVP is central venous pressure, and CO is cardiac output.
- 2. Calculation of resistance (R) of flow through a tube: R = (Pin – Pout)/Flow, where Pin and Pout are inflow and outflow pressures, respectively.

Beer's Law(Lambert-Beer Law)
- 1. Pulse oximetry
- 2. IRAS (Infrared absorption spectroscopy): ETCO2 determinations

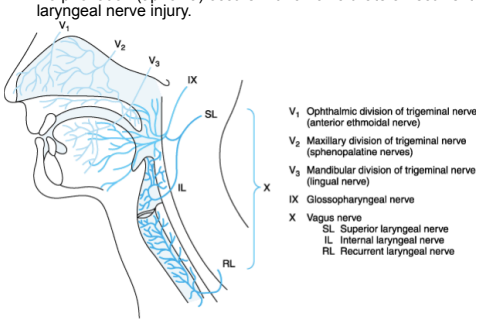
Superimposition Principle
- 1. Pulse pressure increases as the blood pressure wave travels to the periphery.

Le Chatelier's Principle ("A change in any of the factors determining an equilibrium causes the system to adjust to reduce or counteract the effect of the change.")
- 1. The solubility of a salt such as sodium chloride (NaCl) in water increases with increasing temperature.
- 2. The solubility of a gas such as isoflurane, O2, or CO2 increases with decreasing temperature.
- 3. When the concentration of a reactant is increased (or the concentration of a product is decreased), the reaction is driven toward the production of more products. Conversely, when the concentration of a reactant is decreased (or the concentration of a product is increased), the reaction is driven toward the production of less products.

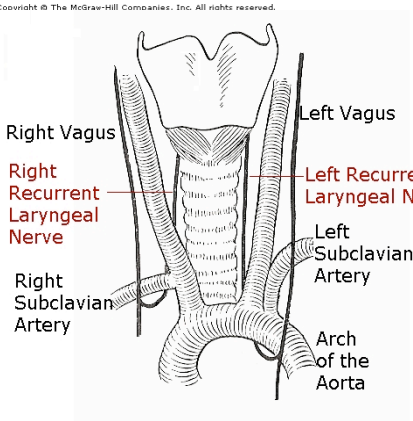
Inverse square law
- 1. The distance from the source determines the amount of exposure of the anesthetist to x-rays. The intensity of cattered radiation is inversely proportional to the square of the distance from the source. This is the inverse square law. The best protection from x-rays is distance from the source.

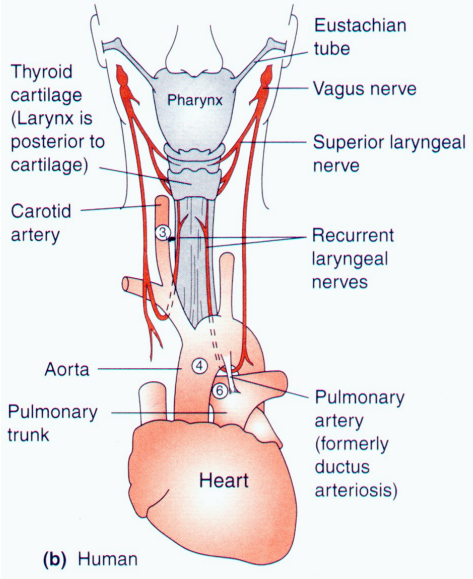
Law of mass action
- 1. When the concentration of a reactant is increased (or the concentration of a product is decreased), the reaction is driven toward the production of more products. Conversely, when the concentration of a reactant is decreased (or the concentration of a product is increased), the reaction is driven toward the production of less products. This is a specific form of Le Chatelier's principle.

- Picture of larynx: know all of the nerves
- Know nerve most injured during thyroid procedure
  - Recurrent Laryngeal
- FYI (MM 417)
  - Hoarseness occurs with unilateral recurrent laryngeal nerve injury.
  - Stridor occurs with acute bilateral recurrent laryngeal nerve injury.
  - No phonation (aphonia) occurs with chronic bilateral recurrent laryngeal nerve injury.



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- HEART VALVE SUMMARY POINTS:
  - REGURG valves: Increase the HR (fast, full, and forward)
  - STENOTIC valves: Decrease the HR
  - Sinus Rhythm is always preferred
  - Preload and Contractility is ALWAYS maintained
  - Know the AFTERLOAD involvement for each lesion
    - AR: decrease afterload (keep moving forward); avoid sudden increases in afterload
    - MR: decrease afterload to improve forward CO
    - AS: MAINTAIN afterload to keep coronary perfusion pressure stable (avoid increase/decreases); May exhibit angina even when CAD is absent
    - MS: MAINTAIN afterload (avoid decreases in SVR and increases in PVR)
- Where to auscultate? (MM 1)
  - Diastolic:
    - AR:
      - Aortic regurgitation is recognized by its diastolic murmur
      - Best heard along the left sternal border
    - MS:
      - Recognized by the characteristic opening snap that occurs early in diastole and by a rumbling diastolic murmur
      - Best heard with the chest piece placed over the cardiac apex
  - Systolic:
    - AS:
      - Recognized by its characteristic systolic murmur
      - Best heard in the second right intercostal space (over the aortic arch) with transmission into the neck.
      - Midsystolic ejection murmur that peaks in late systole
    - MR:
      - Best heard with the chest piece placed over the cardiac apex.
      - Blowing holosystolic, heard throughout systole

- What % is eliminated after 4 ½ half-lives? (fill in the blank)
  - 95%

Number of half-times	Amount Remaining	Amount Gone
0	100 (1.0)	0
1	50 (1/2)	50
2	25 (1/4)	75
3	12.5 (1/8)	87.5
4	6.25 (1/16)	93.8
5	3.125 (1/32)	96.9
6	**1.562 (1/64)**	**98.4**
7	0.781 (1/128)	99.2

4th HL = 93.8
5th HL = 96.9
93.8 + 96.9 average is 95%

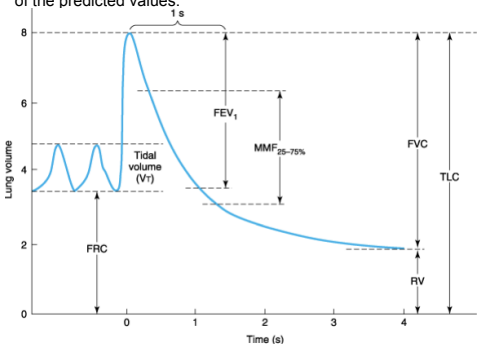
- **Restrictive/Obstructive**

	Normal	Obstructive	Restrictive
FEV	4	1.3	2.8
FVC	5	3.1	3.1

- ** The FEV1/FVC ration is normally → >0.7
- ** Obstructive is
  - BOTH FEV1 and FVC are LOW … and…
  - FEV1/FVC = < 0.7
- ** Restrictive is
  - BOTH FEV1 and FVC are LOW … and…
  - FEV1/FVC = > 0.7 (or equal to)

- Restrictive:
  - Pulmonary fibrosis, pneumothorax, chest wall disease (scoliosis), neuromuscular disease (amyotrophic lateral sclerosis, myasthenia gravis)
- Obstructive: COPD (bronchitis, emphysema), asthma, tracheal stenosis

FYI (M&M)
- Restrictive pulmonary diseases are characterized by decreased lung compliance. Lung volumes are typically reduced, with preservation of normal expiratory flow rates. Thus, both forced expiratory volume in 1 s (FEV1) and forced vital capacity (FVC) are reduced, but the FEV1/FVC ratio is normal.
- Obstructive lung diseases are the most common form of pulmonary dysfunction. They include asthma, emphysema, chronic bronchitis, cystic fibrosis, bronchiectasis, and bronchiolitis. The primary characteristic of these disorders is resistance to airflow. Values for FEF25–75% in adult males and females are normally > 2.0 and > 1.6 L/s, respectively. As the disease progresses, both forced expiratory volume in 1 s (FEV1) and the FEV1/FVC (forced vital capacity) ratio are less than 70% of the predicted values.



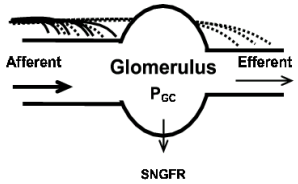
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The normal forced exhalation curve. FEF25–75% is also called the maximum midexpiratory flow rate (MMF25–75%). FRC, functional residual capacity; FEV1, forced expiratory volume in 1 s; FVC, forced vital capacity; RV, residual volume; TLC, total lung capacity.

- Negligence is the failure to use reasonable care, which is that level of care recognized as acceptable and appropriate given the circumstances.
- In a medical negligence action, the plaintiff must prove to the court the following four elements.
  - (1) Duty: reasonable care, as defined by the Standard of Care, is expected.
  - (2) Breach of Duty: failure to meet the Standard of Care.
  - (3) Damage: the plaintiff must have suffered some injury, either physical or emotional.
  - (4) Cause: the damage the plaintiff received was caused by the defendant that deviated from the Standard of Care. There must be a direct or actual cause between the practitioner's actions and the damage the plaintiff received. The plaintiff must also prove proximate cause

---
- Epidural steroid injections are used for (MM 499)
  - acute radiculopathy (nerve root compression)
  - radicular back pain, and
  - degenerative disc disease
- FYI: Epidural steroid injections are most effective when given within 2 weeks of onset of pain and do not appear to provide long-term pain relief beyond 3 months.

- What causes an increase in GFR?
  - Constriction of the EFFERENT arteriole, more than the afferent
  - AFFERENT arteriole dilates more than the EFFERENT



- Cytochrome P450 enzymes aid in all of the following reactions except: (Valley p. 304)
  - Acetylation
  - Hepatic Metabolism
  - Phase I: Oxidation, Reduction, Methylation, Hydrolysis (CP450 is responsible for many of these)
  - Phase II: Glucuronidation, Glutathione Conjugation, Sulfation, Acetylation
- Fyi

* Many exogenous substances, including most drugs, undergo hepatic biotransformation. The end products of these reactions are generally either inactivated or more water-soluble substances that can be readily excreted in bile or urine. Hepatic biotransformations are often categorized as one of two types of reactions. Phase I reactions modify reactive chemical groups through mixed-function oxidases or the cytochrome P-450 enzyme systems, resulting in oxidation, reduction, deamination, sulfoxidation, dealkylation, or methylation. Barbiturates and benzodiazepines are inactivated by phase I reactions. Phase II reactions, which may or may not follow a phase I reaction, involve conjugation of the substance with glucuronide, sulfate, taurine, or glycine. The conjugated compound can then be readily eliminated in urine or bile.

- Cardiopulmonary Bypass
  - Can trigger ARDS
  - ACT > 400 is safe for CABG
  - A cold, 4 degrees C, cardioplegia solution with K+ (10 – 40 mEq/L) is placed into the aortic root which causes cardiac arrest
  - Increased K+ produces myocardial depolarization resulting in cessation of electrical and mechanical activity
  - Hypothermia decreases metabolism of the cardiac muscle

- Chronic Renal Failure – Which drug would have a prolonged elimination?
  - Digoxin, Pancuronium, Morphine, Demerol

fyi

- Digoxin. Elimination ofdigoxin is primarily by the kidneys with approximately 35% of the drug excreted daily (MM 107)

- Quality Assurance
  - **Quality assurance is a system or program that provides an organized procedure for the evaluation of the level of care provided (quality care assessment) and the establishment of mechanisms for improvement. Each hospital department/service is responsible for monitoring QA in their area (required by JCAHO)**
  - A quality assurance program examines adverse outcomes and evaluates the roles of practitioners in the adverse events. Basically, quality assurance documents problems and assigns blame but does not correct or remedy the problems in a systematic fashion.
  - The goal of traditional quality assurance programs identifying deviations from the norm (outliers or "bad apples").
  - **Quality assurance programs:**
    - **(1) define the norm (standard of care),**
    - **(2) determine whether adverse events were caused by deviations from the norm, and**
    - **(3) prevent the occurrence of adverse events**.
  - Note: Thefirst step in a quality assurance program is to define the norm, or set the standards.
  - Quality assurance (QA) has been superseded by quality improvement (QI).

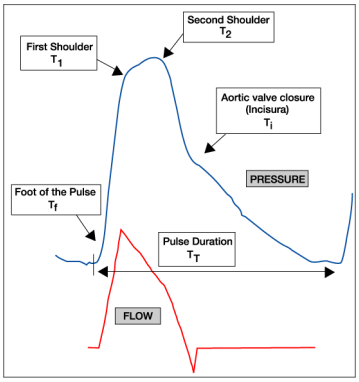
- Quality Improvement:
  - QA has been superseded by QI
  - The problem with QA was the inconsistent reporting of data

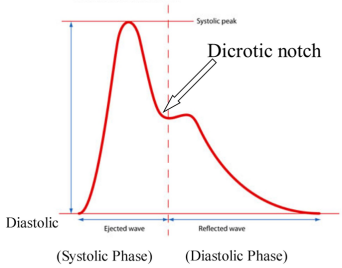
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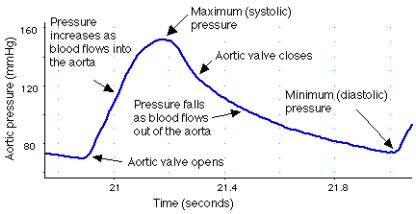
- Opioid with the most side effect with epidurals
  - Epidural morphine, because it has delayed onset, prolonged duration, early & late ventilatory depression
  - Most serious → delayed respiratory depression

- Lipophilic/hydrophilic (What do they have in common)
  - Rostral spread
  - Hydrophilic morphine spreads in much greater quantities than lipophilic
  - Early ventilatory depression
- fyi
  - With regard to intrathecal placement of opioids, hydrophilic opioids (morphine) and lipophilic opioids have what in common?
    - Rostral (headward) spread occurs with either morphine or a lipophilic opioid, but hydrophilic opioid spreads rostrally in much greater quantities than lipophiiic opioids. Late respiratory depression is generally regarded as being due almost exclusively to morphine.

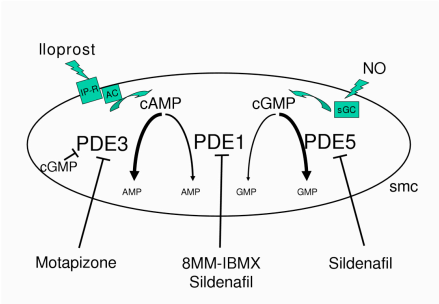
- Arterial waveform where is aortic valve opening?
  - Anacrotic limb (upstroke) is where aortic valve is opening
  - Dicrotic notch is aortic valve closure





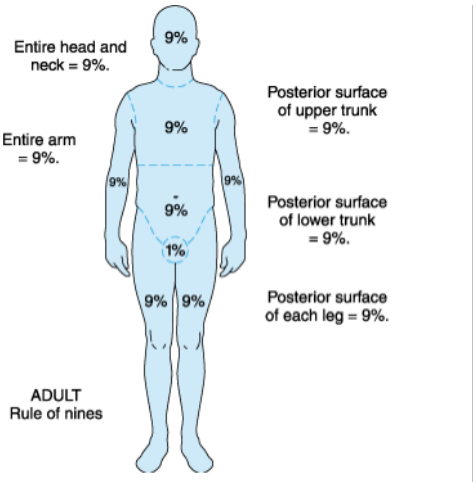


- Name 2 PDE inhibitors
  - Non-Selective: Aminophyline, Caffeine, Theophylline
  - PDE 3: Milrinone
  - PDE 5: Sildenafil, Vardenafil



- 4:2:1 rule:
  - 4 mL/kg/h for the first 10 kg of weight
  - 2 mL/kg/h for the second 10 kg, and
  - 1 mL/kg/h for each remaining kilogram.
- If weight > 20kg, then kg + 40.

- Rule of 9's
  - Adult:
    - Head, both arms (each): 9%
    - Anterior Trunk, Posterior Trunk, Both legs (each): 18 %
    - Perineum: 1%
- Child:
  - Head:18% (9 for front, and 9 for back)
  - Each leg: 13.5 %
- * Everything else is the same as the adult



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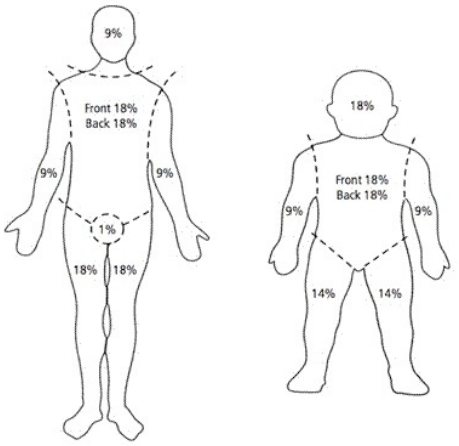
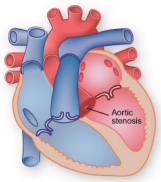


Figure 2: Adult compared to child burn surface areas

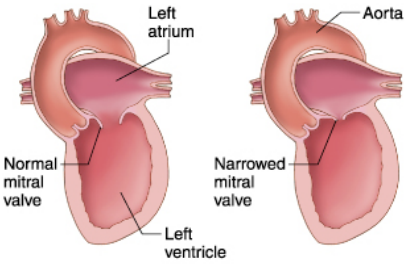
- Train-of-Four (TOF):
  - ND: Fade occurs during TOF & Tetany
  - Depolarizing: No fade
- Fyi
  - phase I depolarizing blockade
    - (1) Decreased twitch height,
    - (2) absence of fade with tetany,
    - (3) slight reduction of twitch height in train-of-four (ratio of T4:TI > 0.7),
    - (4) no post-tetanic potentiation,
    - (5) fasciculations,
    - (6) antagonism of block by nondepolarizing muscle relaxants, and
    - (7) augmentation of block by acetylcholinesterase inhibitors.
  - Phase II block is similar to nondepolarizing block. Characteristics of phase II block include
    - fade in train-of-four,
    - fade during tetany, and
    - posttetanic potentiation (facilitation).

- Aortic Stenosis
  - Low HR (50-70)
  - NSR
  - Maintain preload, SVR, & contractility
  - Aortic Valve area in cm2
    - Mild 1 – 1.5 cm2
    - Moderate: 0.8 – 1 cm2
    - Severe: < 0.8 cm2



fyi: Critical aortic stenosis occurs with an aortic orifice of 0.5–0.7 cm2 (normal is 2.5–3.5 cm2) and a transvalvular pressure gradient of 50 mmHg. (0.7–0.9 cm2 associated with mild to moderate symptoms.)

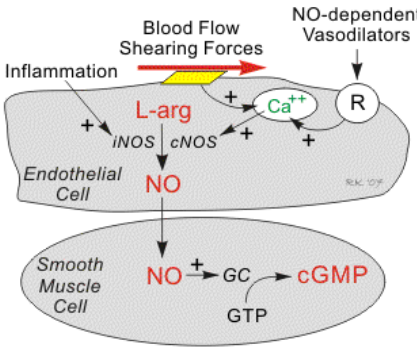
- Mitral Stenosis
  - Same as AS
  - Size of orificace:
    - Normal: 4-6 cm2
    - Severe: < 2 cm



fyi: Mitral orifice is reduced from normal (4–6 cm2) to an opening of less than 2 cm2.

- Don't use ACE inhibitors during pregnancy because of increased receptors in the uterine artery.
- Fyi: Studies have shown that there is a risk associated with the use of ACE inhibitors during the second and third trimesters.

- Nitric Oxide
  - Produced by vascular endothelial cells, diffuses into smooth muscle layer of the vascular wall and activates the enzyme sGC, which triggers the production of 2nd messenger cGMP
  - Causes smooth muscle to relax
- Fyi (MM 208)
  - Nitric oxide (NO) activates the enzyme soluble guanylate cyclase, which increases the production of cyclic guanosine monophosphate (cGMP). Cyclic GMP, a second messenger, relaxes vascular smooth muscle, thereby promoting vasodilation and a decrease in blood pressure.
  - Nitric oxide is formed from L-arginine.
  - The nitric oxide synthase (NOS) family of enzymes catalyzes the release of NO from L-arginine.
  - Nitric oxide stimulates the production of cyclic GMP (cGMP) by activating guanylate cyclase. cGMP activates protein kinase G (PKG) which ultimately decreases cytoplasmic calcium levels.



- What two drugs have synergistic effects?
  - Epi & Cocaine → tachycardia / HTN???

- What LA can you not use topically?
  - Tetracaine

- Spinal/Epidural

Spinal Anesthesia	Epidural Anesthesia
Takes less time to perform	Less risk of postdural puncture headache
Rapid onset	Less Hypotension
Sensory and motor block quality is better	Can prolong block with catheter
Pain during surgery is less	Catheter can be

- Ability to control the spread of the anesthetic by controlling the specific gravity of the solution and the position of the patient

---
- Compartment Syndrome
  - Most associated with lithotomy position, then prone position

---
- Rhabdomyolysis goal is to maintain urinary output > 2mL/kg/hr
  - Caused from myoglobin, which is the O2 carrying pigment of the skeletal muscle, and hemoglobin
- Fyi (MM 420-421)
  - Rhabdomyolysis (literal: "skeletal muscle breakdown") is skeletal muscle necrosis due to muscle tissue injury.
  - The most common causes of rhabdomyolysis are (1) major crush injury, (2) thermal or electrical injury, (3) acute muscle ischemia due to arterial occlusion, (4) acute mmule injury induced by prolonged immobilization, and (5) compartment syndromes, such as occur in hemorrhage or when vascular insufficiency coexists with edema, (6) malignant hyperthermia, (7) extreme lithotomy position, and (8) hyperlordotic position.
  - Myoglobin, the oxygen-carrying pigment of skeletal muscle, and hemo globin are both capable of causing acute renal failure (ARF). Myoglobin seems to be a more potent nephrotoxin because it is more readily filtered at the glomerulus and can be reabsorbed by the renal tubules, where it inhibits nitric oxide (NO) and induces medullary vasoconstriction and ischemia

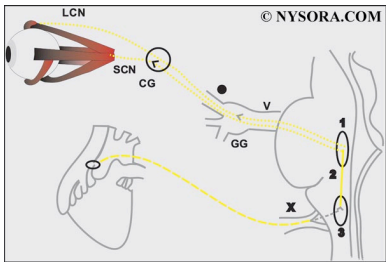
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- Fixed error → failure to correct a machine malfunction during a check??

- IHSS Management Goals:
  - HR, keep near pts baseline
  - NSR
  - **Preload:** keep it full (first line of defense for hypotension)
  - **Contractility:** Maintain or slightly decrease
  - **Afterload:** keep it up (2nd line of defense for hypotension) – pure vasoconstrictor
- Fyi
  - Vasoconstrictors (e.g., phenylephrine), beta-adrenergic receptor blockers (e.g., esrnolol, propranolol), and myocardial depressant anesthetics (e.g., enflurane, halothane) are acceptable for use in the patient with IHSS. These agents reduce the outflow tract obstruction which is reflected by a reduction in the LV-aortic systolic pressure gradient.
  - (1) Volume is first line treatment for IHSS: full full full; (2) The next prescription is administration of an alpha-adrenergic vasoconstrictor such as phenylephrine; (up, up, up).
  - Vasodilators (nitroglycerin, nitroprusside), positive inotropes (digitalis, idiopathic subaortic stenosis? calcium), beta adrenergic agonists, and diuretics are avoided in the patient with idiopathic hypertrophic subaortic stenosis. These agents can worsen the left ventricular outflow obstruction.

- Polyhydramnios is common with what condition?
  - **Esophageal Atresia (MM 485)**
  - Polyhydramnios is the presence of excessive amniotic fluid around the unborn infant. Polyhydramnios occurs when the infant fails to swallow and absorb amniotic fluid in normal amounts. The most common causes are gastrointestinal obstruction and neurological problems. Esophageal atresia, as seen in tracheoesophagealjstula (TEF) is ofgreatest concern for the infant. If polyhydramnios is present, check for TEF by passing a nasogastric tube shortly after delivery. Note: Polyhydramnios also appears antenatally in 30% of congenital diaphragmatic hernia cases. (Barash, Handbook). If both answers appear and no other pertinent information is given, choose tracheoesophageal fistula as the BEST answer.

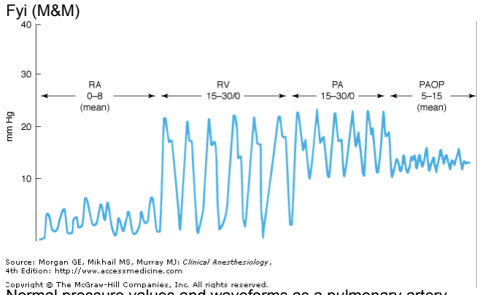
- Prostaglandin inhibitors
  - Inhibit prostaglandin synthesis by inhibiting the enzyme cyclooxygenase which normally converts arachadonic acid to prostaglandin
  - Examples:
    - NSAIDS: ibuprofen, ketorlac, indomethacin
    - ASA

- Occulorcardiac Reflex
  - CN 5 (Trigeminal) Afferent (sensory)
  - CN 10 (Vagus) Efferent (motor)



Insertion Site	To Right Atrium (+10) =>	To Right Ventricle (+15) =>	To Pulmonary Artery
Subclavian	15	25	40
Right IJ	20	30	45
Left IJ	25	35	50
Right AC	40	50	65
Left AC	45	55	70
Femoral	50	60	75

Fyi (M&M)



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Normal pressure values and waveforms as a pulmonary artery catheter is advanced from the right atrium to a "wedged" position in a pulmonary artery. RA, right atrium; RV, right ventricle; PA, pulmonary artery; PAOP, pulmonary artery occlusion pressure.

Variable	Formula	Normal	Units
Cardiac index	$\frac{\text{Cardiac output (L/min)}}{\text{Body surface area } (m^2)}$	2.2–4.2	L/min/m²
Total peripheral resistance	$\frac{(\text{MAP} - \text{CVP}) \times 80}{\text{Cardiac output (L/min)}}$	1200–1500	dynes • s cm^{-5}
Pulmonary vascular resistance	$\frac{(\text{PA} - \text{PAOP}) \times 80}{\text{Cardiac output (L/min)}}$	100–300	dynes • s cm^{-5}
Stroke volume	$\frac{\text{Cardiac output (L/min)} \times 1000}{\text{Heart rate (beats/min)}}$	60–90	mL/beat
Stroke index (SI)	$\frac{\text{Stroke volume (mL/beat)}}{\text{Body surface area } (m^2)}$	20–65	mL/beat/m²
Right ventricular stroke-work index	0.0136 (PA – CVP) × SI	30–65	g-m/beat/m²
Left ventricular stroke-work index	0.0136 (MAP – PAOP) × SI	46–60	g-m/beat/m²

Table 6–5. Pulmonary Artery Occlusion Pressure (PAOP) Can Wrongly Estimate Left Ventricular End-Diastolic Pressure (LVEDP) in Certain Conditions.
- PAOP > LVEDP
  - Mitral stenosis
  - Left atrial myxoma
  - Pulmonary venous obstruction
  - Elevated alveolar pressure
- PAOP < LVEDP
  - Decreased left ventricular compliance (stiff ventricle or LVEDP > 25 mm Hg)
  - Aortic insufficiency

- Kidney Function in Acid/Base Balance ☒
  - Na/H exchange is the fundamental even in kidney's regulation of A/B balance, specifically, it permits HCO3 to be reabsorbed and acids to be excreted.

- What is the best pre-medication for ESRD?
  - Morphine???? --- NO WAY

- What are contraindications to FOI?
  - Hypoxia, heavy airway secretions, bleeding from upper/lower airway, local anesthetic allergy, inability to cooperate

---
- Dibucaine
  - Normal: 70-85% (80%)
  - Heterozygous = 30 – 70% = slightly prolonged
  - Homozygous = 16 – 25% = "Quite" prolonged (6-8 hours)
  - Dibucaine reflects the effectiveness, not the quantity
- fyi
  - Dibucaine depresses the activity of pseudocholinesterase. If pseudocholinesterase is normal, dibucaine will depress the activity of pseudocholinesterase by 70-85%. The percent that dibucaine depresses pseudocholinesterase is called the dibucaine number. The normal dibucaine number is usually said to be 80 (80% depression). The range of dibucaine numbers corresponding to normal is 70-85.
  - Heterozygotes have dibucaine numbers for atypical plasma cholinesterase that range from 30-70 (Morgan and Mikhail state that 40-60 is the range of dibucaine numbers for heterozygotes). Neuromuscular block in heterozygotes is slightly prolonged by succinylcholine.
- Fyi
  - If the dibucaine number is 20, the individual is a homozygote for the abnormal (atypical) pseudocholinesterase.
  - If the dibucaine number is between 30 and 70, the individual is a heterozygote for the abnormal (or atypical) pseudocholinesterase.
  - Homozygotes have a dibucaine number for atypical cholinesterase of about 20 (range: 16-25). Neuromuscular block with succinylcholine is quite prolonged (6-8 hours) in homozygotes
  - The patient who remains paralyzed several hours (2-8 hours) after receiving succinylcholine (or mivacurium) has a dibucaine number of 20 (the patient is homozygous for atypical plasma cholinesterase), and the concentration of plasma cholinesterase is normal
  - The dibucaine number reflects the effectiveness of the enzyme for hydrolyzing succinylcholine, not the quantity or concentration of the enzyme in the plasma

---
- Lithotomy, what nerve is damaged?
  - Common Peroneal

- 5 Nerves that could be damaged in Lithotomy position are?
  - Common peroneal
  - Sciatic
  - Femoral
  - Saphenous
  - Obturator

UNANSWERED

Question on how you know respiratory depression with spinal anesthesia ?????
Best indicator for ischemia in heart surgery when aorta is clamped?
* TEE????
What position decreases venous return the most?
* Reverse T-burg???
Limited Peds & OB questions
Regional Anesthesia complications
Common Nerve Injuries
Lots of Liver Questions
Supine changes → HR down, CO up, decreased FRC ??
???? QA question regarding structure… License/credentialing …. ????
Know what increases and decreased with pregnancy
Regular VS Intubating LMA
Epidural Steroid Injections → MM p. 513 - 514 → Lightening Bolts & Bullseyes
Pick 3 with prolonged propofol use (sedation)
Know Volume of Distribution
READ MM p. 532

- Elderly patient, otherwise healthy, what would indicate they are in good physical condition?
  - ??? VD/VT ???
- Does the VD/VT ratio generally increase, decrease, or remain the same in the geriatric population? [MM p. 489]
  - With advancing age dead space increases (increased VD), therefore the Vd/VT ratio will also increase. Recall that VD/VTis given by the Bohr equation.
  - Normal VD/VT= 150 mL/450 mL = 0.33 (33%).
  - In other words, in the geriatric person, the fraction of tidal volume that is dead space increases.
  - Clinical relevance: increased VD requires increased minute ventilation (VE) and increased work of breathing to maintain alveolar ventilation (VA)

- During and epidural block the patient's blood pressure drops precipitously to 80135, the heart rate falls to 50 bpm, and S,02 falls to 85%. What has probably happened? (MM 431)
  - Severe hypotension, bradycardia, and respiratory insufficiency during an epidural block are signs and symptoms of subdural injection of the anesthetic agent. You have noticed that these are the same signs and symptoms of a "high spinal," that is, sympathetic block with unopposed parasympathetic effects. Other possible signs and symptoms of a subdural injection include patchy and markedly asymmetric extensive spread of analgesia

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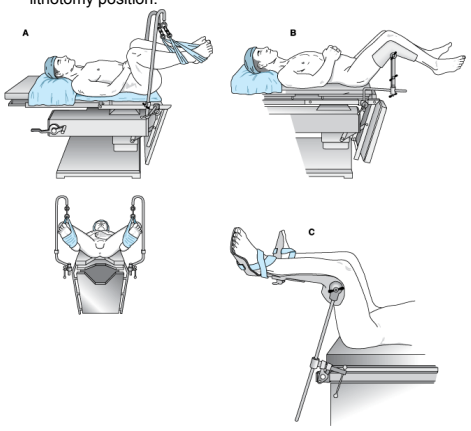
- Pick 2: Pt has hepatic dysfunction, which drugs will be prolonged?
  - Vecuronium and Sux (because plasma cholinesterase is synthesized in liver)
- Fyi (MM 100)
  - Cirrhotic liver disease (and renal failure) often results in an increased volume of distribution and lower plasma concentrations for a given water-soluble drug, such as muscle relaxants. Furthermore, drugs that depend upon hepatic or renal excretion may have a prolonged clearance. Thus, depending on the drug of choice, agreater initial loading dose, but smaller maintenance doses might be required in cirrhotic liver disease.
  - Vecuronium, pancuronium, and rocuronium blockade are prolonged by liver disease.

- In an emergency situation, what 2 drugs can be given IM to a pediatric patient?
  - Succinylcholine & Rocuronium

- Epidural: what two things determine the dermatome level?
  - Volume
  - Level that epidural is placed at

- ** Regional anesthesia can mask compartment syndrome **

- What position would you see compartment syndrome?
  - Lithotomy position
  - Fracture table??? (MM 320)
- Fyi (M&M)
  - A compartment syndrome of the lower extremities with rhabdomyolysis has been reported with prolonged time in the lithotomy position.

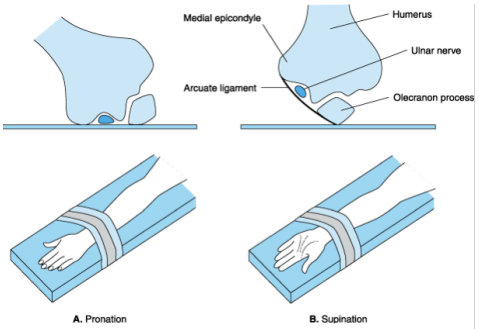


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- Pick 3 Questions: You want to block the Middle and Ring finger for surgery, what nerves would you block?
  - Ulnar, Median, Radial

FYI: PERIPHERAL NERVE INJURY
(M&M Chapter 46. Anesthetic Complications)

- Perioperative nerve injury is a known complication of both regional and general anesthesia
- In most cases, these injuries resolve within 6–12 weeks, but some persist for months or even years.
- The most common peripheral nerve injury is **ulnar neuropathy**.
  - initial symptoms were most frequently noted more than 24 h after a surgical procedure and may have occurred while the patient was on the hospital ward sleeping.
  - Risk factors included male gender, hospital stay greater than 14 days, and very thin or obese body habitus.
  - Anesthetic technique was not implicated as a risk factor



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Pronation of the forearm can cause external compression of the ulnar nerve in the cubital tunnel (A). Forearm supination avoids this problem (B).

Table 46–4. Physiological Effects of Common Patient Positions.

Position	Organ System	Effects
Supine		
Horizontal[1]	Cardiac	Equalization of pressures throughout the arterial system; increased right-sided filling and cardiac output; decreased heart rate and peripheral vascular resistance.
	Respiratory	Gravity increases perfusion of dependent (posterior) lung segments; abdominal viscera displace diaphragm cephalad. Spontaneous ventilation favors dependent lung segments, while controlled ventilation favors independent (anterior) segments. Functional residual capacity decreases and may fall below closing volume in older patients.
Trendelenburg	Cardiac	Activation of baroreceptors, generally causing decreased cardiac output, peripheral vascular resistance, heart rate, and blood pressure.
	Respiratory	Marked decreases in lung capacities from shift of abdominal viscera; increased ventilation/perfusion mismatching and atelectasis; increased likelihood of regurgitation.
	Other	Increase in intracranial pressure and decrease in cerebral blood flow because of cerebral venous congestion; increased intraocular pressure in patients with glaucoma.
Reverse Trendelenburg	Cardiac	Preload, cardiac output, and arterial pressure decrease. Baroreflexes increase sympathetic tone, heart rate, and peripheral vascular resistance.
	Respiratory	Spontaneous respiration requires less work; functional residual capacity increases.
	Other	Cerebral perfusion pressure and blood flow may decrease.
Lithotomy	Cardiac	Autotransfusion from leg vessels increases circulating blood volume and preload; lowering legs has opposite effect. Effect on blood pressure and cardiac output depends on volume status.
	Respiratory	Decreases vital capacity; increases likelihood of aspiration.
Prone	Cardiac	Pooling of blood in extremities and compression of abdominal muscles may decrease preload, cardiac output, and blood pressure.[2]
	Respiratory	Compression of abdomen and thorax decreases total lung compliance and increases work of breathing.
	Other	Extreme head rotation may decrease cerebral venous drainage and cerebral blood flow.
Lateral decubitus	Cardiac	Cardiac output unchanged unless venous return obstructed (eg, kidney rest). Arterial blood pressure may fall as a result of decreased vascular resistance (right side > left side).
	Respiratory	Decreased volume of dependent lung; increased perfusion of dependent lung. Increased ventilation of dependent lung in awake patients (no $\dot{V}/\dot{Q}$ mismatch); decreased ventilation of dependent lung in anesthetized patients ($\dot{V}/\dot{Q}$ mismatch). Further decreases in dependent lung ventilation with paralysis and an open chest (see Chapter 24).
Sitting	Cardiac	Pooling blood in lower body decreases central blood volume. Cardiac output and arterial blood pressure fall despite rise in heart rate and systemic vascular resistance.
	Respiratory	Lung volumes and functional residual capacity increase; work of breathing increases.
	Other	Cerebral blood flow decreases.

[1]The effects described for the horizontal position are in comparison with a patient standing erect. All other positions are compared with the horizontal position.

[2]Changes associated with the prone position are exaggerated by the convex saddle frame used in posterior spinal surgery and minimized by the prone jackknife position.

Table 46–5. Complications Associated with Patient Positioning.

Complication	Position	Prevention
Air embolism	Sitting, prone, reverse Trendelenburg	Maintain venous pressure above 0 at the wound (see Chapter 26).
Alopecia	Supine, lithotomy, Trendelenburg	Normotension, padding, and occasional head turning.
Backache	Any	Lumbar support, padding, and slight hip flexion.
Compartment syndrome	Especially lithotomy	Maintain perfusion pressure and avoid external compression.
Corneal abrasion	Especially prone	Taping and/or lubricating eye.
Digit amputation	Any	Check for protruding digits before changing table configuration.
Nerve palsies		
Brachial plexus	Any	Avoid stretching or direct compression at neck or axilla.
Common peroneal	Lithotomy, lateral decubitus	Pad lateral aspect of upper fibula.
Radial	Any	Avoid compression of lateral humerus.
Ulnar	Any	Padding at elbow, forearm supination.
Retinal ischemia	Prone, sitting	Avoid pressure on globe.
Skin necrosis	Any	Padding over bony prominences.

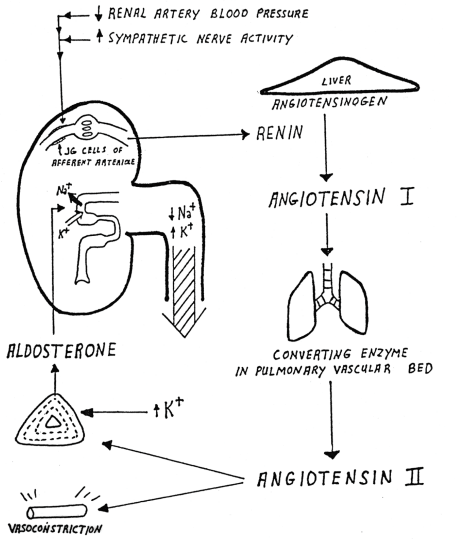
- Pick 4 LOW Pressure systems [MM p. 270]
- Flow meter tubes, vaporizers, check valves, & common gas outlet
- Just FYI
  - Components found in the **LOW PRESSURE** system include: (1) flow indicators, (2)vaporizers, (3) vaporizer circuit control valves, (4) back pressure safety devices, (5) low-pressure safety devices, and (6) the common gas outlet.
  - The four components of the anesthesia gas machine that are exposed to **HIGH-PRESSURE** (cylinder pressure) are: (1) hanger yoke, (2) yoke block with check valves, (3) cylinder pressure gauge, and (4) cylinder pressure regulators (MM 252).
  - The eight components of the anesthesia gas machine that are exposed to **intermediate pressures** (pipeline pressure, 50 psi) are: (1) pipeline inlets, (2) check valves, (3) pressure gauges, (4) ventilator power inlet, (5) oxy gen pressure-failure device, (6) flowmeter valve, (7) oxygen second-stage regulator, and (8) flush valve (MM 253).

- How does muscle weakness of the patient with myasthenic (Eaton-Lambert) syndrome differ from the muscle weakness of the patient with myasthenia gravis?
  - Muscle strength deteriorates rapidly with repeated effort in the patient with myasthenia gravis but improves with repeated effort in the patient with Lambert-Eaton myasthenic syndrome (LEMS).
  - Muscle strength improves with anticholinesterase therapy in the patient with myasthenia gravis but is unaffected in the patient with Lambert-Eaton myasthenic syndrome.
- Fyi:
  - The Lambert-Eaton myasthenic syndrome (LEMS) is characterized by proximal skeletal muscle weakness that typically affects the lower extremities.
  - The Lambert-Eaton myasthenic syndrome (LEMS) is usually associated with small-cell carcinoma of the lung, or less commonly, other occult malignancies, or autoimmune disorders.
  - A defect in the motor nerve terminal causes a decreased release of acetylcholine. The decreased release of acetylcholine is thought to be due to antibodies directed against the voltage-gated calcium channel.
  - VERY sensitive to NDNMB & Depolarizing agents. (Only give small doses)

- Mediastinoscopy is associated with **LEFT Reccurent Laryngeal Nerve** damage [M&M]
- Fyi: Complications associated with mediastinoscopy include
  - (1) vagally mediated reflex bradycardia from compression of the trachea or the great vessels,
  - (2) excessive hemorrhage (see above),
  - (3) cerebral ischemia from compression of the innominate artery (detected with a plethysmograph or pulse oximeter on the right hand),
  - (4) pneumothorax (usually presents postoperatively),
  - (5) air embolism (because of a 30° head elevation, the risk is greatest during spontaneous ventilation),
  - (6) recurrent laryngeal nerve damage, and
  - (7) phrenic nerve injury.
- Fyi (MM 375): List the three most frequently encountered complications of mediastinoscopy, on order of greatest to least incidence.
  - The most common complication of rnediastinoscopy is hemorrhage because of the proximity of the vessels and the vascularity of certain tumors. The second most common complication of rnediastinoscopy is pneumo thorax, usually right-sided. The third most common complication is recurrent laryngeal nerve injury, and is permanent in up to 50% of cases.

- List the one absolute contraindication for performing anesthesia for mediastinoscopy.
  - Previous rnediastinoscopy.

- Renin–Angiotensin–Aldosterone (M&M)
  - Renin secretion increases the formation of angiotensin II. The latter increases the secretion of aldosterone and has some direct effect in enhancing Na+ reabsorption in the proximal renal tubules. Angiotensin II is also a potent direct vasoconstrictor and potentiates the actions of norepinephrine. Secretion of aldosterone enhances Na+ reabsorption in the distal nephron (see Chapter 31) and is a major determinant of urinary Na+ excretion.
- Renin–Angiotensin–Aldosterone (Valley 2010, p. 42)
  - 1. 85% of resting blood pressure is controlled by renin.
  - 2. Renin, an enzyme released from juxtaglomerular (JG) cells of the afferent arteriole (see upper portion of figure), converts angiotensinogen (a protein released into the circulation from the liver) to angiotensin I.
  - 3. Angiotensin I is converted to angiotensin II by angiotensin converting enzyme (ACE); angiotensin converting enzyme is found on the endothelial surface of pulmonary capillaries.
  - 4. Angiotensin II promotes vasoconstriction and aldosterone release (note: the two important stimuli for aldosterone release are angiotensin II and high serum potassium; two other less potent stimuli for aldosterone release are low serum sodium and ACTH); note that while angiotensin II is a potent vasoconstrictor, antidiuretic hormone (ADH, vasopressin) is a more potent vasoconstrictor. Note: this statement is controversial; Guyton states: "Vasopressin...is even more powerful than angiotensin II..." (2006, p202). Barash, p1126, states: "Angiotensin II is the most potent vasoconstrictor produced by the body."
  - 5. Aldosterone increases potassium secretion (and excretion) and increases sodium reabsorption, which promotes sodium retention; sodium retention promotes volume expansion.
  - 6. Renin release occurs in response to decreased renal blood pressure (e.g., renal artery stenosis) and/or increased sympathetic nervous system activity, and Cl– .



- Malignant Hyperthermia Protocol (M&M)
  - 1. Discontinue volatile anesthetic and succinylcholine. Call for help!
  - 2. Hyperventilate with 100% O2 at high flows.
  - 3. Administer sodium bicarbonate, 1–2 mEq/kg intravenously.
  - 4. Mix dantrolene sodium with sterile distilled water and administer 2.5 mg/kg intravenously as soon as possible.
  - 5. Institute cooling measures (lavage, cooling blanket, cold intravenous solutions).
  - 6. Administer inotropes and antiarrhythmic agents as necessary.
  - 7. Administer additional doses of dantrolene if needed.
  - 8. Change anesthetic tubing and soda lime.
  - 9. Monitor urinary output, K+, Ca2+, blood gases, end-tidal carbon dioxide; perform clotting studies.
  - 10. Treat severe hyperkalemia with dextrose, 25–50 g intravenously, and regular insulin, 10–20 U intravenously (adult dose).
  - 11. Consider invasive monitoring of arterial blood pressure and central venous pressure.
  - 12. If necessary, consult on-call physicians at the 24-hour MHAUS hotline, 1-800-MH-HYPER.
- Fyi (Dantrolene, M&M)
  - Dantrolene, a hydantoin derivative, directly interferes with muscle contraction by binding the Ryr1 receptor calcium channel and inhibiting calcium ion release from the sarcoplasmic reticulum. This intracellular dissociation of excitation–contraction coupling contrasts with the depolarizing and nondepolarizing muscle relaxants that antagonize the extracellular neuromuscular junction.
  - The dose is 2.5 mg/kg intravenously every 5 min until the episode is terminated. The upper limit of dantrolene therapy is generally 10 mg/kg.
  - Dantrolene is packaged as 20 mg of lyophilized powder to be dissolved in 60 mL of sterile water. Depending on the dose
  - required, reconstitution can be time consuming.
  - The effective half-life of dantrolene is about 6 h.
  - After initial control, dantrolene 1 mg/kg intravenously is recommended every 6 h for 24–48 h to prevent relapse because MH can recur within 24 h.
  - It should be noted that dantrolene is not specific for MH; it also decreases temperature in thyroid storm and neuroleptic malignant syndrome. Dantrolene is a relatively safe drug.
  - Although chronic therapy for spastic disorders has been associated with hepatic dysfunction, the most serious complication following acute administration is generalized muscle weakness that may result in respiratory insufficiency or aspiration pneumonia. Dantrolene can cause phlebitis in small peripheral veins and should be given through a central venous line if one is available. The safety and efficacy of dantrolene call for its immediate use in this potentially life-threatening disease.

- How do you know that you are successful at mask-ventilating a patient?
  - No air leak around sides of mask??

- You've given a dose of propofol…How do you know when it is time to put in an LMA?
  - When pt doesn't respond to jaw thrust ??

- Board Question: Suspect OB patient has high spinal, what do you do last?
  - Choices Were: Fluid, vasopressor, left lateral tilt, reverse trendelenburg
- Of the following, what is the last thing you would do to treat the patient who has signs and symptoms of high spinal: (1) mechanical ventilation with 100% oxygen; (2) place the patient in Trendelenburg position, (3) give ephedrine, (4) give fluids?
  - Because this is an emergency situation, immediate treatment includes positive pressure ventilation with 100% oxygen, infusion of fluids, and administration of a vasopressors such as ephedrine and a positive chronotrope such as atropine. It appears the last action is to place the patient in the head down (Trendelenburg) position.

- Pick 2: most important monitors for accidental extubation? [MM p. 294]
  - **ETCO2**, O2 analyzer, loss of pressure, **pulse ox**
- Disconnection?
  - (1) Pulse Ox, (2) Mass Spectrometer, (3) Capnograph, (4) Stethoscope, (5) Spirometer
  - What is the best monitor to detect a disconnection?
    - BEST: **capnography and spirometry**
    - NEXT BEST: pulse oximetry and the stethoscope
- Esophageal Intubation?
  - (1) Pulse Ox, (2) Mass Spectrometer, (3) Capnograph, (4) Stethoscope
- Bronchial Intubation?
  - **(1) Pulse Ox**, (2) **Stethoscope**

Fyi

- What is the most reliable evidence that the endotracheal tube is placed in the trachea and not the esophagus?
  - The most reliable method of verification is ***direct visualization of the laryngeal inlet*** with the endotracheal tube positively identified in it. Detection of **expired carbon dioxide** with a **capnograph or mass spectrometer** also provides reliable evidence of tracheal rather than esophageal intubation.
- List 13 specific problems that can be detected by mass spectrometry.
  - (1) Error in gas delivery (C02, 02, N2, and inhaled agent analysis);
  - (2) anesthesia machine malfunction (C02, 02, N2, and inhaled agent
  - analysis);
  - (3) disconnection (C02, 02, N2, and inhaled agent analysis);
  - (4) vaporizer malfunction or contamination (inhaled agent analysis);
  - (5) anesthesia circuit leaks (C02 and N2 analysis);
  - (6) endotracheal tube cuff leak (CO2 and N2 analysis);
  - (7) poor mask or LMA fit (C02 and N2 analysis);
  - (8) hypoventilation (COz analysis);
  - (9) malignant hyperthermia (COz analysis);
  - (10) airway obstruction including kinked endotracheal tube (CO2 analysis);
  - (11) air embolism (CO2 and N2 analysis);
  - (12) circuit hypoxia (02 analysis);
  - (13) vaporizer overdose (agent analysis).
- Fyi (M&M)
  - esophageal intubation
    - visualization of the tip of the TT passing through the vocal cords, careful auscultation for the presence of bilateral breath sounds and the absence of gastric gurgling while ventilating through the TT, analysis of exhaled gas for the presence of CO2 (the most reliable method), chest radiography, or use of an FOB.
  - Clues to the diagnosis of bronchial intubation include unilateral breath sounds, unexpected hypoxia with pulse oximetry (unreliable with high inspired oxygen concentrations), inability to palpate the TT cuff in the sternal notch during cuff inflation, and decreased breathing-bag compliance (high peak inspiratory pressures).
  - Capnography rapidly and reliably indicates esophageal intubation—a common cause of anesthetic catastrophe—but does not detect bronchial intubation.
  - Bronchial intubation will usually go undetected by pulse oximetry in the absence of lung disease or low fraction of inspired oxygen concentrations (FIO2).

- ANEMIA
- Fyi (M&M)
  - What Physiological Effects Result from Severe Anemia?
    - Assuming the maintenance of normovolemia and the absence of preexisting major end-organ dysfunction, most patients tolerate severe anemia surprisingly well. Decreased blood viscosity and vasodilation lower systemic vascular resistance and increase blood flow. Augmentation of stroke volume increases cardiac output, allowing arterial blood pressure and heart rate to remain relatively unchanged. Coronary and cerebral blood flows increase in the absence of coronary artery disease and carotid artery stenosis. A decrease in venous oxygen saturation reflects an increase in tissue oxygen extraction. Oozing from surgical wounds as a result of dilutional coagulopathy may accompany extreme degrees of anemia.
-

- ***Late physiologic compensation for pt with anemia?***
  - Choices were: decreased blood viscocity, decreased plasma volume, and 2 others
- [MM p. 112, # 2]: What four physiological compensatory mechanisms of anemia tend to restore tissue oxygenation in the anemic patient? Flow, and hence oxygen deliver, to the tissues is increased in the anemic patient because there is:
  - (1) **a decrease in blood viscosity**,
  - (2) dilation of peripheral vasculature (decreased SVR), and
  - (3) an increased stroke volume (increased cardiac output). In addition,
  - (4) tissues extract more oxygen.
- [# 3]: What are 5 compensatory mechanisms to increase oxygen delivery in chronic anemia?
  - (1) Increased cardiac output,
  - (2) increased red blood cell 2,3 DPG levels,
  - (3) increased Pm (rightward shift of oxyhemoglobin dissociation curve),
  - (4) increased dissolved 02 in the plasma volume, and
  - (5) **decreased blood viscosity** (causing increased blood flow through tissues).

- Pt refused anesthesia plan in pre-op? Next action?
  - Choices were: Get a signature of declined informed consent???? Talk to family members???

- Benzacaine is a weak acid with pKa 3.5

- Closure of PDA in relation to prostaglandins, toradol, steroids
  - NSAIDS are not recommended due to suppression of uterine contractions and they promote closure of fetal ductus arteriosus

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- A right-to-left intracardiac shunt is present in the patient with a ventricular septal defect (VSD with Eisenmenger's syndrome). What hemodynamic alterations may **worsen** (increase shunt flow) the right-to-left shunt of VSD with Eisenmenger's syndrome?
  - An abrupt increase in pulmonary vascular resistance (PVR) or a decrease in systemic vascular resistance (SVR) is poorly tolerated in the patient with ventricular septal defect (VSD).
  - Avoid interventions that may increase PVR or decrease SVR in the patient with a right-to-left intracardiac shunt.
- List three conditions that increase right-to-left shunt (tetralogy of Fallot).
  - Acidosis, hypercarbia, and hypotension (decreased SVR) increase right-to-left shunt. In general, increases in pulmonary vascular resistance or decreases in systemic vascular resistance (caused by acidosis or hypercarbia) increase right-to-left shunt. Volatile anesthetics and histamine release decrease SVR.

fyi

- List four congenital heart defects involved with tetralogy of Fallot (cyanotic heart disease).
  - (1) Ventricular septal defect, (2) right ventricular outflow tract obstruction (pulmonary stenosis), (3) right ventricular hypertrophy, and (4) dextroposition (to the right) of the aorta with overriding of the ventricular septal defect
- Does blood shunt right-to-left or left-to-right through the ventricular septal defect in tetralogy of Fallot (cyanotic heart disease)?
  - Blood shunts right-to-left, permitting unoxygenated blood to mix with oxygenated blood, resulting in cyanosis.
- What pharmacologic agent decreases a right-to-left shunt?
  - A decrease in the magnitude of a right-to-left shunt (tetralogy of Fallot is a right-to-left shunt) occurs if systemic vascular resistance (SVR) increases. **Phenylephrine** increases SVR and decreases a right-to-left shunt.

- Sharp, prickling pain → **somatic** or visceral pain?
  - Somatic pain (A Delta) refers to pain that is well localized, sharp in nature, and generally hurts in the area of stimulus
  - Visceral pain (C Fibers) is diffuse, can be referred to another area, and is often described as dull and vague in nature
- Fyi (MM 445): Describe nociceptive pain.
  - There are two general categories of pain, nociceptive and neuropathic. Nociceptive pain is pain associated with the stimulation of specific nociceptors (pain receptors). Nociceptive pain may be either somatic or visceral.
    - Somatic pain refers to pain that is well localized, sharp in nature, and generally hurts in the area of stimulus.
    - Visceral pain is diffuse, can be referred to another area, and is often described as dull and vague in nature.
    - Pain originating from cutaneous nociceptors is carried by A-delta and C fibers.
    - Nociceptive pain is "physiological" pain. Nociception is the process of transduction, transmission, perception, and modulation of pain.
    - Main point: Nociceptive pain arises when tissue trauma stimulates somatic or visceral pain receptors called nociceptors.

- What happens to red blood cells and hematocrit early (within the first 24 hours) in the burn patient? (MM 425)
  - Red blood cells are destroyed, but hematocrit increases due to a rapid loss of plasma volume.
  - *** Hemolysis with a normal HCT ***

- ALLOWABLE BLOOD LOSS CALCULATION [MM p. 307, # 4] The loss what volume of RBCs will cause the HCT to drop from 40% to 30% in a 70kg adult male? What is the allowable blood loss.
  - Figure out EBV → 70 kg X 75 mL/kg = 5250 mL
  - EBV X Starting HCT (40% or 0.4) = 2100 mL
  - EBV X Ending HCT (30% or 0.3) = 1575 mL
  - Subtract the two … 2100 – 1575 = 525 mL = RBC loss
  - Then multiply by 3 … 525 X 3 = 1575 = ABL
  - ALLOWABLE BLOOD LOSS = 1575 ML

- What is the Nurse Practice Act?
  - The Nurse Practice Act codifies (sets forth in legal terms) the duties and practices of advanced practice nursing in **state** statutes. The Nurse Practice Act in each state (the guidelines may vary from state to state) provides a legal framework for public protection of patients receiving nurse anesthesia services and protects against charges of "practicing medicine."

- To be successful in a lawsuit, the patient-plantiff must prove what 4 things?
  - [1] **Duty**: that the anesthetist owed him/her a duty
  - [2] **Breach of Duty**: that the anesthetist failed to fulfill his/her duty
  - [3] **Causation**: that a reasonably close causal relation exists between the anesthetist's acts and the resultant injury
  - [4] **Damages**: that actual damage resulted because of the acts of the anesthetist

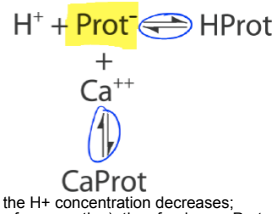
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- Obese pt was on 70% O2 and O2 saturation were 93%, which ones will increase O2 tension the most?
  - Choices were: **PEEP**, Increase TV, Increased RR, Increased O2
  - CPAP wasn't an option

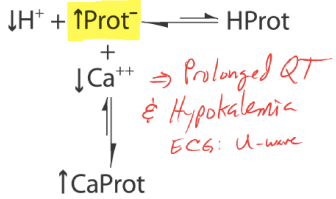
- Hyperventilation → hypOcalcemia

- How Hyperventilation Lowers Ionized Calcium

1. Both H^+ and Ca^{++} bind to plasma proteins (Prot) NORMAL



2. With hyperventilation, the H+ concentration decreases; proteins release H+ (law of mass action), thus freeing up Prot– which can bind ionized calcium (Ca++) and correspondingly lower plasma concentration of ionized calcium (Ca++).



- Know: With acute respiratory alkalosis (hyperventilation), free ionized calcium concentration decreases in plasma: signs and symptoms of hypocalcemia may be manifested.

- Question: Signs and symptoms of what two electrolyte abnormalities may be manifested in the hyperventilating patient?
- Answer: Signs and symptoms of hypokalemia and hypocalcemia may be manifested in the patient who is hyperventilating. Remember, however, that with hyperventilation there may develop a true hypokalemia but a true hypocalcemia does not develop (total calcium, ionized plus non-ionized, does not change). Signs and symptoms of hypocalcemia may develop during hyperventilation because of the decrease in free ionized calcium.

- Addison's disease → Primary adrenal insufficiency
  - Decreased: BP, Na++, Blood glucose
  - Increased: K+
- Fyi
  - Addison's disease (primary adrenal insufficiency) is caused by autoimmune destruction of the adrenal cortex, causing both glucocorticoid and mineralocorticoid deficiency.
  - Signs and symptoms of Addison's disease include: (1) hypotension, (2) hyponatremia, (3) hyperkalemia, (4) hypoglycemia, (5) hemoconcentration, (6) skin pigmentation, (7) weight loss, (8) skeletal muscle weakness, and (9) abdominal or back pain. Note: The four signs and symptoms to remember are hypotension, hyponatremia, hyperkalemia, and hypoglycemia. Hypotension is almost always present, while hyponatremia, hyperkalemia and hypoglycemia are often, but not always, present.

- Hypoglycemia is the result of diminished cortisol secretion from the adrenal cortex. Recall that cortisol normally increases plasma glucose levels by preventing entry of glucose into cells of skeletal muscle and adipose tissue.
- The hyponatremia and hyperkalemia are attributable to a deficiency of aldosterone. In Addison's disease, secretion of mineralocorticoids (namely, aldosterone) and glucocorticoids (namely, cortisol) from the adrenal cortex is markedly reduced. In the absence of aldosterone, sodium reabsorption by the renal tubules is diminished, sodium excretion increases, and hyponatremia develops. Also, in the absence of aldosterone, potassium secretion is diminished, potassium retention occurs, and hyperkalemia develops.

- O2 FLOW RATES

Table 49–2. Oxygen Delivery Devices and Systems.

Device/System	Oxygen Flow Rate (L/min)	FIO2 Range
Nasal cannula	1	0.21–0.24
	2	0.23–0.28
	3	0.27–0.34
	4	0.31–0.38
	5–6	0.32–0.44
Simple masks	5–6	0.30–0.45
	7–8	0.40–0.60
Masks with reservoirs	5	0.35–0.50
Partial rebreather mask-bag	7	0.35–0.75
	15	0.65–1.00
Nonrebreathing mask-bag	7–15	0.40–1.00
Venturi masks and jet nebulizers	4–6 (total flow = 15)	0.24
	4–6 (total flow = 45)	0.28
	8–10 (total flow = 45)	0.35
	8–10 (total flow = 33)	0.40
	8–12 (total flow = 33)	0.50

- Facemask 6-7L ???

- ?? Anterior spinal cord is more vulnerable to ischemia??
- Fyi (MM 381)
  - The anterior cord is vulnerable to ischemia. The posterior arteries descend along the length of the cord and supply the posterior one-third of the cord. There is good collateral flow posteriorly. The anterior spinal artery descends anteriorly along the cord's midline. The anterior portion of the cord is more prone to ischemia than the posterior columns because it is supplied by the single anterior spinal artery.
  - A major complication of clamping the thoracic aorta is spinal cord ischemia and paraplegia. The incidence of transient postoperative deficits and postoperative paraplegia are 11% and 6%, respectively. Higher rates are associated with cross-clamping periods longer than 30 minutes, extensive surgical dissections, and emergency procedures. The classic deficit is an anterior spinal artery syndrome with loss of motor function and pinprick sensation but preservation of proprioception and vibration.
  - Anterior spinal artery syndrome is loss of motor function secondary to ischemia of the anterior spinal cord. Recall that somatic motor nerves originate in the anterior horn of the spinal cord. Prolonged ischemia can result in paraplegia, rectal or urinary incontinence, and loss of pain and temperature sensation.
  - A well-known, large and important anterior radicular artery is the artery of Adamkiewicz, which is also referred to as the artery radicularis magna. The artery of Adarnkiewicz (artery radicularis magna) enters the vertebral canal from the left side (it is not bilateral) in the lower thoracic region or upper lumbar region. The importance of this artery lies in the fact that it may be the major source of blood to the lower two-thirds of the spinal cord. Interruption of flow through this vessel can lead to paraplegia.
  - With cross-clamp of the descending thoracic aorta, you would be most concerned about interrupting flow through what vessel supplying the spinal cord? Why? You should be most concerned about interrupting flow through the artery ofAdamkiewicz. Aortic cross-clamping, which either incorporates the artery of Adamkiewicz between clamps or exposes it to hypoperfusion, ay lead to paraplegia. Note: If the descending aorta is cross-clamped, flow through the anterior spinal artery and the posterior spinal arteries is not interrupted

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- Immunity question → plasma cells ??

- Drag & Drop Question: Mag Levels/Toxicity

Plasma Mag (mEq/L)	Effects
1.5 – 2.0	Normal Plasma Level
4.0 – 8.0	Therapeutic Range
5.0 – 10	Electrocardiographic changes (P-Q interval prolonged, QRS complex widens)
10	**Loss of Deep Tendon Reflexes**
15	Sinoatrial and Atrioventricular Node Block
15	Respiratory Paralysis
25	Cardiac Arrest

- Max dose of Lidocaine with epinephrine in a 70kg patient:
  - Lidocaine MAX DOSE: 4.5 mg/kg = 315 mg
  - Lidocaine with EPI MAX Dose: 7 mg/kg = 490 mg

- Describe the perioperative management of the patient with hyperparathyroidism (MM 87)
  - Hypercalcemia (serum Ca2+c oncentration greater than 5.5 mEqlL and ionized calcium concentration greater than 2.5 mEq/L) is a hallmark of patients with hyperparathyroidism.
  - The perioperative goal is to correct intravascular volume and electrolyte abnormalities.
    - Normal saline and a loop diuretic (usuallyfurosemide) are administered to increase calcium excretion by means of hydration and diuresis.
    - Perioperatively, avoid fluids with calcium, e.g., lactated Ringers. Also, increased dosing of vecuronium and probably all nondepolarizing muscle relaxants will be required.

- ??? 70 kg patient. How much blood loss for Level III shock ???
  - 50% of EBV????

- FYI: Stages of hypovolemic shock (Source: Wiki)
  - The 4 stages are sometimes known as the "Tennis" staging of hypovolemic shock, as the 4 stages of % volume of blood loss mimic the scores in a game of tennis: 15, 15-30, 30-40, 40. It is basically the same as used in classifying bleeding by blood loss.

- Stage 1 (Initial Stage)
  - Up to 15% blood volume loss (750 mL)
  - Compensated by constriction of vascular bed
  - Blood pressure maintained
  - Normal respiratory rate
  - Pallor of the skin
  - Normal mental status to slight anxiety
  - Normal capillary refill
  - Normal urine output

- Stage 2 (Compensatory Stage)
  - 15–30% blood volume loss (750–1500 mL)
  - Cardiac output cannot be maintained by arterial constriction
  - Tachycardia >100bpm
  - Increased respiratory rate
  - Blood pressure maintained
  - Increased diastolic pressure
  - Narrow pulse pressure
  - Sweating from sympathetic stimulation
  - Mildly anxious/Restless
  - Delayed capillary refill
  - Urine output of 20-30 milliliters/hour

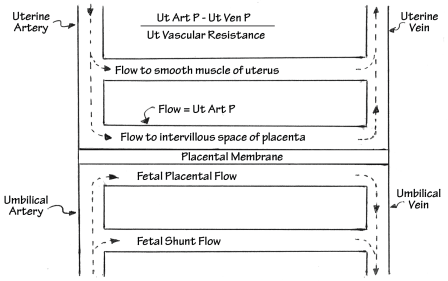
- Stage 3 (Progressive Shock)
  - **30–40% blood volume loss (1500–2000 mL)**
  - Systolic BP falls to 100mmHg or less
  - Classic signs of hypovolemic shock
  - Marked tachycardia >120 bpm
  - Marked tachypnea >30 bpm
  - Decreased systolic pressure
  - Alteration in mental status (confusion,[12] anxiety, agitation)
  - Sweating with cool, pale skin
  - Delayed capillary refill
  - Urine output of approximately 20 milliliters/hour

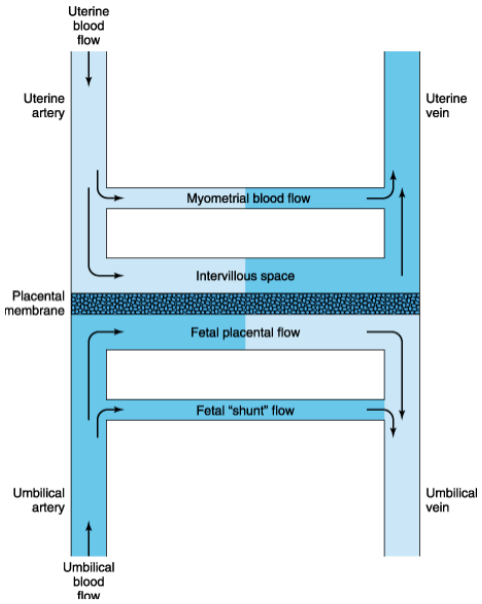
- Stage 4
  - Loss greater than 40% (>2000 mL)
  - Extreme tachycardia (>140[12]) with weak pulse
  - Pronounced tachypnea
  - Significantly decreased systolic blood pressure of 70 mmHg or less
  - Decreased level of consciousness, lethargy, coma
  - Skin is sweaty, cool, and extremely pale (moribund)
  - Absent capillary refill
  - Negligible urine output

- Drag & Drop Question: blood flow from placenta to fetus
- Uterine Artery → Intervillous Space → Umbilical Vein → BABY → Umbilical Artery → Intervillous Space → Uterine Vein

- Or … Umbilical Vein → Ductus Venosis → Foramen Ovale → PDA??????

-
- Fyi (MM 462)
  - The maternal blood is carried initially in the uterine arteries. Blood is spurted into the intervillous space. Blood in the intervillous space passes fetal villi before draining back to the veins of the uterine wall (myometrium).





Source: Morgan GE, Mikhail MS, Murray MJ: *Clinical Anesthesiology*, 4th Edition: http://www.accessmedicine.com

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- Know which NEUROTRANSMITTERS are EXCITATORY & which are INHIBITORY

- Table 18–4. Major Neurotransmitters Mediating or Modulating Pain.

Neurotransmitter	Receptor1	Effect on Nociception
Substance P	NK–1	Excitatory
Calcitonin gene-related peptide		Excitatory
Glutamate	NMDA, AMPA, kainite, quisqualate	Excitatory
Aspartate	NMDA, AMPA, kainite, quisqualate	Excitatory
Adenosine triphosphate (ATP)	P1, P2	Excitatory
Somatostatin		Inhibitory
Acetylcholine	Muscarinic	Inhibitory
Enkephalins	μ, δ, κ	Inhibitory
β-Endorphin	μ, δ, κ	Inhibitory
Norepinephrine	α2	Inhibitory
Adenosine	A1	Inhibitory
Serotonin	5-HT1 (5-HT3)	Inhibitory
Y-Aminobutyric acid (GABA)	A, B	Inhibitory
Glycine		Inhibitory

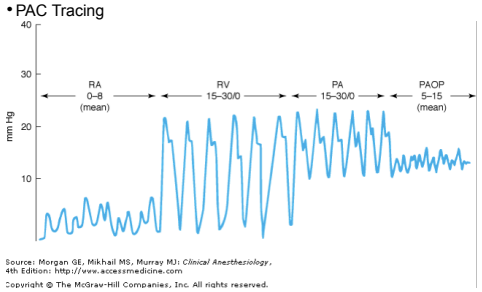
- IBW Calcuation
  - What is the IBW of a male patient who is 170 cm tall?
    - Convert 170 cm to inches
    - 1 in = 2.54 cm
    - 170 cm/2.54 = 66.9 inches
    - Then calculate base on IBW calcuation

- Pulmonary Artery Wedge
- Placement and pressure
  - Used to assess left ventricular function
  - Length is 60 (child) to 110 (adult) cm and 4.0 to 7.5 French in size
  - Balloon holds about 0.5 ml in pediatric Swan and up to 1.5 ml in adult Swan with diameters ranging from 8-13 mm in the pediatric patient and adult, respectively
  - Shaft marked at 10 cm increments
- Pulmonary artery catheter distances (centimeters)

Insertion Site	To Right Atrium	+10 =	To Right Ventricle	+15 =	To Pulmonary Artery
Subclavian	15		25		40
Right Internal Jugular	20		30		45
Left Internal Jugular	25		35		50
Right Antecubital	40		50		65
Left Antecubital	45		55		70
Femoral	50		60		75

All distances in centimeters (cm)

- PAC Tracing



Source: Morgan GE, Mikhail MS, Murray MJ: *Clinical Anesthesiology*, 4th Edition: http://www.accessmedicine.com

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- Pressure tracings
  - RIGHT ATRIUM has a pressure range of 1-8 mm Hg and a characteristic low amplitude wave. The right atrial pressure equals the central venous pressure. The a-wave represents atrial contraction and the c-wave early ventricular contraction. A low pressure wave is seen.



  - RIGHT VENTRICLE has a systolic pressure of 15-25 mm Hg and a diastolic pressure (RVEDP) of 1-8 mm Hg. The wave form is larger than the atrial wave form due to the increase in pressure. The upstroke of the pressure curve is 2-3 times greater than that of the atrium. The downstroke is without a dicrotic notch. Elevated pressure is usually due to inability to pump blood due to out flow resistance, ventricle damage, hypovolemia or narrow pulse pressure. A higher pressure wave is seen.



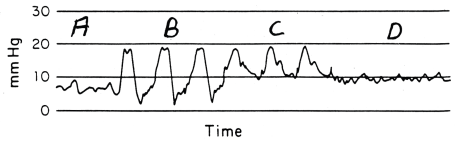
  - PULMONARY ARTERY has a systolic pressure of 15-25 mm Hg, diastolic of 8-15 mm Hg and a mean pressure of 10-20 mm Hg. Wave form is generally smaller than the ventricular wave form. There is a dicrotic notch, which results from pulmonic valve closure. The diastolic pressure should be greater than the right ventricular end-diastolic pressure. Right ventricular systolic pressure should equal pulmonary artery systolic pressure.



  - RAP = CVP = RVEDP = 0 – 8 mmHg
  - RVS = PAS = 15 – 25 mmHg
  - PAD = 8 – 15 mmHg
  - Pulmonary capillary wedge pressure is 6–12 mm Hg, and it measures the “back pressure” from the pulmonary veins. The wave form is low in amplitude but should be elevated from baseline. The maximum wedge time is about 15 seconds.
  - The pulmonary artery wedge pressure is never higher than pulmonary artery diastolic pressure.
  - Increases in pressure are seen with volume overload, LV failure, mitral stenosis, and cardiac tamponade.
  - Decreases in pressure indicate hypovolemia.

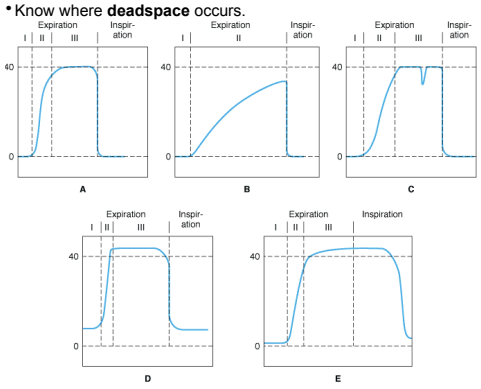
Patient Status	CVP	PCWP
Hypovolemia	Low	Low
Left ventricular failure	Normal ***or*** high	High
Right ventricular failure	High	Normal
Pulmonary embolism	High	Normal
Chronic pulmonary hypertension	High	Normal
Cardiac tamponade	High	High

Abbreviations: CVP, central venous pressure; PCWP, pulmonary capillary wedge pressure; PADP, pulmonary artery diastolic pressure.



- A. the cannula tip is located in the right atrium and has been inserted a distance of 18–22 cm (20 cm)
- B. the cannula tip is located in the right ventricle and has been inserted a distance of 28–32 cm (30 cm)
- C. the cannula tip is located in the pulmonary artery and has been inserted a distance of 40–50 cm (45 cm)
- D. the cannula tip is wedged in the pulmonary artery and has been inserted a distance of 45–50 cm.

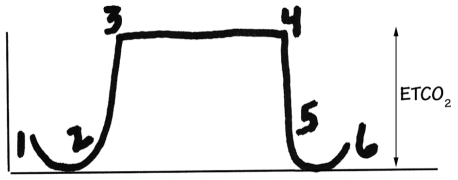
- Know where **deadspace** occurs.



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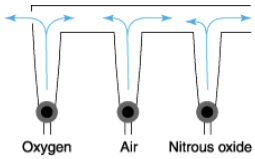
A: A normal capnograph demonstrating the three phases of expiration:
- phase I—dead space;
- phase II—mixture of dead space and alveolar gas;
- phase III—alveolar gas plateau.

B: Capnograph of a patient with severe chronic obstructive pulmonary disease. No plateau is reached before the next inspiration. The gradient between end-tidal CO2 and arterial CO2 is increased.
C: Depression during phase III indicates spontaneous respiratory effort.
D: Failure of the inspired CO2 to return to zero may represent an incompetent expiratory valve or exhausted CO2 absorbent. E: The persistence of exhaled gas during part of the inspiratory cycle signals the presence of an incompetent inspiratory valve.

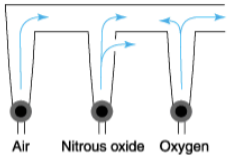


1 = baseline
2 = begin expiration
2–3 = combination of dead space and alveolar gases
3–4 = alveolar plateau, also called alveolar gas plateau
4 = actual alveolar $ETCO_2$: patient begins to inspire CO_2-free gases
4–5 = inspiratory limb, inspiration to baseline of 0 (zero), unless there is rebreathing
6 = end of inspiration

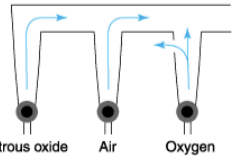
- Picture of flow meter → pick O2???
- FYI (M&M)



Incorrect sequence



Datex-Ohmeda sequence



Draeger sequence

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Sequence of flowmeters in a three-gas machine. A: An unsafe sequence. B: Typical Datex-Ohmeda sequence. C: Typical Draeger sequence. Note that regardless of sequence a leak in the oxygen tube or further downstream can result in delivery of a hypoxic mixture.

- Ketamine → NMDA receptors

- fyi (Valley)
  - Ketamine produces amnesia and profound analgesia by interacting with the NMDA-type glutamate receptor

---

- List four requirements of the Joint Commission on Accreditation of Healthcare Organizations (JCAHO) for a hospital's quality improvement (QI) program.
  - (1) **design a process or function**;
  - (2) measure performance;
  - (3) assess performance (by statistical analysis or comparison with other data sources); and
  - (4) improve process orfunction.

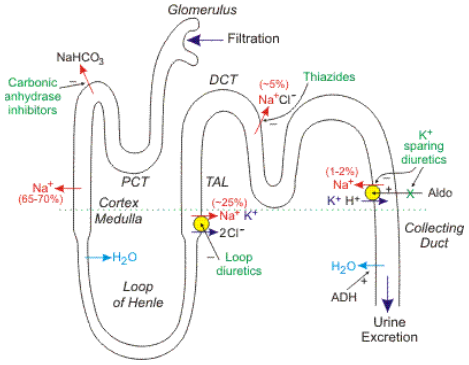
- Founder of the AANA → Agatha Hodgins

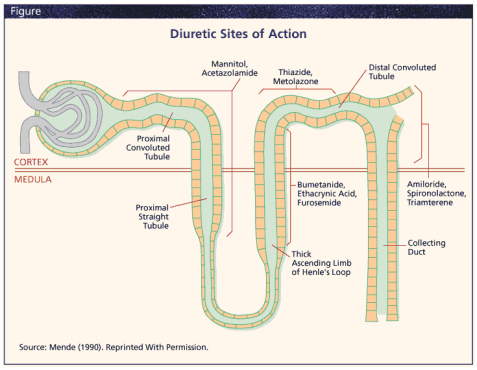
- Patient with WPW syndrome in SVT → give Adenosine
- Fyi (MM 13)
  - The patient has Wolff-Parkinson-White syndrome. Atrial fibrillation develops. How should the atrial fibrillation be treated? What drugs should be avoided in this situation?
  - If rapid ventricular response during atrial fibrillation results in lifethreatening hypotension, electrical cardioversion is necessary. If the atrial fibrillation is tolerated, give IV procainamide, which prolongs the refractory period of accessory fibers. Avoid verapamil or digitalis because either may accelerate conduction through the accessory pathway

---

- Drag & Drop: Diuretics

Furosemide (Lasix)	Loop of Henle
Spironolactone (Aldactone)	Collecting Duct
Acetazolamide (Diamox)	Proximal Tubule
HCTZ (Esidrix, Hydrodiuril)	Distal Tubule





- During an aorto-femoral bypass, the patient becomes hypothermic; what cardiac arrhythmia is likely? (MM 134)
  - Hypothermia prolongs the refractory period of excitable tissues. In the heart, a prolonged refractory period leads to sinus bradycardia and conduction deficits, which may progress to atrioventricular block and eventually ventricular fibrillation.

- Drag & Drop question: 4 Types of Allergic Reactions
  - Anaphylaxis — Type I
  - Transfusion Reaction — Type II
  - Serum Sickness — Type III
  - Dermatitis — Type IV
- FYI (M&M)
- Table 46–7. Hypersensitivity Reactions.
  - Type I (immediate)
    - Atopy
    - Urticaria—angioedema
    - Anaphylaxis
  - Type II (cytotoxic)
    - Hemolytic transfusion reactions
    - Autoimmune hemolytic anemia
    - Heparin-induced thrombocytopenia
  - Type III (immune complex)
    - Arthus reaction
    - Serum sickness
    - Acute hypersensitivity pneumonitis
  - Type IV (delayed, cell-mediated)
    - Contact dermatitis
    - Tuberculin-type hypersensitivity
    - Chronic hypersensitivity pneumonitis

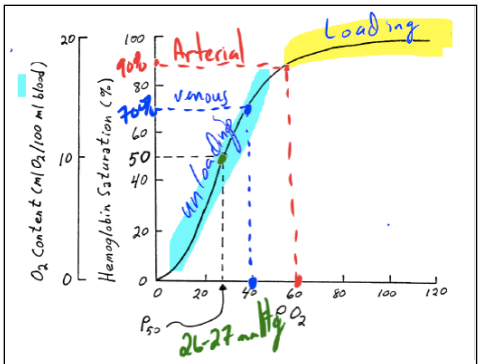
- The most common cause of chronic liver disease → non-alcoholic fatty liver???

- Fyi (internet):
  - Common causes of chronic liver disease in the US include hepatitis C infection and long-term alcohol abuse. Hepatitis C is now the most common reason for liver transplantation in the US.
  - Alcohol abuse is the most common cause of liver disease in North America.

- Largest epidural space → L2 – L3
- Fyi (MM 55): Where is the epidural space between the ligamentum flavum and the dura mater the largest?
  - The epidural space between the ligamentum flavum and the dura mater **is largest at L2-L3**, approximately 4-6 mm.

- Largest interlaminar interspace of the vertebral column → L5 – S1
- Fyi (MM 337): Where is the largest interlaminar interspace of the vertebral column found? The approach using this interspace to the subarachnoid compartment is a variation of what approach?
  - The L5-S1 interspace is the largest interlaminar interspace of the vertebral column. This lumbosacral approach to the subarachnoid space is a variation on the paramedian approach.

Oxyhemoglobin Dissociation Curve	
Left	**RIGHT**
Decreased Temperature	Increased PCO2
Decreased H+ concentration (Increased pH)	Increased 2,3-DPG
Decreased PCO2	Increased H+ concentration
Decreased 2,3-DPG	Increased Temperature
Fetal Hgb (HgbF)	Sickle Cell
Carboxyhemoglobin	Maternal HGB
Methemoglobin	



Questions (Q) and Answers (A)

Q: What is P50, and what is the normal value for P50?

A: P50 is the partial pressure of oxygen at which hemoglobin is 50% saturated by oxygen. The normal P50 is 26–27 mm Hg.

Q: What SO2s correspond to PO2s of 40 and 60 mm Hg?

A: SO2s of 70 and 90% correspond to PO2s of 40 and 60 mm Hg, respectively. (See following table.)

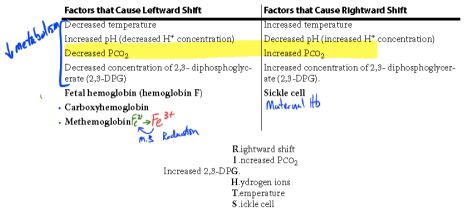
PO_2	O_2	Source
40 mm Hg	70%	Mixed Venous Blood
60 mm Hg	90%	Arterial Blood

Q: What is the significance of the flat portion of the oxyhemoglobin dissociation curve?

A: The flat portion of the oxyhemoglobin dissociation curve facilitates the loading of oxygen by the blood because, in the flat portion of this curve, large changes in the partial pressure of oxygen in arterial blood (PaO2) produce only small changes in oxygen saturation (SaO2).

Q: What is the significance of the steep portion of the oxyhemoglobin dissociation curve?

A: The steep portion of the oxyhemoglobin dissociation curve facilitates unloading of oxygen at tissues because large amounts of oxygen are unloaded from hemoglobin (large decrease in oxygen saturation) in response to a small change in the partial pressure of oxygen.



- What is not an ether derivative?
  - Choices were: Desflurane, Isoflurane, Enflurane, **Halothane**
- Fyi
  - Halothane is a halogenated alkane derivative.
  - Enflurane, isoflurane, and desflurane are methyl ethyl ethers.
  - Sevoflurane is fluorinated methyl isopropyl ether.
  - Enflurane and isoflurane are isomers of each other.
  - Halothane is the only brominated modern volatile anesthetic agent.
  - Desflurane and sevoflurane are halogenated exclusively with fluorine. Volatile agents halogenated exclusively with fluorine are less soluble in blood.
  - Desflurane differs from isoflurane only by the substitution of a fluorine atom for the chlorine atom. This one change increases the vapor pressure dramatically (669 mm-Hg for desflurane compared to 240 mm-Hg for isoflurane).

- Caudal → Bupivacaine 0.125% to 0.25% → 0.5 – **1 mL**/kg

- Calculate the partial pressure of O2 → 760 X 0.21 = 159.6 mmHg

---

- What is the highest form of evaluative criteria for nurse anesthesia clinical practice, as set forth by the AANA?
  - Practice standards are the highest mandate for clinical behavior.
  - Practice Standards >> then, Practice Guidelines >> then, Position Statements
- Fyi (MM 516)
  - Position statements are the least forceful criteria and usually represent emerging trends, address economically driven practice modalities, or discuss procedural policies. Practice guidelines are the next highest form of evaluative criteria and are guides to provider behavior and critical decision making that are commonly accepted within the anesthesia profession. Practice standards are the highest form of evaluative criteria. A standard represents behaviors that must be exercised by the prudent nurse anesthetist in similar circumstances.

- Fluid calculation for 3 hour case
- Know how to do NPO, MIVF, Evaporative loss

Hour	MIVF	NPO Loss	Evaporative Loss	Blood Loss	Total fluid / hour
1		Replace 1/2			
2		Replace 1/4			
3		Replace 1/4			
4					

- Maintenance IVF: 4-2-1 rule
  - 4 mL/kg for the first 10 kg
  - 2 mL/kg for the 2nd 10 kg
  - 1 mL/kg for the remaining weight

- Evaporative Loss (M&M)

Table 29–6. Redistribution and Evaporative Surgical Fluid Losses.

Degree of Tissue Trauma	Additional Fluid Requirement
Minimal (eg, herniorrhaphy)	0–2 mL/kg
Moderate (eg, cholecystectomy)	2–4 mL/kg
Severe (eg, bowel resection)	4–8 mL/kg

- Be able to calculate ANION GAP!
- Anion Gap = Major measure Cations – Major measured Anions
- Anion Gap = (Na + K) – (Cl + HCO3)
- Or … Without K: Na – (Cl + HCO3)
- Normal is 9 – 15 mEq
- High anion metabolic acidosis → Lactic acidosis or Ketoacidosis

Fyi: Table 30–4. Causes of Metabolic Acidosis (M&M)

- **INCREASED ANION GAP**
  - **Increased production of endogenous nonvolatile acids**
    - Renal failure
    - Ketoacidosis
      - Diabetic
      - Starvation
    - Lactic acidosis
    - Mixed
      - Nonketotic hyperosmolar coma
      - Alcoholic
    - Inborn errors of metabolism
  - **Ingestion of toxin**
    - Salicylate
    - Methanol
    - Ethylene glycol
    - Paraldehyde
    - Toluene
    - Sulfur
  - **Rhabdomyolysis**
- **NORMAL ANION GAP (HYPERCHLOREMIC)**
  - **Increased gastrointestinal losses of HCO3–**
    - Diarrhea
    - Anion exchange resins (cholestyramine)
    - Ingestion of CaCl2, MgCl2
    - Fistulas (pancreatic, biliary, or small bowel)
    - Ureterosigmoidostomy or obstructed ileal loop
  - **Increased renal losses of HCO3–**
    - Renal tubular acidosis
    - Carbonic anhydrase inhibitors
    - Hypoaldosteronism
  - **Dilutional**
    - Large amount of bicarbonate-free fluids
  - **Total parenteral nutrition (Cl– salts of amino acids)**
  - **Increased intake of chloride-containing acids**
    - Ammonium chloride
    - Lysine hydrochloride
    - Arginine hydrochloride

- PDA repair → Reccurent Laryngeal Nerve & Phrenic nerve damge…. Maybe Vagus? ???

- Popliteal fossa → Tibia Nerve AND **Common Peroneal Nerve**

- Popliteal Nerve block nerve stimulator response
  - Common Peroneal — Dorisflexion & Eversion
  - Tibial — Plantarflexion & Inversion

- 5 risk factors for difficult mask ventilation, from greatest to least
- Presence of beard → BMI > 26 → Lack of Teeth → Age > 55 → History of Snoring

- Fyi (MM 325)
  - State five (5) risk factor for difficult mask ventilation, from greatest risk to least.
    - (1) presence of a beard,
    - (2) body mass index >26 kg/m2,
    - (3) lack of teeth (edentulous),
    - (4) age > 55 years, and
    - (5) history of snoring.

- An infant has a life-threatening succinylcholine-induced hyperkalemia: what is the definitive treatment? (MM 461)
- The definitive treatment of succinylcholine-induced hyperkalemia is IV **calcium** (10 mg/kg calcium chloride or 30 mg/kg calcium gluconate or more). This restores the gap between the resting membrane potential of the cardiac cells and the threshold potential for depolarization. Repeated doses of calcium must be administered together with cardiopulmonary resuscitation, epinephrine, sodium bicarbonate, glucose and insulin, andhyperventilation until the arrhythmias abate.

- What sites give the most reliable approximation of core body temperature?
  - Lower 25% of the esophagus gives a reliable approximation of blood and cerebral temperature. (Not an option on test)
  - The **tympanic membrane** and aural canal temperatures correlate well with esophageal temperature.
  - Most accurate is PAC, not used routinely for this purpose

---

- What stimulates renin secretion?
  - Decreased renal perfusion (hemorrhage, hypovolemia, renal artery stenosis)
  - HypOnatremia,
  - Sympathetic nervous system stimulation of beta-1 adrenergic receptors of JG cells

---

- You have a patient in Right Lateral Decubitus position. The RIGHT radial arterial line read 120/80 and the LEFT NIBP cuff is located 20 cm about the Right radial A-Line. 1 cm=0.75 mmHg. What is the BP reading in the Left Cuff NIBP?
  - 0.75mmHg X 20 cm = 15 mmHg
  - Subtract 15 from SBP & DBP
  - **NIBP cuff is reading 105/65**

	Heart Rate	Rhythm	Preload	Afterload	Contractility
IHSS	Not Too Slow Not Too Fast	Sinus	Full	Up	Maintain Slightly decrease
Aortic Regurg	Increased (80-100bpm)	Sinus	Increase	Decrease	Maintain
Mitral Regurg	Increased	Sinus	Maintain	Decrease	Maintain
Aortic Stenosis	Low (50-70 bpm)	Sinus	Maintain	Maintain	Maintain

Mitral Stenosis	Low	Sinus	Maintain	Maintain	Maintain

- PCWP overestimates LVEDP in the patient with → Mitral Stenosis

---

- What do you NOT see with patients with Rheumatoid Arthritis?
  - MR

- Fyi: Patients with RA have
  - Cricoarytenoid involvement
  - Pericardial effusion, Aortic Regurg, conduction abnormalities, cardiac valve fibrosis, coronary artery arteritis
  - Pulmonary fibrosis, pleural effusion
  - Anemia
  - Keratoconjuctivitis sicca

- **Intrathecal** Injection:
  - Morphine: LATE (6-12 hours) respiratory depression
    - Slow Onset
    - Long DOA
  - Fentanyl, Alfentanil, Sufentanil: EARLY (w/i 2 hours) respiratory depression
    - Rapid Onset
    - Short Duration

- **Epidural** Injection
  - Morphine: EARLY (w/in 2 hours, but unlikely) and LATE respiratory depression
    - Slow Onset
    - Long DOA
  - Fentanyl, Alfentanil, Sufentanil: EARLY (w/in 2 hours) respiratory depression
    - Rapid Onset
    - Short Duration

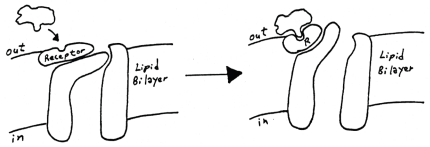
- Adverse Effects of Vagus Nerve Stimulator? (Only found this online, not in any of our books….)
  - Horseness, Throat pain, coughing, dyspnea, paresthesia, muscular pain, voice alteration, chest pain, N/V, dyspnea, paresthesia, infection

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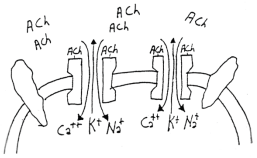
- Pt who recently had a pain pump implanted. Paresthesias are better, but are having increased back pain?
  - Choices were: Spinal cord damage, spinal hematoma, **infection**???

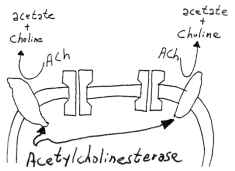
- Post-synaptic NMJ nicotinic receptor
  - **Ligand gated**? G-coupled?

- Fyi (valley)
  - Receptors that control ion channels (gates). When a ligand (chemical, drug, neurotransmitter, hormone) attaches to its receptor, the channel opens and substances specific to the channel diffuse down their concentration gradients through it. At the neuromuscular junction, for example, acetylcholine binds to the nicotinic receptor and the channel opens to sodium ions (which diffuse in), potassium ions (which diffuse out), and calcium ions (which diffuse in).

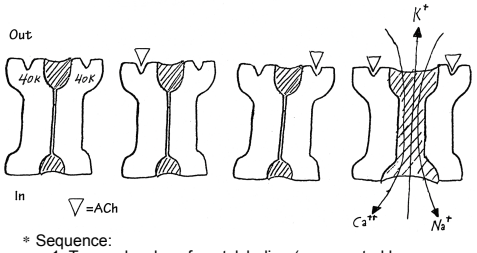


  - Events at the Postsynaptic Membrane
    - 1. ACh combines with nicotinic receptors of the protein channel.
    - 2. When both alpha subunits of the nicotinic receptor channel are occupied by ACh, the channel snaps open, and sodium, calcium and potassium ions diffuse through the channel. Note: Sodium and calcium ions diffuse into the cell and potassium ions diffuse out to the extracellular space.
    - 3. The diffusion of these three types of ions through the channel causes the motor end-plate to depolarize.
    - 4. At a critical level of depolarization (threshold), an action potential is initiated.
    - 5. The action potential sweeps across the skeletal muscle cell and triggers contraction.





  - Postsynaptic Receptors and Channels
    - Note that the channel operated by the nicotinic receptor has five subunits. Two of the five subunits, the 40K subunits, are identical. An acetylcholine molecule must attach to each of these two identical subunits where the nicotinic receptors are located in order to open the channel. Stated another way: 2 ACh molecules are needed to open each nicotinic acetylcholine receptor(nAChR). Acetylcholinesterase (designated as AChE in the figure) is anchored to the external (extracellular) surface of the cell membrane.



  - Sequence:
    - 1. Two molecules of acetylcholine (represented by open triangles above) combine with two nicotinic receptors on the channel.
    - 2. The ion channel opens and becomes permeable to Na+, K+, Ca++.
    - 3. The motor end-plate depolarizes; a local, sub-threshold depolarization occurs at the end-plate.
    - 4. When threshold is reached, an all-or-none action potential is initiated in the muscle fiber.
    - 5. An action potential passes over the muscle cell and into the transverse tubules and triggers the contraction.

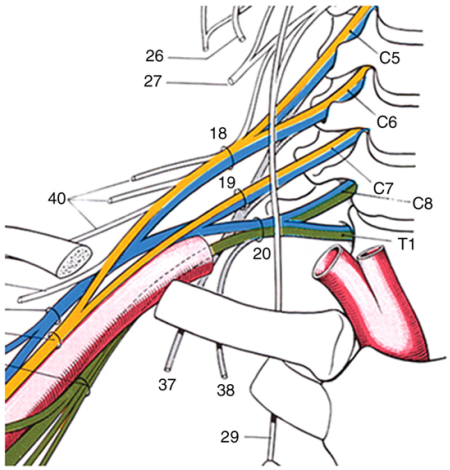
- Reflex decrease in HR when you go from standing/sitting → supine

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- What would increase the likelihood of local anesthetic toxicity?
  - Choices were: Infection, Increased CO, **hypERcapnia (vasodilation)**??
  - Hypoventilation → hypercapnia → vasodilation of cerebral blood flow → increased cerebral blood flow → more LA to the brain → increased toxicity

---

- Know the Brachial Plexus



Roots Trunks Division Cords Branches

- "Rob Taylor Drinks Cold Beer": Roots-Trunks-Division-Cords-Branches

- ROOTS = emerge between the scalenus medius and scalenus anterior to lie on the floor of the posterior triangle of the neck.

- TRUNKS= behind the scapula/clavicle, at the apex of the axilla.

- DIVISIONS = within the axilla according to the relationship of axillary artery.

- Drag & Drop: Ways to lose heat…
- Radiation (60%) >> Convection (15-30%) >> Evaporation >> Conduction (< 5%)

- Will a L-to-R shunt (ASD/VSD/PDA) slow or accelerate INHALATION induction?
  - A L→R shunt should ACCELERATE the speed of inhalation induction because the rate of transfer of anesthetic agent from the lungs to the blood is increased.
  - However, this effect is rarely clinically evident because decreased delivery of anesthetic to the target tissues negates the increased uptake of agent with a L→ R shunt
- What about IV induction?
  - Should be slowed by L→ R shunt;
  - However, unless cardiac output is very poor, the effect is clinically irrelevant.

- Ach → What causes to open?
- Choices were: VG Cl Channels, VG K Channels, Chemically gated Ca channels, **Chemically gated Na channels**???

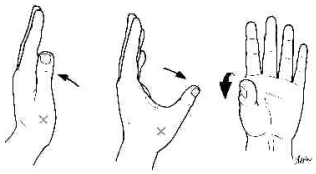
- Cardiopulmonary changes in adult with obesity hypoventiation syndrome (OHS)? (Pickwickian Syndrome)
  - HypERcapnia, severe hypoxemia, somnolence (secondary to night-time OSA), periodic respiration, pulmonary hypertension, systemic hypertension, right & left ventricular hypertrophy (but particularly RVH), dependent edema, polycythemia, rales, pulmonary edema
- Difference between OSA & Pickwickian Syndrome?
  - Awake hypERcarbia with pickwickian syndrome

No answer

A LOT of Risk Management/QI/QA questions ☒ MM p. 529 - 536
Lots of Geriatric questions ☒ Look in Valley (most helpful)
Positioning ☒ CV changes in supine position ☒ increased preload and CO ??
Read over Geriatric / Obesity/ Positioning
??? Which drug would increase effects? Something to do with OB? ☒ Methergine ???
Methergine in ob pt with pre-eclampsia ☒ what would augment HTN?
??? What would you not see with MS? Volumes/pressure ☒ no increase LVEDV & LVEDP ???
??? Coagulation factors, proteins ???
??? Increases CNS activity ???
* morphine 6, something…
* normeperidine
? about what inhibits CP450 the most?
* Choices were: Barbs, Des, Ropivacaine ????
Complex Regional Pain Syndrome from Mixed Review 50
Least traumatic positioning for ulnar nerve injury?
* Supination ???

just fyi

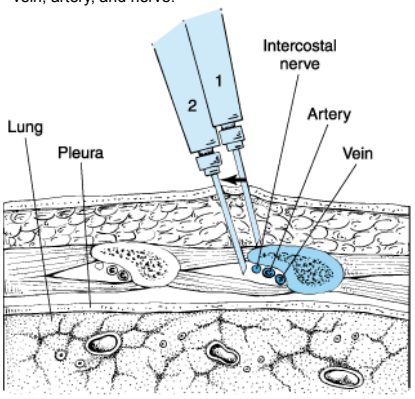
- this analysis requires knowledge of the anion gap, the formula for which is: [Na+] - ([Cl-] - [HCO3-]). In this instance, the anion gap is 23 mEq/L. The normal anion gap is 7 to 12 mEq/L. Causes of increased anion gap acidosis include renal failure, ketoacidosis, lactic acidosis, hyperosmolar nonketotic coma, alcohol toxicity, and ingestion of salicylates, methanol, ethylene glycol, paraldehyde, toluene, and sulfur. Although the other options listed above can result in metabolic acidosis, alcohol toxicity is the only option associated with an increased anion gap.
- Because creatinine is produced at a nearly constant level and creatinine is not reabsorbed by the kidneys, creatinine clearance is used as a method of evaluating renal function. A creatinine clearance less than 25 mL/min indicates overt renal failure, 25-40 mL/min is moderate renal dysfunction, and 40-60 mL/min is mild renal dysfunction.
- The reaction that occurs within the CO2 absorber between soda lime and carbon dioxide occurs in two steps. The first is CO2 + H2O --> H2CO3. The second portion of the reaction is H2CO3 + 2NaOH --> Na2CO3 + 2H2O + heat. Therefore, H2CO3 is an intermediate product of the reaction, but not a final product as it degraded by the secondary reaction. The final products are Na2CO3, H2O, and heat.
- double-burst stimulation (DBS) is characterized by two short sequences of 50 H electrical impulses separated by 750 milliseconds. The primary use for DBS is detect residual blockade by a nondepolarizing neuromuscular relaxant. Fade on the second burst of DBS is easier to detect than fade on a train of four. Tetanus consists of repeated electrical impulses at a rapid frequency of 50, 100, or even 200 Hertz. In the case of a depolarizing block, the amplitude of the contraction is decreased and fade is exhibited over the course of the five-second recommended duration of stimulus. A train-of-four stimulus consists of four single pulses of equal intensity at intervals of 0.5 seconds. When a partial depolarizing block is present, all four contractions will be diminished while in a partial nondepolarizing block, there will be a fade in intensity of the contraction over the course of the stimulations, with the potential absence of some or all of the contractions depending on the intensity of the block.
- According to the standards set by the ASTM, 3-liter reservoir bags (also called breathing bags) that are distended to four times their normal size should not exert a pressure less than 35 cm H2O or greater than 60 cm H2O. Bags that are 1.5 L in size or smaller shall not exert a pressure less than 30 cm H2O or greater than 50 cm H2O when distended to this extent. The lowest pressure to be exerted is to ensure that the bag is capable of exerting enough minimum pressure to ventilate a patient and the highest pressure is to limit the amount of barotrauma that a patient may be exposed to when using the breathing bag.
- If nitrous oxide is also used, the maximum exposure to halogenated agents allowed by OSHA is 0.5 ppm. If nitrous oxide is not used, then the maximum exposure is 2 ppm. The maximum exposure to nitrous alone is 25 ppm.
- Pressing the oxygen flush valve during inspiration can result in over-distention of the lungs and barotrauma because the flush valve delivers 35 to 75 L/min of oxygen at 50 psi.
- Three types of damages may be sought for in a civil lawsuit: general damages, special damages, and punitive damages. General damages result directly from the injury and include reimbursement for pain and suffering. Special damages that occur as a result of the injury, such as lost wages, medical expenses, and funeral expenses. Punitive damages are rare in medical malpractice cases, but are intended to punish the wrongdoer for the negligence that led to the damage to the plaintif
- Except in cases where consent is implied (such as emergency situations where it is assumed the patient would consent to life-saving treatment) informed consent must be obtained prior to administering anesthesia. Failure to do so is considered a breach of duty. It is not necessary to inform a patient of all possible complications, but it is prudent to inform them that some complications may be life-threatening. Although the closest relative can provide consent for an unconscious adult, only the parent or legal guardian can provide or withhold consent for an unconscious child.
- Radial nerve = thumb **Ab**duction



- Ulnar nerve = thumb Adduction



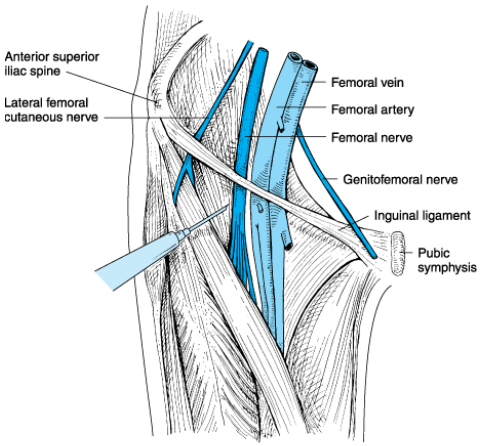
- Brainstem auditory evoked potentials assess the integrity of the 8th cranial nerve and the auditory pathways above the pons and are best for monitoring during posterior fossa surgery.
- Somatosensory evoked potentials assess the dorsal spinal column and sensory cortex and are best for monitoring during spinal surgery, carotid endarterectomy, and aortic surgery. Visual evoked potentials assess the integrity of the optic nerve and upper brainstem and are best for monitoring during pituitary resection. The wake-up test assesses movement and evaluates the integrity of the motor tracts in the ventral spinal column.
- A sensory level of T2-T4 can cause loss of sensory perception of abdominal wall and intercostal muscle movement resulting in a sensation of dyspnea despite normal respiration. A level of C2-C3 can result in phrenic nerve and intercostal muscle paralysis resulting in hypoxia, hypercarbia, or respiratory arrest.
- EMLA cream is a topical local anesthetic that consists of a 1:1 oil in water mixture of 2.5% lidocaine and 2.5% prilocaine. For adequate anesthesia, it typically requires a 60 minute contact time and usually achieves a penetration depth of 3-5 mm with a duration of action of 1-2 hours.
- Because EMLA cream contains prilocaine, it can result in methemoglobinemia. Methemoglobinemia has equal absorbent coefficient ratios for both red and infrared wavelengths and results in a false pulse oximetry reading that tends to equal 85%. This can result in falsely low pulse oximetry readings when the actual oxygen saturation is higher than 85% and falsely high readings when the oxygen saturation is lower than 85%.
- Most allergic reactions to local anesthetics are to ester anesthetics because of the metabolite para-aminobenzoic acid. Although methylparaben, the preservative found in amide anesthetics is similar to para-aminobenzoic acid, allergic reactions to it are not as common. Thymol is the preservative found in halothane and propylene glycol is the preservative found in lorazepam and diazepam.
- Unlike spinal anesthesia which relies more on the dose in milligrams of the local anesthetic, the baricity of the solution, and the patient's position , the segmental spread of an epidural is reliant primarily upon the volume of local anesthetic solution used.
- A field block is the injection of a typically large volume of local anesthetic in the subcutaneous tissue in order to block cutaneous nerves. A superficial cervical plexus block is an example of a field block that may be used to provide anesthesia for carotid endarterectomies.
- The intercostal neurovascular bundle travels on the inferior surface of the rib. From superior to inferior in the bundle are the vein, artery, and nerve.



Source: Morgan GE, Mikhail MS, Murray MJ: *Clinical Anesthesiology*, 4th Edition: http://www.accessmedicine.com

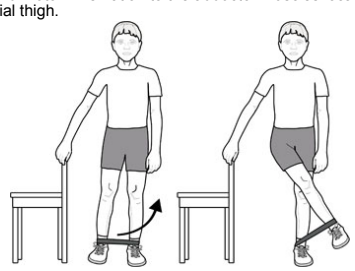
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- The femoral nerve maintains a lateral position in the femoral canal. In order from lateral to medial the structures are remembered by the acronym NAVEL (nerve, artery, vein, empty space, and lymphatics).

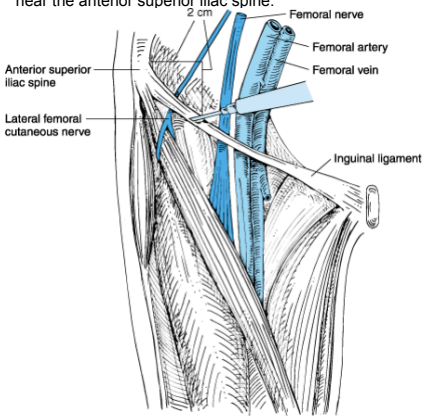


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- The musculocutaneous, medial brachial cutaneous, and intercostobrachial nerves may need to be blocked separately as they exit the brachial plexus sheath high in the axilla to adequately prevent pain from an arm tourniquet.
- To avoid a rapid bolus of intravenous local anesthetic that could result in local anesthetic toxicity, the tourniquet should remain inflated for a minimum of 15 to 20 minutes as was the case in this procedure.
- A femoral nerve block will provide anesthesia for the anterior thigh, knee, and a portion of the medial aspect of the foot. A lateral femoral cutaneous nerve block would be more appropriate for a lateral thigh procedure while an obturator nerve block would be appropriate for procedures on the medial thigh such excision of a medial thigh mass or a release procedure on the medially located adductor muscles.
- The obturator nerve provides sensation to the medial aspect of the thigh and motor innervation to the adductor muscles located in the medial thigh.



- The lateral femoral cutaneous nerve emerges underneath the inguinal ligament and becomes superficial inside the fascia lata near the anterior superior iliac spine.



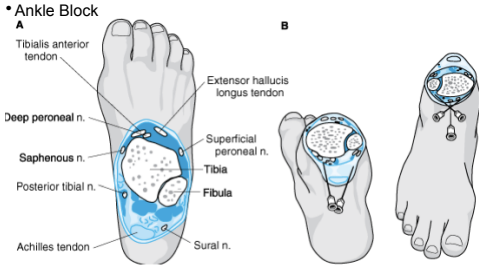
Source: Morgan GE, Mikhail MS, Murray MJ: *Clinical Anesthesiology*, 4th Edition: http://www.accessmedicine.com

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- A sciatic nerve block will provide anesthesia for the knee and all structures distal, but will not provide anesthesia for the placement of a tourniquet on the thigh. A common remedy for this is to combine a "three-in-one" block of the obturator, femoral, and lateral cutaneous nerves with a sciatic block to provide anesthesia for the entire extremity.
-

- The popliteal nerve is a branch of the sciatic nerve. In fact, all of the nerves supplying the foot and ankle are branches of the sciatic nerve except for the saphenous nerve which is a branch of the femoral nerve. The saphenous nerve supplies sensation to the anteromedial foot and is located anterior to the medial malleolus. A nerve block in the popliteal fossa will anesthetize both the common peroneal and tibial branches of the sciatic nerve which terminate as the superficial peroneal, deep peroneal, sural, and posterior tibial nerves.

- Ankle Block



Source: Morgan GE, Mikhail MS, Murray MJ: *Clinical Anesthesiology*, 4th Edition: http://www.accessmedicine.com

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- The deep peroneal, superficial peroneal, and saphenous nerves can be blocked by infiltrating local anesthetic on a malleolar-level line across the anterior aspect of the foot. The deep peroneal nerve is blocked by identifying the groove formed proximally by the extensor hallicus longus tendon and the extensor digitorum longus tendons. Then 5–8 mL of local anesthetic is delivered at the apex of this groove as a field block with a 25-gauge needle inserted subcutaneously and then advanced to just contact the periosteum. From this insertion site, a subcutaneous infiltration with 5–8 mL of local anesthetic directed toward the lateral malleolus is used to block the superficial peroneal nerve. From the initial insertion site, redirecting the needle toward the medial malleolus, a subcutaneous infiltration with 5–8 mL of local anesthetic will block the saphenous nerve (Figure 17–32B). The tibial nerve travels posterior to the medial malleolus and lies adjacent to the posterior tibial artery. The block is performed by injecting 5–8 mL of local anesthetic in a fanlike manner posterior to the medial malleolus and at a right angle to the course direction of the nerve. The sural nerve subcutaneously courses posterior to the lateral malleolus. A field block performed posterior to the malleolus with 5–8 mL of local anesthetic will block this nerve.
- The injection of local anesthetic superficially near the medial malleolus will anesthetize the saphenous nerve while the continued injection toward the extensor digitorum longus tendon will anesthetize the superficial peroneal nerve.

- thoracic paravertebral block, the most common complication is pneumothorax.

- The blood supply to the spinal cord is derived from a single anterior and paired posterior arteries. The anterior spinal artery is formed from the vertebral artery and supplies the anterior 2/3 of the cord while the posterior spinal arteries are derived from the posterior inferior cerebellar arteries and supply the posterior 1/3 of the spinal cord.

- With neuraxial anesthesia, a differential blockade is produced which typically results in sympathetic blockade which is 2 segments higher than sensory blockade which in turn, is 2 segments higher than motor blockade

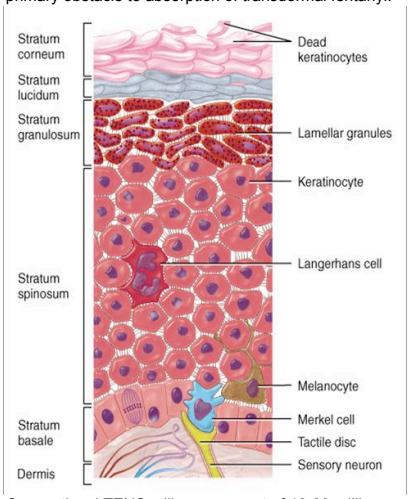
- The sympathectomy associated with a spinal anesthetic produces hypotension due principally to venous pooling from dilation of the venous capacitance vessels, although arteriolar dilation may contribute to the hypotension by a lesser degree. Bradycardia and decreased contractility do not likely contribute to hypotension in this scenario. Blockade of the cardioacceleratory fibers at the T1 to T4 level however, could result in bradycardia and decreased contractility. Since a sympathetic blockade is typically 2 segments above a sensory block, a sensory block at T6 or higher could result in blockade of the cardioacceleratory fibers.
- Spinal blockade, even at the cervical level is reported not to be high enough to block the large A-alpha fibers of the phrenic nerve. Apnea appears to be the result of decreased perfusion to the brainstem as apnea typically resolves with hemodynamic resuscitation.
- The sympathectomy produced by spinal anesthesia results in unopposed vagal activity. This increased parasympathetic activity results in increased peristalsis and a contracted gut. Although vital capacity can diminish due to decreased intercostal muscle activity, tidal volume is left unchanged. The blockade of the autonomic fibers to the bladder results in a decreased ability to void, so urinary retention is a common complication but urinary incontinence is not. Sympathectomy to the adrenal glands results in a decreased stress response to surgical stimulation which, by limiting the release of stress hormones, also limits any resulting increase in blood sugar.
- Conditions resulting in left ventricular outflow obstruction such as aortic stenosis or hypertrophic subaortic stenosis restrict the ability of the heart to increase cardiac output as compensation for hypotension due to the sympathectomy induced by neuraxial anesthesia. In these circumstances, the hypotension can become refractory to treatment.
- With the patients arms relaxed at the sides, the inferior angle of the scapula lies at the level of T7.
- In adults, the subarachnoid space extends from the foramen magnum to S2 and extends to S3 in children. The spinal cord itself extends to L1 in adults and L3 in children.
- When performing spinal anesthesia, two 'pops' are often felt. The first 'pop' is the needle penetrating the ligamentum flavum and the second is the needle penetrating the dura mater.
- CSF has a specific gravity of 1.003 to 1.008. Plain 2% lidocaine has a specific gravity between 1.0004 and 1.0066.
- Normally, in the supine position, the most dependent portion of the spinal column occurs at T4-T8 with the peak of the thoracolumbar curvature occurring at T4. This is important in relation to hyperbaric spinal solutions which will tend to pool in the T4-T8 curvature but limit their ascent above T4 in the supine position.
- Conditions that result in increased abdominal pressure or engorgement of epidural veins such as pregnancy, ascites, abdominal tumors, kyphoscoliosis, or increased age reduce the CSF volume and increase the height of spinal blockade. In pregnant patients, the dose of spinal anesthetic to reach acceptable levels is one-third that of non-pregnant patients.

- The standard test dose for epidural anesthesia contains 3 mL of 1.5% lidocaine and 1:200,000 epinephrine or 15 mg/mL of lidocaine and 5 mcg/mL of epinephrine. Thus the total test dose contains 45 mg lidocaine and 15 mcg epinephrine.
- Epidural blood patches should be administered in 15-20 mL doses (autologous blood) at the level of the dural puncture or 1 level below it.
- If LMWH is planned postoperatively the epidural catheter should be removed no sooner than 2 hours before the first dose. If the patient is already on LMWH, the catheter should be removed at least 10 hours after the most recent dose and at least 2 hours before the next dose.
- To answer this question, you must first determine the patients body mass index (BMI) which is the weight in kilograms divided by the height in meters squared. To obtain the weight in kilograms, divide the weight in pounds by 2.2. The answer is 100 kilograms. Next, multiply the height in inches (69) by 0.0254 to convert it to meters. Take this result (1.75m) and square it to arrive at 3.0 meters squared. Divide the weight in kilograms (100) by 3.0 meters squared to arrive at a BMI of 33.3. Ideal body weight falls in a BMI range between 20 and 25, mild obesity is 26 to 29, moderate obesity is 30 to 35, severe obesity is 35 to 39, and morbid or extreme obesity is a BMI of 40 or above.
- The maximum amount of hetastarch that may be administered to avoid dilutional coagulopathy, factor VIII inhibition, and inhibited platelet adhesiveness is 20 mL/kg. It is important to remember, however, that this value should be calculated based on the ideal body weight in obese individuals. The ideal body weight for a female is the height in centimeters minus 105 (subtract 100 from the height in centimeters for males). In this example, multiply the height in inches (64) by 2.54 to arrive at a height of 162.65 centimeters. Subtracting 105 from this value leaves an ideal body weight of 57.65 kg. Multiplying this value by 20 mL/kg yields 1153 or 1150 mL. Increasing the tidal volume can improve oxygenation but should be calculated up to 15-20 mL/kg of ideal bodyweight not actual bodyweight.
- In the obese patient, liver enzymes may be elevated due to fatty liver infiltrates. The cardiac output increases approximately 0.1 L/min for each additional kilogram of body fat. Alveolar ventilation increases accordingly with the increased metabolic demand and CO2 production associated with an increased body mass. Chest wall compliance is unchanged.
- Autonomic neuropathy has been demonstrated to be the best predictor of silent coronary artery disease in diabetic patients.
- the site and type of surgery are the strongest predictors of postoperative pulmonary complications with thoracic and upper abdominal surgery associated with the highest risk.
- Hypomagnesemia is common with chronic alcoholism as is hypokalemia and metabolic alkalosis (from frequent vomiting). As the liver becomes damaged, the prothrombin time may become increased.
- Patients with Down syndrome and rheumatoid arthritis have an increased risk for cervical spine instability while patients with diabetes can develop limited-mobility joint syndrome and a decreased range of motion in the cervical spine.

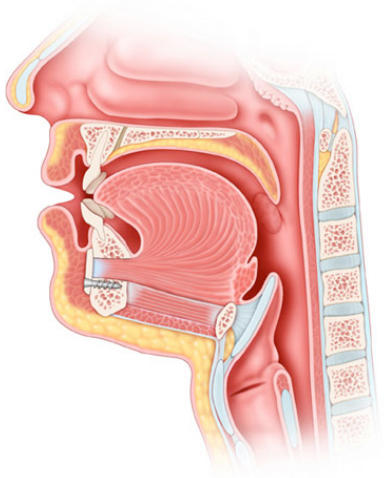
- Serum thyroxine levels less than 1 mcg/dL are indicative of severe hypothyroidism and warrant delay of an elective case or treatment with thyroid hormone prior to emergency surgery. The normal serum thyroxine levels in adult patients is 5-12 mcg/dL. This (and other clinical laboratory reference data) refutes the statement found in Clinical Anesthesiology (Morgan, Mikhail, and Murray) that severe hypothyroidism occurs with thyroxine levels below 1 mg/dL.
- Myasthenia gravis is characterized by skeletal muscle weakness and is treated with anticholinesterase drugs. An overdose of anticholinesterases can also produce excessive weakness. The administration of edrophonium 10 mg IV can help distinguish between the two conditions. If the patient continues to exhibit weakness, then the pathology is cholinergic crisis. If the weakness resolves then the patient has myasthenic crisis.
- the anion gap, the formula for which is: [Na+] - ([Cl-] - [HCO3-]). In this instance, the anion gap is 23 mEq/L. The normal anion gap is 7 to 12 mEq/L. Causes of increased anion gap acidosis include renal failure, ketoacidosis, lactic acidosis, hyperosmolar nonketotic coma, alcohol toxicity, and ingestion of salicylates, methanol, ethylene glycol, paraldehyde, toluene, and sulfur. Although the other options listed above can result in metabolic acidosis, alcohol toxicity is the only option associated with an increased anion gap.
- Because creatinine is produced at a nearly constant level and creatinine is not reabsorbed by the kidneys, creatinine clearance is used as a method of evaluating renal function. A creatinine clearance less than 25 mL/min indicates overt renal failure, 25-40 mL/min is moderate renal dysfunction, and 40-60 mL/min is mild renal dysfunction.
- ALT is located primarily in the liver and is therefore more specific for liver damage
- Alkaline phosphatase is a family of enzymes found in many organ systems. In healthy persons, most of it originates in liver or bone. In the liver, it is concentrated in the bile canaliculi and the sinusoidal surface of hepatocytes. 5' nucleotidase is a form of alkaline phosphatase that is present in most tissues, but elevated serum levels are always hepatobiliary in origin and are markedly elevated in intrahepatic or extrahepatic obstruction.
- This patient has a low serum sodium concentration but a high serum osmolarity. Hyponatremia with a normal or high serum osmolarity result from the presence of non-sodium solutes such as glucose or mannitol.
- Increased protein intake can elevate the BUN level leading to an inaccurate diagnosis of renal dysfunction. Other causes of altered BUN levels include altered intravascular fluid volume and co-existing diseases.
- Urinary sodium > 40 mEq/L is associated with damage to the renal tubules. Orthostatic proteinuria occurs in 5-10% of the population and is not associated with renal damage. Glucosuria is associated with increased serum glucose levels. Decreased hematocrit is associated with anemia and increased intravascular fluid volume.
- Bumetanide is a loop diuretic associated with decreased serum potassium levels. Drugs such as triamterene, spironolactone, NSAIDs, ACE inhibitors, and beta blockers are associated with increased potassium levels.
- The treatment of high cholesterol through the use of HMG-CoA reductase inhibitors, dietary changes, and smoking cessation is considered adequate if LDL levels are reduced to below 130 mg/dL. Treatment is considered optimal, however, if LDL levels are less than 100 mg/dL.

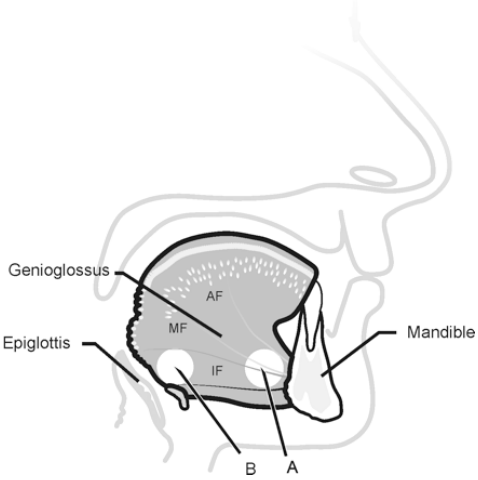
- Glycogen storage disease results in a lack of the enzyme glucose-6-phosphatase. As a result, glycogen can be hydrolyzed in hepatocytes and other cells and becomes inappropriately stored in the intracellular space. Hypoglycemia can be severe and metabolic acidosis develops as a consequence.
- The wake up test is performed by reducing the patients intraoperative anesthetic level sufficiently to allow the patient to follow a command such as squeezing a hand to assess the integrity of the motor pathways of the ventral cord. Somatosensory evoked potentials are typically used to assess the sensory neurons of the dorsal cord.
- Relative contraindications to performing controlled hypotension include severe anemia, cardiovascular and cerebrovascular disease, renal or hepatic insufficiency, uncontrolled glaucoma, and hypovolemia. Chronic hypertension is not a relative contraindication to controlled hypotension, but it warrants closer monitoring of blood pressure. Because the alteration of autoregulation of cerebral blood flow in chronically hypertensive individuals, the mean arterial blood pressure should not be lowered more than 20-30% below baseline.
- Although all of the methods above are acceptable of producing controlled hypotension, the rapid onset and short duration of action of sodium nitroprusside and nitroglycerin make them most suitable for controlled hypotensive techniques.
- Although core temperatures as low as 20 degrees Celsius have been used to successfully provide neurologic protection during cases of total circulatory arrest, even mild hypothermia (33-35 degrees Celsius) demonstrates some potential for neurologic protection by maintaining ATP levels at normal even when oxygen levels are at half normal. It should be noted, however, that several studies have demonstrated no improvement in overall mortality when controlled hypothermia is used and the use of controlled hypothermia may be waning.
- The most effective method of providing neurologic protection is through hypothermia. Profound hypothermia has been used successfully to perform surgeries as long as 1 hour during total circulatory arrest. No anesthetic agent has been shown to protect the brain against global ischemia.
- Because of the proximity of the injection needle to the pleura, pneumothorax is the most common complication of thoracic paravertebral blocks. Intravascular injection, epidural, subdural and subarachnoid injection are also potential complications.
- The celiac ganglia are generally grouped around the level of L1 just to the left of the aorta. Blockade of this plexus relieves pain arising from the abdominal viscera. Thoracic paravertebral blocks relieve pain arising from the abdominal or chest wall at the dermatome level of the block. A thoracic sympathetic chain block comes with a significant risk of pneumothorax and is no longer performed routinely. A trans-sacral nerve block is performed to relieve pelvic and perineal pain.
- Phenol and alcohol are both used to perform neurolytic blocks. These blocks are occasionally used to treat refractory neuralgia and peripheral vascular disease, but the chief indication is for intractable cancer pain.
- Antidepressants provide analgesia by inhibiting the presynaptic reuptake of norepinephrine, serotonin, or both.

- Both non-selective and selective COX inhibitors inhibit prostaglandin synthesis and are highly protein bound. Unlike non-selective COX inhibitors, COX-2 inhibitors have a lower incidence of gastrointestinal side effects and do not inhibit platelet aggregation.
- The stratum corneum, the topmost layer of the epidermis, is the primary obstacle to absorption of transdermal fentanyl.



- Conventional TENS utilizes a current of 10-30 milliamps, a pulse width of 50-80 microseconds, and a frequency of 80-100 Hz.
- The nerve fibers involved in transmitting pain from the head are carried by the trigeminal, facial, vagus, and glossopharyngeal nerves.
- A cricothyrotomy is performed by inserting a large-bore intravenous catheter or cannula into the cricothyroid membrane which lies between the inferior border of the thyroid cartilage and the superior border of the cricoid cartilage.
- The mucous membranes of the nose are innervated by the opthalmic division of the trigeminal nerve (V1) anteriorly and the maxillary division of the trigeminal nerve (V2) posteriorly. These nerves are also known as the anterior ethmoidal and sphenopalatine nerves respectively.
- The recurrent laryngeal nerve provides sensation to the larynx below the vocal cords. The recurrent laryngeal nerve also provides motor innervation to all of the laryngeal muscles except for the cricothyroid muscle, which is innervated by the external laryngeal nerve.
- The airway between the epiglottis and the vocal cords derives sensory innervation from the internal branch of the superior laryngeal nerve. The external branch of the superior laryngeal nerve is motor only and innervates the cricothyroid muscle.
- Unilateral paralysis of the recurrent laryngeal nerve can result in changes in phonation and hoarseness. The internal branch of the superior laryngeal nerve, not the recurrent laryngeal nerve provides sensation between the epiglottis and the vocal cords. Stridor and acute airway compromise can result from bilateral paralysis of the recurrent laryngeal nerve.
- Loss of motor tone to the **genioglossus** allows the tongue to fall back against the posterior pharynx resulting in airway obstruction.





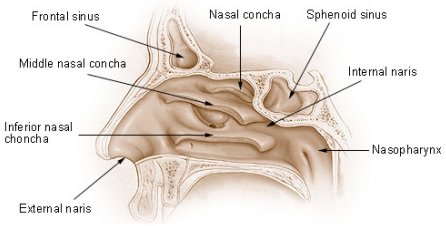
- Contraction of the genioglossus stabilizes and enlarges the portion of the upper airway that is most vulnerable to collapse. A relaxation of the genioglossus and geniohyoideus muscles, especially during REM sleep, is implicated in Obstructive Sleep Apnea (OSA.)
- Peripheral damage to the hypoglossal nerve can result in deviation of the tongue to the damaged side.

- An LMA protects the airway from pharyngeal secretions (but not gastric secretions), is contraindicated in pharyngeal pathology such as tumor or abscess, is contraindicated in situations requiring positive pressures greater than 30 cm H2O such as decreased pulmonary compliance, and is less likely to produce bronchospasm than endotracheal intubation.

- For a full-term infant, a 3.5 mm internal diameter tube is recommended. For older children, the formula: 4 + age/4 is a useful predictor for the appropriate endotracheal tube size.

- When performing a nasotracheal intubation, the tracheal tube should be inserted into the nares at an angle perpendicular to the face with the bevel directed away from the turbinates.

Nose and Nasal Cavities



- Nasotracheal intubation may be contraindicated in patient with severe mid-facial trauma (such as a maxillary fracture) which may represent an existing cribriform plate fracture. If the cribriform plate is unstable, the tracheal tube could potentially pass through it into the cranial vault.

- Clinical signs of hypovolemia include: hypernatremia, urine osmolality > 450 mOsm/kg, urinary sodium < 10 mEq/L, elevated hematocrit, metabolic acidosis, BUN:creatinine ratio > 10:1, and a PaOP < 8mmHg. A CVP > 12 mmHg is considered indicative of overhydration in an otherwise healthy patient.

- The intravascular half-life of crystalloid solutions is 20-30 mins compared to colloidal solutions which remain in the bloodstream for 3-6 hours.

- Colloids such as hetastarch and dextran are generally used to replace blood loss in a ratio of 1mL to 1 mL. Crystalloids are generally used to replace blood loss in a 3:1 ratio.

- Maintenance fluid requirements are calculated as 4 mL/Kg/hour for the first 10 kilograms, 2 mL/Kg/hour for the second 10 kilograms, and 1 mL/Kg/hour thereafter. For a 60 Kg patient, this would equal 40 + 20 + 40 or 100 mL/hour. Over a period of 12 hours, the NPO deficit is 1.2 liters.

- Doses of dextran 40 above 20 mL/Kg in a 24 hour period can result in prolonged bleeding times due to an antiplatelet effect, interferes with blood typing, is associated with renal failure, and can result in allergic reactions. In an 70 Kg patient, the maximum dose is 70 Kg X 20 mL/Kg or 1400 mL.

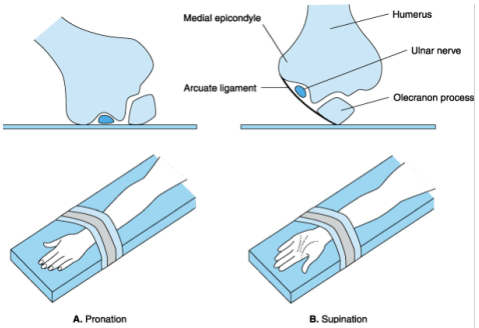
- A single, fully-soaked 4X4 sponge will hold approximately 10 mL, thus 5 sponges will hold 50 mL.
- A patient with Type A blood has anti-B antibodies. Patients with Type B blood has anti-A antibodies. Patients with Type O blood has both anti-A antibodies and anti-B antibodies and patients with Type AB have neither.
- D51/4 Normal Saline has an tonicity of 355 mOsm/L and, although it is actually slightly hypertonic, is considered to be isotonic. The normal serum osmolality is between 280 and 290 mOsm/L. 1/2 Normal Saline has an tonicity of 154 mOsm/L and is hypotonic, as is D5W which has a tonicity of 253 mOsm/L. D5NS is hypertonic with a tonicity of 586 mOsm/L.
- Lactated Ringer's and D5LR both contain 4 mEq/L of potassium and Plasmalyte contains 5 mEq/L of potassium. 7.5% NaHCO3 is extraordinarily hypertonic with 893 mEq/L of sodium and 893 mEq/L of HCO3, but it contains no potassium.
- Steep reverse trendelenburg position elevates the head, which displaces the abdominal contents toward the pelvis resulting in a decreased mean thoracic pressure. It also results in venous pooling in the lower extremities which decreases preload and cardiac output. The downward displacement of the abdominal contents decreases the work of breathing while increasing functional residual capacity and total lung volume.

- Although lithotomy position can result in damage to any of these nerves, this patient's presentation is most consistent with damage to the saphenous nerve, which innervates the medial calf. This is most likely to occur with medially placed strap supports. Extreme flexion at the thigh can result in injury to the sciatic, obturator, and femoral nerves while injury to the common peroneal nerve can occur if the lateral thigh is allowed to rest on strap supports.



- The most commonly occurring nerve injuries due to lithotomy position involve the common peroneal nerve when the head of the fibula is compressed against the leg support device, the sciatic nerve due to excessive flexion of the hips, the femoral nerve which can become compressed due to extreme flexion and abduction of the highs, the saphenous nerve which may be compressed when the medial tibial condyle rests against a strap support, and the obturator nerve which can be overstretched by thigh flexion. The brachial plexus is at risk for injury if the arms are extended greater than 90 degrees, but this arm position is not directly related to lithotomy position. This answer is in contrast to the one found in Clinical Anesthesiology which states that the brachial plexus is most commonly at risk. As this statement is not supported in any other textbooks, or elsewhere in the Clinical Anesthesiology text, it is believed to be a typographical error.
- Loss of dorsiflexion of the foot is consistent with injury to the common peroneal nerve which can occur in the lithotomy position if the lateral thigh is allowed to rest on strap supports. The femoral nerve innervates the anterior thigh and knee and the saphenous nerve, which is a branch of the femoral nerve, provides sensation to the medial aspect of the foot.

- The ulnar nerve is most at risk for compression damage where it passes under the medial epicondyle. Pronation of the hand positions the ulnar nerve directly against the table surface, making it a high-risk position. Supinating the hand shifts the ulnar nerve above the table surface and allows the olecranon to bear the weight of the arm. Tucking the arms at the sides can still leave the ulnar nerves at risk and padding should be placed around the elbows to protect the ulnar nerve. Abducting the arms less than 90 degrees provides a degree of protection for stretch injury to the brachial plexus, but has little bearing on the ulnar nerve.

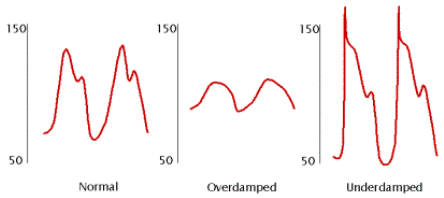


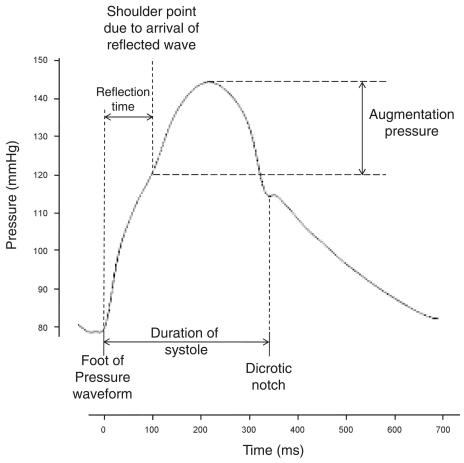
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Pronation of the forearm can cause external compression of the ulnar nerve in the cubital tunnel (A). Forearm supination avoids this problem (B).

- Factors associated with an increased risk of complications due to intra-arterial monitoring such as thrombosis, vasospasm, infection, hematoma, nerve damage, and loss of digits include female gender, hyperlipidemia, prolonged cannulation, repeated insertion attempts, hyperlipidemia, and use of vasopressors.

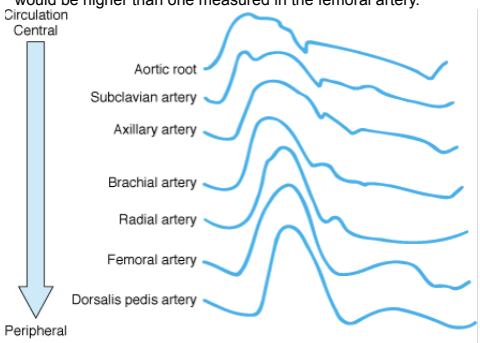
- The presence of air in the transducer tubing and the addition of stopcocks and longer tubing can decrease the frequency of the monitoring system and lead to over-damping which will tend to underestimate the systolic blood pressure.





- If the blood pressure readings between two extremities vary significantly in patients with peripheral vascular disease, you should record the higher pressure. Inserting an arterial line would provide the advantage of continuous blood pressure monitoring, but may increase the risk for vascular occlusion in these patients.

- As the pulse wave moves from centrally to peripherally, the systolic waveform increases resulting in a wider pulse pressure than in the more central locations. Thus, the systolic pressure as measured in an arterial line placed in the dorsalis pedis artery would be higher than one measured in the femoral artery.



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As a pulse moves peripherally through the arterial tree, wave reflection distorts the pressure waveform, leading to an exaggeration

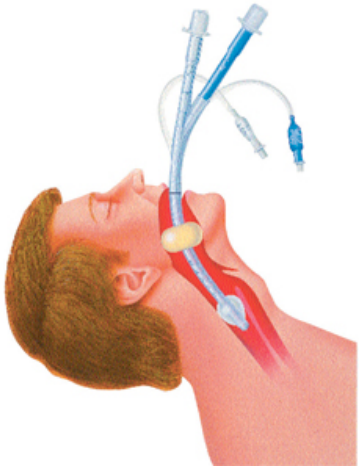
of systolic and pulse pressures.

- Because air reflects sound waves, placement of a coupling gel eliminates any layer of air between the skin and the Doppler probe and thereby increases the reliability of assessment.

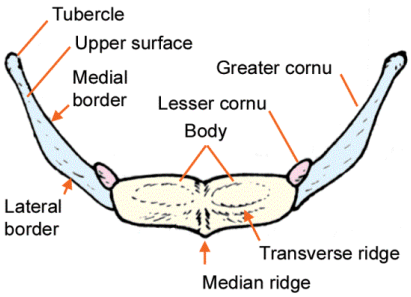
- When assessing the blood pressure using a blood pressure cuff and a stethoscope, the onset of Korotkoff sounds indicates the systolic blood pressure and the disappearance of the sounds indicates diastolic blood pressure.

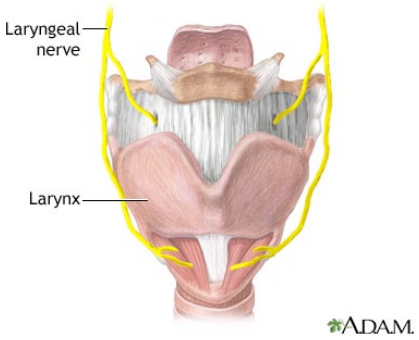
- The LMA comes in several sizes: 1 for infants, 2 for children 6.5-20 Kg, 2.5 for children 20-30 Kg, 3 for children and small adults greater than 30 Kg, and size 4-5 for larger adults.

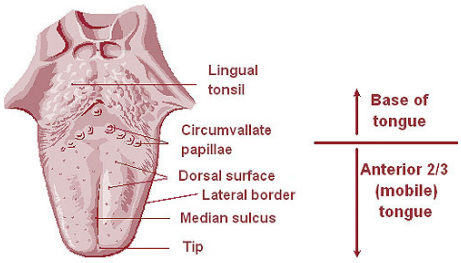
- After inserting a Combitube, you should inflate both the proximal and distal cuffs. The Combitube will be placed within the esophagus in 95% of insertions in which case ventilation through the blue tube will force air through the side perforations and into the larynx. If the Combitube enters the trachea, ventilation can proceed by connecting the circuit to the shorter, clear tube.



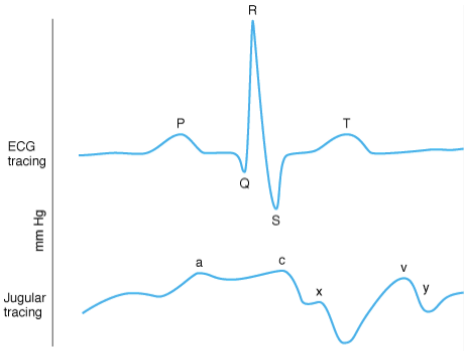
- Injection of 3 mL at the cornu on either side of the hyoid bone anesthetizes the superior laryngeal nerve, which supplies sensation to the airway below the epiglottis. The posterior 1/3 of the tongue and tonsillar pillars can be anesthetized by blocking the lingual and pharyngeal branches of the glossopharyngeal nerve with 2 mL of local anesthetic at the base of the palatoglossal arch.







- An auscultatory gap is noted during blood pressure assessment when Korotkoff sounds cannot be heard through a portion of the range between the systolic and diastolic pressures. This is most common in patients with hypertension and may lead to inaccurate reporting of diastolic blood pressure.



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- A central venous pressure tracing exhibits several waveforms worth noting: the a wave represents atrial contraction, the c wave represents elevation of the tricuspid valve during ventricular contraction, the v wave represents venous return against a closed tricuspid valve, and the x and y descents represent downward displacement of the ventricle during systole and opening of the tricuspid valve during diastole.

- A central venous pressure tracing exhibits several waveforms worth noting: the a wave represents atrial contraction, the c wave represent elevation of the tricuspid valve during ventricular contraction, the v wave represents venous return against a closed tricuspid valve, and the x and y descents represent downward displacement of the ventricle during systole and opening of the tricuspid valve during diastole. Because it represents atrial contraction, the a wave would be absent in patients with atrial fibrillation.

- When an O2 sensor indicates a hypoxic breathing mixture, the first step you should take is to call for help. Then, you should turn on the O2 tank supply AND disconnect the O2 supply hose from the wall. If the inspired O2 content doesn't increase rapidly then ventilate the patient with an Ambu bag with room air until a functioning O2 tank can be obtained.

- If the circuit exhibits sustained high pressures only during mechanical ventilation and not during manual ventilation, then the problem is likely a malfunctioning ventilator relief valve. If the scavenger was obstructed or the endotracheal tube is obstructed with secretions then the circuit pressures would remain elevated during mechanical or manual ventilation.

- Patients with von Willebrand's disease who do not respond to treatment with DDAVP (desmopressin) should be given cryoprecipitate which contains factor VIII (both the procoagulant and von Willebrand's factor), fibrinogen, and factor XIII.

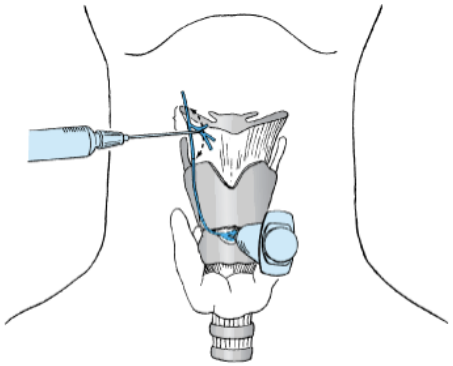
- What is the minimum size filter that can be used when administering blood transfusions to patients with a history of febrile transfusion reactions to trap the white blood cell and platelet contaminants ? A 20-40 micron filter can be used to trap the white blood cell and platelet debris that is most commonly associated with causing febrile reactions to blood transfusions (occurs in 1% of patients receiving transfusion). Many standard blood administration sets utilize a 170 micron filter.

- EMLA cream consists of a 1:1 mixture of 2.5% lidocaine and 2.5% prilocaine in an oil-in-water emulsion. It requires a contact time of at least one hour under an occlusive dressing, reaches a depth of analgesia of about 3-5 mm, and has a duration of about 1-2 hours.

- When performing a brachial plexus block using the axillary approach, it is often necessary to perform separate blocks of the medial brachial cutaneous and intercostobrachial nerves because the former exits the sheath just below the clavicle and the latter doesn't travel in the sheath at all. These two nerves provide sensation to the skin of the medial and posterior proximal arm.

- In combined spinal and epidural analgesia, the spinal and epidural needles may be placed at the same site or at different sites. The epidural space should be identified with the use of air because the use of saline may create a potential confusion between saline and CSF.

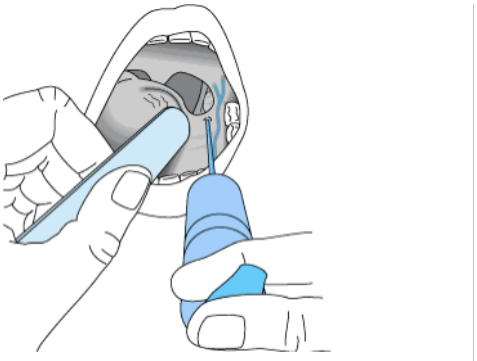
- Injection of 3mL of 2% lidocaine bilaterally 1 cm below each greater cornu of the hyoid is a superior laryngeal nerve block which anesthetizes the airway below the epiglottis. A transtracheal block is performed by injecting 4mL of 4% lidocaine through a needle penetrating the cricothyroid membrane into the trachea. The lingual and pharyngeal branches of the glossopharyngeal nerve are blocked by infiltrating local anesthetic into the base of the palatoglossal arch.



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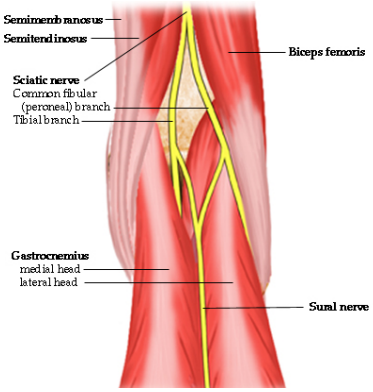
SLN block and Transtracheal block



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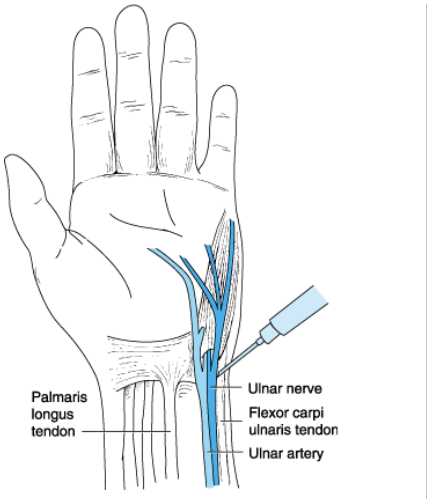
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- The glossopharyngeal block is performed by having the patient open their mouth while use a 25-gauge needle to inject 2 mL of local anesthetic bilaterally at the base of the palatoglossal arch (also called the anterior tonsillar pillar). This blocks the lingual and some of the pharyngeal branches of the glossopharyngeal nerve that provide sensation to the posterior third of the tongue and oropharynx. A transtracheal block is performed by injecting 4 mL of 4% lidocaine into the trachea by passing a needle through the cricothyroid membrane and into the trachea. The local anesthetic should be injected at the end of expiration, at which point the patient will inhale deeply and cough as a result of the irritation and distribute the local anesthetic through the trachea. A bilateral superior laryngeal nerve block can be used to provide anesthesia to the airway below the level of the epiglottis. Injecting 3 mL of 2% lidocaine 1 cm below the left and right greater cornu of the hyoid bone will block the internal branches of the superior laryngeal nerve.
- The popliteal fossa is bounded laterally by the biceps femoris tendon and medially by the semimembranosus and semitendinosis tendons. The popliteal artery is found lateral to the semitendinosus tendon.



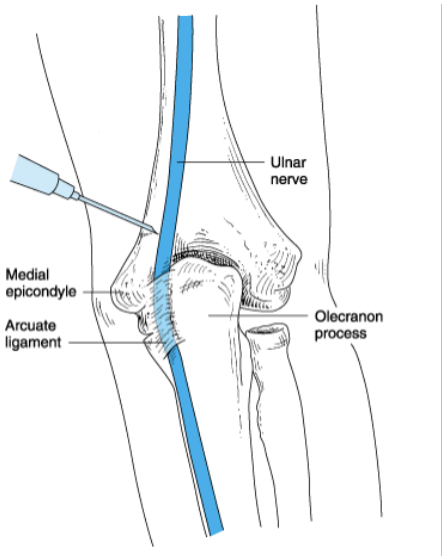
- Epidural steroids have been shown to improve sensory function and flexibility, but only at a three week assessment. The lumbar and caudal routes are equivalent in their effectiveness. Diabetics may require lower doses of epidural steroids to reduce the risk for steroid-induced hyperglycemia.
- Typically, the patient in hypovolemic shock presents with a low pulmonary capillary wedge pressure, low cardiac output, and a high systemic vascular resistance
- Tic douloureux (trigeminal neuralgia) is a painful condition involving the trigeminal nerve (usually the V2 or V3 branches). Like other forms of neuropathic pain such as shingles, reflex sympathetic dystrophy, phantom limb pain, and central postherpetic pain, it results from abnormalities in or damage to the nervous system. Nociceptive pain is produced by activation of normal pain fibers. Psychobehavioral pain results from anxiety, depression, neurosis, or hysteria. Allodynia is the sensation of pain from a stimulus that is normally not painful.
- Causalgia, the former name of complex regional pain syndrome type II, means burning pain. It is usually caused by high velocity injuries to large nerves (most commonly the brachial plexus). It has an immediate onset and is associated with allodynia, hyperpathia, and vasomotor and sudomotor dysfunction in the region of pain, resulting in erythema, edema, and warmth in the area of pain. Anything that increases sympathetic activity worsens the pain. In fact, one of the effective treatments in the early stages of the disease is sympathetic blockade.
- Epinephrine should not be added to the local anesthetic used for wrist blocks as it could compromise distal circulation. The ulnar nerve can blocked as it passes between the flexor carpi ulnaris tendon and the ulnar artery. The median nerve may be blocked as it passes between the palmaris longus and flexor carpi radialis tendons although it should be noted that the palmaris longus may be absent in some patients. The radial nerve block is considered a field block because of its less predictable anatomic location and division into numerous smaller branches.

Ulnar N. Block



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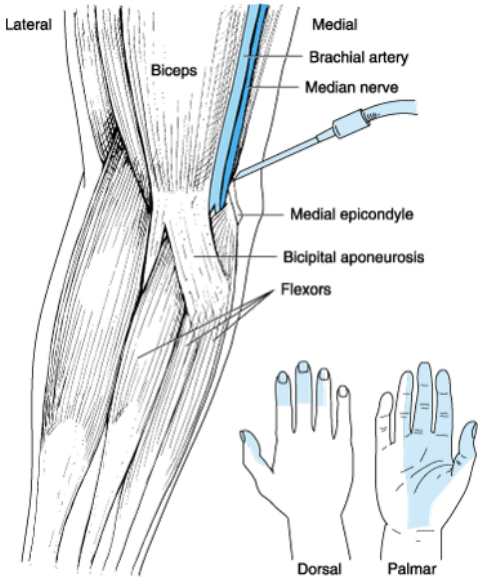
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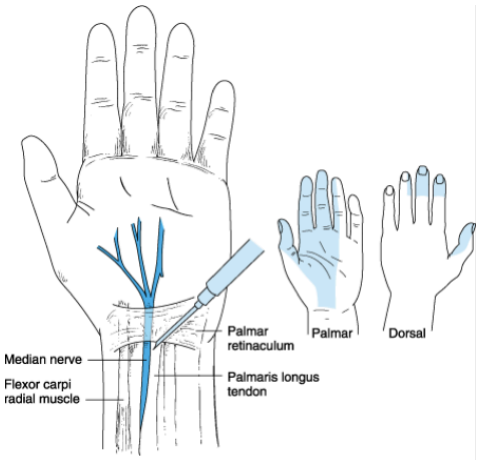
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Median N. Block



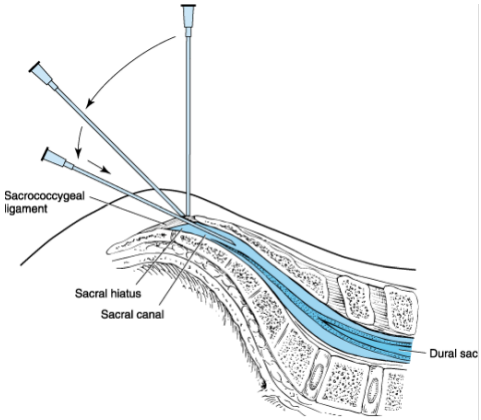
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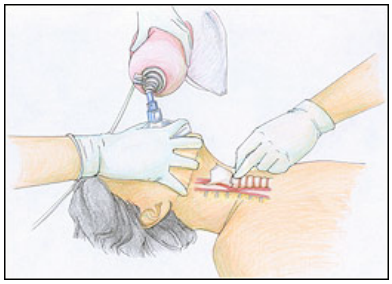
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- Three nerves innervate the hand and can be blocked in three steps: 1) The ulnar nerve is identified where it runs between the ulnar artery and the flexor carpi ulnaris at the level of the styloid process on the volar aspect of the forearm. A skin wheal is raised and 3 mL of local anesthetic is injected between these two landmarks. 2) Also at the level of the styloid process on the volar aspect of the arm is the median nerve, which can be found between the palmaris longus and flexor carpi radialis tendons. After placing a skin wheal, a needle is inserted until it pierces the deep fascia and 3 mL of local anesthetic is injected. If only the palmaris longus tendon is appreciable, then the nerve is just to the radial side of the tendon. 3.) By the time it reaches the hand, the radial nerve branches widely and requires the injection of 3 mL of local anesthetic where it passes through the anatomical snuffbox formed by the extensor pollicis longus and the extensor pollicis brevis tendons. A skin wheal is then carried over the dorsum of the wrist for another 3 to 4 cm to block the branches of the radial nerve.
- The epidural sac extends to S1 in adults and S3 in children making inadvertant intrathecal injection more likely in children. Caudal anesthesia is commonly used for procedures below the diaphragm including genitourinary procedures, rectal, inguinal, and lower extremity surgery. The caudal epidural needle is inserted through the sacral hiatus which is found between the unfused S4 and S5 vertebrae.



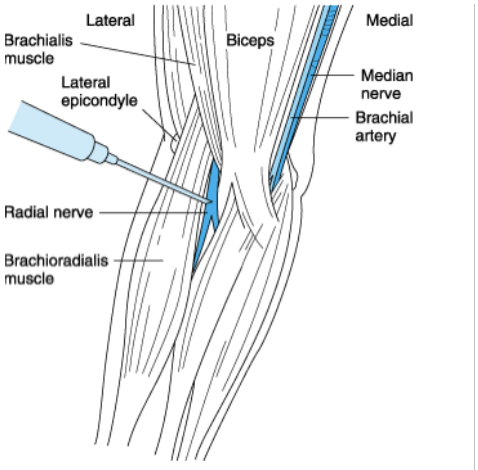
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to cause pain. Another more practical method that has been proposed is the BURP method, in which Backward, Upward, and Rightward Pressure is applied to the cricoid cartilage with the thumb and forefinger with enough force to blanch the fingers. Pressure should be maintained on the cricoid cartilage until the position of the endotracheal tube has been verified.

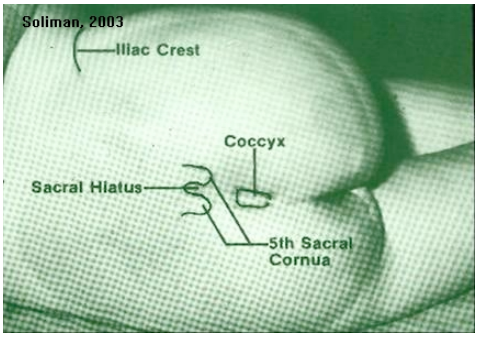


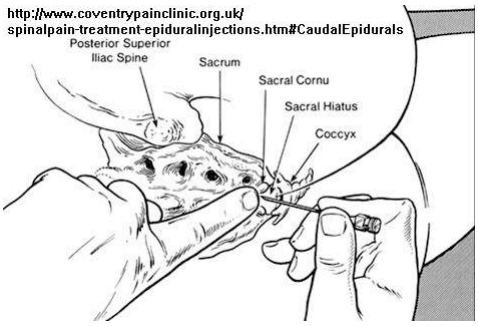
- Injecting local anesthetic between the flexor hallucis longus and the extensor digitorum tendons will anesthetize the deep peroneal nerve.

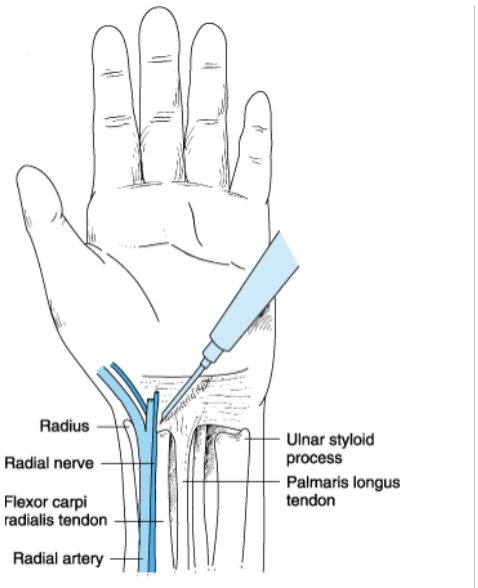
Radial N. Block



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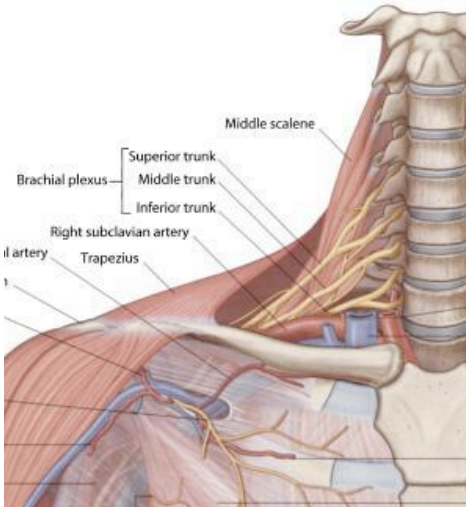
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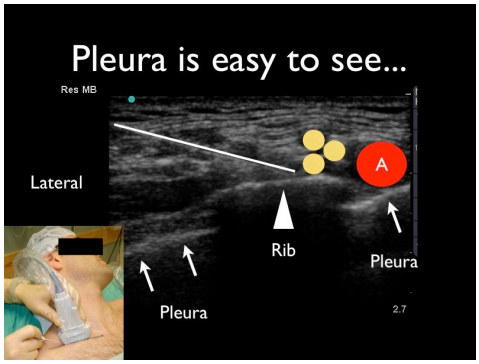
- The most common cause of postoperative bleeding is inadequate hemostasis at a suture line or the site of traumatized tissue. It should be noted, however, that elevated postoperative blood pressure, abnormal preoperative clotting, and even the administration of postoperative antibiotics have all been linked to an increase risk for postoperative bleeding.
- When applying the Sellick maneuver, it is recommended that about 9-10 pounds of pressure be applied to the cricoid cartilage. Some practitioner's estimate this amount of pressure as the minimum amount that one could apply to the bridge of the nose

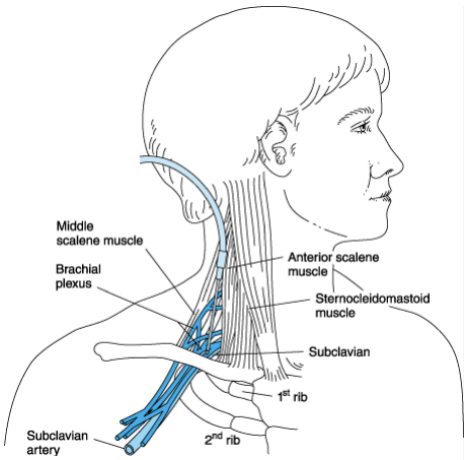
- The saphenous nerve is the largest cutaneous branch of the femoral nerve and can be blocked by injecting 5 mL of anesthetic anterior to the medial malleolus. The superficial peroneal nerve is located lateral to the extensor digitorum longus.

- Horner's syndrome and phrenic nerve block often occur with supraclavicular blocks, but the most serious complication with this approach is a 1-6% chance of pneumothorax.

- Just posterior to the subclavian artery between the clavicle and the first rib, the brachial plexus is tightly compacted. Because of this, a block performed here will achieve adequate anesthesia of the entire arm and hand. The patient is positioned supine with the head turned to the opposite side A skin wheal is raised posterior to the subclavian artery at the most inferior point of the interscalene groove. A 22-gauge, 1.5 inch needle enters the wheal posterior to the pulse and is directed caudally until a paresthesia or muscle contraction (if using a nerve stimulator) is elicited. If blood is aspirated, redirect the needle posteriorly and laterally. If the first rib is encountered, walk the needle in an anterior to posterior manner.







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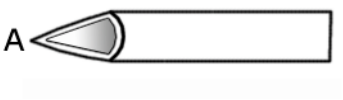
- Combined spinal-epidural analgesia is a regional anesthesia technique that has proved useful in laboring parturients. The spinal anesthetic may be given to alleve early labor pain while the epidural can be activated after the patient is in active labor. Clinicians using this technique often inject an opioid such as fentanyl 25 mcg or sufentanil 10 mcg intrathecally during the first stage of labor. The opioid may be injected alone or with a small amount of local anesthetic such as 2.5 mg of preservative free bupivacaine or 3-4 mg of ropivacaine. The addition of 100 mcg of epinephrine can prolong the block, but should not be used with opioids alone. It is not necessary to place the spinal and epidural at the same interspace, although many clinicians do. The use of 24 to 27 gauge pencil-point needles (Whitacre, Sprotte, or Gertie Marx) helps reduce the incidence of postdural puncture headache. Also, the use of smaller gauge spinal needles reduce the likelihood that the epidural catheter will pass through the dural hole into the subarachnoid space.

- The steroid injection (methylprednisolone 40-80 mg or triamcinolone 40-80 mg) usually takes 12-48 hours to achieve its full anti-inflammatory effect. Local anesthetics are typically mixed with the steroid to provide pain control and relief from muscle spasm until the steroid takes effect. Caudal epidural injections are preferable in some patients as they bypass the scarring and fusion hardware that may be present in patients who have had lumbar surgery. The injection is most effective if administered at the site of the pain, however, and spread of the injection to the affected nerve may not be as effective when using the caudal route. Subarachnoid injection is to be avoided as the ethylene glycol preservative in the steroid has been implicated in adhesive arachnoiditis.

- For patients whose weight is greater than 20 Kg, a shortcut to calculating the hourly fluid requirement is to add 40 to the patient's weight. Thus, for an 80 Kg patient, the hourly fluid requirement is 120 mL. Multiplying by the 9 hours the patient has not taken fluids will demonstrate an NPO deficit of 1080 mL.
- Remember that for a patient that weighs more than 20 Kg, you may simply add 40 to their weight to arrive at their hourly fluid requirement. For a 70 Kg patient, the hourly fluid requirement is 110 mL. Over 8 hours, the total NPO deficit would be 880 mL. However, because the patient has had an IV running at 100 mL/hour for 4.5 hours, they have already received 450 mL of their fluid requirement, leaving a deficit of 880-450, or 430 mL.

- When moving from the erect to supine position, there is a considerable increase in central blood volume. The resulting stretch of baroreceptors in the central circulation result in a decrease in MAP, heart rate, and peripheral vascular resistance whereas cardiac output and stroke volume increase.
- When a surgical patient is placed in the sitting position, the stroke volume and cardiac output are decreased by about 12-20% and cerebral perfusion pressure is reduced by about 15% without a significant change in heart rate.
- In the Trendelenburg position, the MAP, pulmonary capillary wedge pressure, and systemic vascular resistance increase whereas the cardiac index, oxygen delivery, and oxygen consumption remain unchanged. Despite these changes, the significant increase in venous and cerebrospinal fluid pressure create an offset that results in decreased arterial blood flow to the brain
- In the lateral decubitus position, the vital capacity is decreased by about 10%. The FRC is also reduced, as is the mobility of the diaphragm. The mediastinum is shifted into the dependent hemithorax and can potentially result in a decrease in venous return, especially in the kidney position. The dependent lung receives and increase in pulmonary blood flow due to the effects of gravity.
- The anode (or armored) tube contains an embedded wire that is designed to prevent kinking when the tube is bent. Nasal and oral RAE tubes possess pre-formed bends that direct the tube away from the surgical field and are especially useful in oral and facial surgeries. Double-lumen tubes and bronchial blocker tubes possess the ability to direct gas flow to one or another lung during surgery.
- A laryngeal mask airway normally produces a gas leak when air pressures reach 20 cm H2O, provide no protection against laryngospasm or aspiration, and may be difficult to position.
- Proper denitrogenation should be performed using a high flow of oxygen with a fully-open pop-off valve, a leak-free mask fit, and the allowance of tidal breathing for 2-3 minutes or a series of 4 vital capacity breaths. Signs that denitrogenation is being accomplished include: noticing the reservoir bag emptying and refilling, end-tidal CO2 approaches 40 mmHg, and end-tidal O2 approaches 85-90%.
- The administration of hetastarch should be limited to a volume of 20 mL/kg as it can impair coagulation. A newer version of hetastarch, called Hextend, that contains electrolytes, physiologic glucose, and a lactate buffer has been used in volumes greater than 20 mL/kg with no effect on coagulation.
- Normal Saline contains 154 mEq/L of sodium and 154 mEq/L of chloride. Lactated Ringer's solution contains 130 mEq/L of sodium, 109 mEq/L of chloride, 4 mEq/L of potassium, 3 mEq/L of calcium, and 28 mEq/L of lactate. D51/2NS contains 77 mEq/L of sodium, 77 mEq/L of chloride, and 50 grams/L of glucose.
- A-alpha fibers are heavily myelinated, have a diameter of 15-20 microns, and are responsible for somatic motor impulses and proprioception. A-beta fibers are moderately myelinated, have a diameter of 5-12 microns, and transmit touch and pressure impulses. A-gamma fibers are also moderately myelinated, have a diameter of 5-10 microns, and transmit proprioception impulses. B fibers are lightly myelinated, have a diameter of 1-4 microns, and transmit autonomic impulses. C fibers have a diameter of 0.5-1 micron and transmit pain and temperature impulses.
- One of the hallmark gastrointestinal symptoms of hypothyroidism is delayed gastric emptying. Gastrointestinal symptoms associated with hyperparathyroidism include anorexia, nausea, vomiting, constipation, and epigastric pain. Hyperthyroidism is commonly associated with diarrhea.
- Patients with placenta previa who have had one previous c-section are 20-25% more likely to have placenta accreta, (penetration of the myometrium by placental villi). After four or more c-sections, the incidence of placenta accreta increases by more than 67%.
- A properly inserted LMA will exhibit an air leak at 20 cm H2O. Procedures (or patient types) that may require higher pressures to produce adequate ventilation such as abdominal insufflation, obesity, or steep Trendelenburg position are not suitable for the use of an LMA.
- Contraindications to blind nasal intubation include nasal pathology, coagulopathy, thrombocytopenia, severe midface trauma, and previous transphenoidal surgery (the latter two of which could permit intracranial intubation). Blind nasal intubation is not contraindicated but likely to be less successful in patients with laryngeal masses, edema, or scarring from a previous surgery. Blind nasal intubation is possible in an awake patient, and can even be easier as a patient encouraged to breathe rapidly may facilitate introduction of the endotracheal tube through the larynx.
- Conditions that warrant postponement of elective procedures for further evaluation include new onset atrial fibrillation, right bundle branch block if accompanied by pulmonary symptoms, symptomatic bradycardia, high-grade heart block, and new onset left bundle branch block regardless of other symptoms as it exhibits a high association with coronary artery disease.
- Core body temperature usually decreases 1-2 degrees Celsius during the first hour of anesthesia (Phase I), then gradually decreases over the next 3-4 hours (Phase II), then a consistent, steady-state or equilibrium temperature is reached (Phase III).
- When performing a caudal anesthetic, the needle or intravenous catheter is advanced toward the sacral canal at a 45 degree angle towards the head until a pop is felt as it pierces the sacrococcygeal ligament. The angle of the needle is then flattened and advanced.
- Retrobulbar block may result in bradycardia as there is a risk of eliciting the oculocardiac reflex during injection. Because of potential trauma resulting from puncture or perforation of the globe or hemorrhage from puncture of a vessel located behind the globe, blindness is a potential risk of retrobulbar block. The injection is typically only performed below the globe as most of the vessels are positioned superiorly to the globe. Although the larger volume of anesthetic injected when performing peribulbar blocks results in akinesia of the eyelids, retrobulbar blocks typically leave the orbicularis oculi muscles intact, requiring a separate block of the facial nerve branches supplying these muscles to achieve akinesia.
- Epinephrine 0.2 to 0.3 mg, clonidine 75 to 100 mcg, or phenylephrine 2 to 5 mg can be added to a spinal anesthetic to prolong the duration without resulting in any significant cardiovascular changes. Sodium bicarbonate 0.2 mL or a 0.42% solution is thought to shorten the onset of a spinal anesthetic, but does not prolong the duration of block.
- Factors associated with a higher risk for developing postoperative blindness following prone procedures include prolonged surgical duration, anemia or massive blood loss, and hypotension. Some studies have also implicated the knee-chest position and the use of a horseshoe headrest as possible contributing agents.
- On injecting the local anesthetic for a Bier block, the arm discolors and the patient experiences a 'pins and needles' sensation over the next five minutes. You should reassure the patient that this is normal and proceed as planned.
- The addition of clonidine to an epidural has been shown to significantly increase the quality and duration of pain relief. As it absorbs into the systemic circulation, it can produce bradycardia, hypotension, and sedation.
- Shivering occurs in about 40% of patients postoperatively. It is the primary means of thermogenesis and can produce increases in metabolic demand, intraocular pressure, and intracranial pressure.

- When incompatible blood is administered, antibodies and complement attack antigens on the donor red blood cell. The antigen-antibody complex activates Hageman factor (factor XII) which stimulates the kinin system to produce bradykinin. Bradykinin then increases capillary permeability and dilates arterioles causing hypotension. Activation of the complement system results in the release of histamine and serotonin, resulting in bronchospasm.
- Cryoprecipitate contains fibrinogen (factor I), factor VIII, von Willebrand's factor, fibronectin, and factor XIII.
- The hourly fluid maintenance rate is calculated as 4 mL/kg/hour for the first 10 kg, 2 mL/kg/hour for the second 10 kilograms, and 1 mL/kg/hour thereafter. In this instance, the child weighs 13 Kg which you can think of as 10 kg + 3 Kg. The first 10 kilograms warrant 40 mL/hour and the remaining 3 kilograms warrant 6 additional mL/hour for a total of 46 mL/hour.
- The approach for a high thoracic epidural should be made at the C7-T1 interspace. The C7 spinous process is the most prominent and allows for easy determination of the C7-T1 level. A midthoracic approach is typically made using the root of the scapula, which lies at about T3 and the inferior angle of the scapula which lies at about T7.
- The primary difference noticed between the Quincke and its two counterparts are that it has a cutting needle while the latter both have blunt, non-cutting tips.



- The nerve distribution shown is that of the median nerve, which can be blocked by the injection of local anesthetic at the wrist between the flexor carpi radialis and the palmaris longus tendons.



- An injection of local anesthetic both at the wrist between the ulnar artery and flexor carpi ulnaris tendon or at the elbow just lateral to the medial epicondyle of the humerus will anesthetize the ulnar nerve.

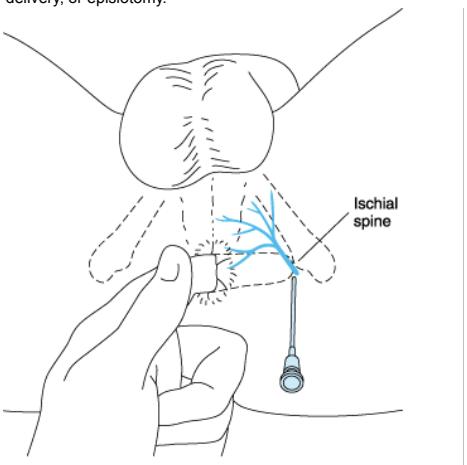
- The main aim of of washing red blood cells in isotonic saline solution is to remove plasma proteins, along with some white blood cells and platelets to prevent severe allergic transfusion reactions mediated by the antibodies of the recipient to the donor's plasma proteins.
- Whole blood and packed red blood cells need to be matched to the recipient's blood type whereas FFP, cryoprecipitate, and platelets do not (although it is preferable when time permits).
- 10% Dextran 40, able to retain 30 mL of fluid for each gram administered, exerts the highest colloidal osmotic effect, followed by 25% albumin, 6% hetastarch, and finally, plasma.
- Ulnar neuropathy occurs more often patients with a high BMI (>38) and occurs more commonly in men than in women. It is believed to be more common in men because the tubercle of the coronoid process is 1.5 times larger in males, the cubital retinaculum tends to be thicker, and there is less adipose tissue over the medial aspect of the elbow. Elbow flexion > 110 degrees can result in ulnar neuropathy as the aponeurosis of the flexor carpi ulnaris muscle and the cubital tunnel compress the nerve in this position. A significant number of patients exhibit mild ulnar neuropathies preoperatively in both arms that can worsen despite conventional positioning and padding. For this reason, it is important to ask the patient about elbow problems, 'funny bone' sensitivity, or neuropathies in the hands and fingers and document them carefully.
- perineal crush injury: A traction table places the patient supine with one leg elevated and the affected leg placed in traction. A perineal post is positioned between the patient's legs to keep them stationary while the traction pulls on the leg. The elevated leg suffers the risk of hypoperfusion while the perineal post poses a risk of crushing the perineum.

- West et al, defined three lung zones based on the relationship between pressures in the alveoli, arterioles, and venules.
- In Zone I, the pressure in the alveoli exceeds that of the arterioles and the venules. Zone 1 is rarely present normally, but can occur in pulmonary hypertension, excessive PEEP, or large tidal volumes during positive-pressure ventilation.
- In Zone 2, the pressure in the arterioles exceeds that of the alveoli and the venules. This occurs in the nondependent portions of the lung and perfusion is the result of the oscillating balance between arterial and alveolar pressures.
- In Zone 3, the pressure caused by hydrostatic forces (often due to gravity) produce venous congestion and perfusion is dictated by the difference in arterial pressure and venous pressure.

Match the area of the lung with the location of West Zone III based upon the patient's position.

Drop answer in this column		Click and drag from this column
Dorsal body wall	Dorsal recumbent	
Apices	Trendelenburg	
Lung bases	Reverse Trendelenburg	

- According to the standards set forth by the ASA, the patient's oxygenation, ventilation, circulation, and temperature shall be continually monitored. This includes pulse oximetry, ECG, and blood pressure (which must be assessed at least every five minutes). During monitored anesthesia care cases, capnography is not required, but ventilation must be assessed, at least, by continual observation of qualitative clinical signs (e.g. chest rise and breath sounds).
- A popliteal fossa blockade can anesthetize the nerves of the lower leg. However, a supplementary femoral nerve block is required to block its terminal saphenous branch which innervates the medial anterior calf and the dorsum of the foot.
- The sensory and motor fibers of the cervical plexus originate from the cervical 2, 3, and 4 nerve roots and are unique in the the sensory fibers separate from the motor fibers early and can be blocked separately. The main potential risk of this procedure is vertebral artery injection, but if the needle is advanced too far medially into the vertebral foramen, epidural or even subarachnoid anesthesia is a potential risk as well, particularly because the the dural sleeves of the cervical nerves are longer on these branches. A cervical plexus block does not protect against the carotid sinus reflex and local injection of the carotid sinus at the point where the carotid artery bifurcates is often necessary.
- Proper application of the Sellick maneuver calls for application of the BURP method (Back, Upward, and Rightward Pressure) directly on top of the cricoid cartilage until the fingers blanch. The reason for the directional component of this technique is because in about 75% of patients, the esophagus is slightly to the right of the trachea, so moving the cricoid cartilage downward, upward, and to the right should occlude the esophagus in most patients.
- Confirmation of appropriate placement of an Eschmann stylet (gum elastic bougie) is made when you feel the stylet travel across the bumpy surface of the tracheal rings. Although the stylet may strike a surface and stop, this can occur in both the esophagus and the trachea and is not confirmation.
- The pudendal nerves, which arise from the S2-S4 nerve roots, innervate the perineum, rectum, vaginal vault, and parts of the bladder. These nerves can be blocked by the injection of 10 mL of dilute local anesthetic behind the sacrospinous ligament. A pudendal block will provide short-term anesthesia (maximum of 90 minutes) that is suitable for perineal procedures, forceps delivery, or episiotomy.



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- Hoarseness is a common complaint following intubation. The use of a smaller endotracheal tubes, low-pressure cuffs, and lubricating the tube with lidocaine jelly will decrease the incidence of postoperative hoarseness due to intubation.
- The administration of one unit of packed red blood cells to a 70 Kg patient will typically increase the hemoglobin by 1 mg/dL and the hematocrit by 3%. Thus, the administration of two units should increase the hemoglobin by 2 mg/dl and the hematocrit by about 6%.
- Should braces placed medially (close to the neck) are associated with a high incidence of brachial plexus injury at the root level. If at all possible, shoulder braces should be avoided. If they are used, placing them laterally over the acromioclavicular joint reduces the risk of injury to the brachial plexus.
- Compartment syndrome occurs primarily as a result of inadequate perfusion to an extremity for any reason. Factors that contribute to the development of compartment syndrome include systemic hypotension, pressure against the extremity by an extremity holder or pressure by any compressive wrap or sheets used to secure an extremity, excessive flexion of the extremity (as occurs in lithotomy position), elevation of the extremity, or surgery length greater than five hours.
- An interscalene brachial plexus block with 40-50 mL of local anesthetic ensures a nearly 100% incidence of ipsilateral hemidiaphragmatic paresis. Reducing the volume of local anesthetic to 30 mL, however, does not reduce the incidence of paresis and its pulmonary effects significantly. The paresis is typically complete within 15 minutes of injection and normal hemidiaphragmatic motion typically returns in about 3-4 hours. The tip of the scapula corresponds with T7. The most protuberant cervical vertebra is at the level of C7. The origin of the scapular spine corresponds with T3. The tip of the 12th rib corresponds with L1.
- The performance of a Bier block is contraindicated in patient's with severe crush injuries to the arm as hypoxia may worsen the potential outcome, uncontrolled hypertension because it would increase the risk that arterial flow may not be occluded sufficiently, and Raynaud's disease or other forms of peripheral vascular disease affecting the surgical limb. Heterozygous sickle cell anemia is not a contraindication, however, homozygous

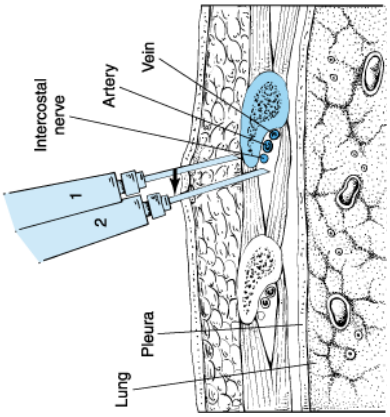
sickle cell is a contraindication to the procedure. Malignant hyperthermia is not a contraindication to this procedure.

- Injection of 4-5 mL of local anesthetic at the location indicated below would result in numbness to the posterior portion of the heel and sole of the foot.



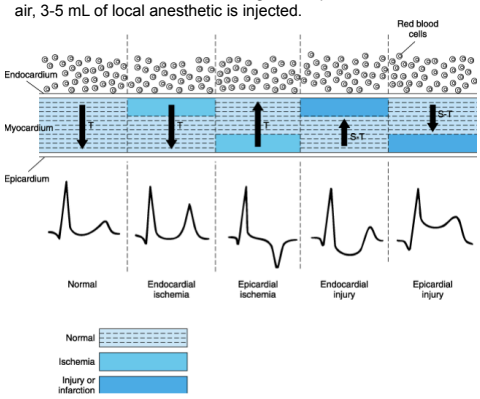
- The sural nerve travels behind the lateral malleolus and provides sensory innervation to the posterior portion of the heel and sole of the foot. The saphenous nerve provides sensory innervation to the medial aspect of the foot. The deep peroneal nerve innervates the lateral side of the hallux and the medial side of the second digit. The posterior tibial nerve, which passes behind the medial malleolus, innervates the distal aspect of the sole of the foot.
- The incidence of acquiring hepatitis B is about 1 in 205,000 units while the risk of acquiring hepatitis C or HIV is about 1 in 2 million units. Banked blood is not tested in the laboratory for the West Nile virus, but donors are interviewed regarding exposure to endemic areas and history of fevers. The risk of acquiring West Nile virus is dependent, therefore, on the immune status of the patient, presence of antibodies in the donor, and the endemic prevalence of the disease. Screening for syphilis began in 1958 and the last case of syphilis acquired by transfusion was 1966. The risk of bacterial contamination by any pathogenic organism is about 0.2% or about 1 in 500.
- Factor IX concentrate is available for the treatment of hemophilia B (factor IX deficiency). Hemophilia A is characterized by factor VIII deficiency. von Willebrand disease is characterized by a deficiency of von Willebrand factor.
- A transtracheal block is an effective method of anesthetizing the airway for awake airway procedures. It is performed by injecting 5 mL of 2% lidocaine through the cricothyroid membrane using a 23-gauge needle or 24-gauge angiocatheter. Aspiration should be performed continually while the needle is directed caudally. When air bubbles are seen in the syringe, the tip of the needle is in the trachea. On deep inspiration, the local anesthetic should be injected into the trachea. The patient will cough, which will spread the local anesthetic throughout the trachea.
- The incidence of postdural puncture headache increases with the use of non-pencil point needles, a cutting needle inserted with the bevel of the needle perpendicular to the long axis of the body, and multiple attempts or punctures. It is also more likely to occur in young patients than elderly patients and more common in female patients.
- The sympatholysis associated with spinal and epidural anesthesia results in unopposed parasympathetic activity, which results in generalized constriction of the bowel, increased gastrointestinal blood flow, increased peristalsis, and increased gastric intraluminal pressure.
- The duration of a spinal anesthetic depends primarily on the local anesthetic selected and the total dose administered. Volume has little if any effect on the duration of the block, but increasing the total dose injected, for example by increasing the dose of hyperbaric bupivacaine from 10 mg to 15 mg will prolong the block. The addition of opioids will prolong the duration of a spinal anesthetic. The addition of vasopressors such as epinephrine prolong the duration of the block by causing vasoconstriction, acting directly on nociceptors, or a combination of both. Highly protein-bound anesthetics such as tetracaine, bupivacaine, and ropivacaine have longer durations than less protein-bound agents such as mepivacaine and lidocaine.
- Because different neurons have a different susceptibility to local anesthetics, specific sensations exhibit inhibition at different dermatome levels. This is referred to as differential block. With regards to spinal anesthesia, kinesthetic sense is typically blocked at a higher dermatome level than light touch or cold sensation, which in turn is inhibited at a higher dermatome level than pinprick sensation.
- Epidural opioids cross from the epidural space through the dura to the opioid receptors in Laminas II and III, which together are referred to as the substantia gelatinosa. Lamina VIII consists of motor interneurons. Lamina X consists of central gray matter. The periaqueductal gray region is a portion of the paleospinothalamic pathway surrounding the aqueduct of Sylvius.
-

- From superior to inferior, the structures in the intercostal neurovascular bundle are positioned: Vein, artery, nerve.



- A patient is undergoing surgery on the lower leg and wishes to avoid general anesthesia. You have successfully performed a popliteal block. What other block should be performed to provide complete anesthesia distal to the knee? A popliteal nerve block can provide complete anesthesia of the leg distal to the knee if combined with a saphenous nerve block. The saphenous nerve is the terminal branch of the femoral nerve and provides sensory innervation along the medial aspect of the lower leg from the knee to the medial malleolus.
- The average blood volume for a female patient is calculated as 65 mL/Kg. In this case, the patient weighs 68 kilograms, so 65 mL/kg X 68 Kg = 4420 mL
- Beta-blockers should be continued on the morning of surgery if they are taken to treat angina, symptomatic arrhythmias, or hypertension. Anxiety and akathisia (restlessness) are also indications for beta-blocker use, but adequate anesthetic pre-medication is sufficient to treat these symptoms.
- Autonomic neuropathy is the best predictor of silent coronary artery disease in the diabetic patient.
- The saphenous nerve, which is a branch of the femoral nerve, innervates the medial aspect of the foot. Loss of dorsiflexion of the foot is consistent with injury to the common peroneal nerve.
- Normal saline (308 mOsm/L), D51/4NS (355 mOsm/L), Lactated Ringer's solution (273 mOsm/L), and Plasmalyte (295 mOsm/L) are all considered to be isotonic. D5W (253 mOsm/L) is considered to be hypotonic.
- The glucose content of D5W is 50 grams per liter.
- Hypocalcemia and hyperkalemia are the most common electrolyte abnormalities seen in chronic renal failure.
- Extreme flexion at the thigh can result in injury to the sciatic, obturator, and femoral nerves.
- A cricothyrotomy can be suitable for establishing an airway during can't intubate/can't ventilate scenarios, as a definitive airway during surgery of the face or neck, and when traumatic face and neck injuries make traditional airway management hazardous or impossible.
- Although aspiration and tooth damage can occur due to tonsillectomy, the most common complication following tonsillectomy is bleeding which occurs in 1 out of every 25 patients.
- The two sacral cornua, the coccyx, and the posterior superior iliac spines can all be used as anatomic landmarks to identify the correct placement of the needle for a caudal block.
- The standard test dose for epidural anesthesia contains 3 mL of 1.5% lidocaine and 1:200,000 epinephrine or 15 mg/mL of lidocaine and 5 mcg/mL of epinephrine. A test dose contains a total of 45 mg lidocaine and 15 mcg epinephrine. A brachial plexus block is usually indicated for procedures on the shoulder, upper arm and forearm. It is usually not sufficient for procedures on the hand, however. The C8 and T1 nerves which innervate the ulnar aspect of the forearm are frequently missed as well.
- Patients with diabetes have an increased risk for gastroesophageal reflux disease and may require intubation. They also have an increased risk of developing limited-mobility joint syndrome and a decreased range of motion in the cervical spine which can make intubation difficult.
- The injection of local anesthetic superficially near the medial malleolus will anesthetize the saphenous nerve while the continued injection toward the extensor digitorum longus tendon will anesthetize the superficial peroneal nerve.
- The sciatic nerve may be combined with a lumbar plexus block, femoral nerve block, or saphenous nerve block to provide complete anesthesia for lower-extremity surgery.
- A patient unable to bring the mandibular incisors anterior to the maxillary incisors (mandibular malocclusion) has an increased incidence of difficult intubation. Patients with Down syndrome (Trisomy 21) have an increased incidence of cervical spine instability which can make intubation difficult. A thyromental distance less than 7 cm (about three fingerbreadths) is associated with difficult intubation. Autonomic neuropathy is is the best predictor of silent coronary artery disease in diabetic patients, but does not have a known relationship to airway difficulty.
- Leukoreduction is the reduction of white blood cells in transfused blood through the use of filters. It is performed to reduce the incidence of nonhemolytic transfusion reactions.
- The storage of blood is associated with a progressive decrease in adenosine triphosphate (ATP) and 2,3 DPG levels. This results in a leftward shift in the oxyhemoglobin dissociation curve in the red blood cells. Even though the administration of hemoglobin will increase the ability of the blood to transport oxygen, the longer the blood is stored and the more of it that is given, the more the patient's oxyhemoglobin dissociation curve shifts to the left, thereby decreasing the effectiveness of the blood given. The potassium present in the plasma in packed red blood cells stored for 21 days can reach levels as high as 35 mEq/L. When large doses of blood are administered through rapid infusion devices, the sudden increase in potassium can be hazardous and sudden cardiac arrests have been attributed to it.
- Because the malaria parasite resides within the red blood cell, PRBCs or whole blood would be most likely to transmit the disease if present in the donor.
- A significant number of patients exhibit mild ulnar neuropathies preoperatively in both arms that can worsen despite conventional positioning and padding. For this reason, it is important to ask the patient about elbow problems, 'funny bone' sensitivity, or neuropathies in the hands and fingers and document them carefully.
- Total spinal anesthesia occurs when the local anesthetic spreads throughout the subarachnoid space high enough to block sympathetic outflow from the entire spinal cord and possibly even the brainstem. It results in respiratory arrest, profound hypotension, and bradycardia from sympathetic blockade. Spinal blockade, even at the cervical level is reported not to be high enough to block the large A-alpha fibers of the phrenic nerve. Apnea appears to be the result of decreased perfusion to the brainstem as apnea typically resolves with hemodynamic resuscitation.
- The glossopharyngeal block is performed by having the patient open their mouth while using a 25-gauge needle to inject 2 mL of local anesthetic bilaterally at the base of the palatoglossal arch (also called the anterior tonsillar pillar). This blocks the lingual and some of the pharyngeal branches of the glossopharyngeal nerve that provide sensation to the posterior third of the tongue and oropharynx.
- The anesthetic drug used and the total dose given are the principal determinants of the duration of action of a spinal anesthetic.
- The incidence of cardiac arrest in spinal anesthesia is about 1 in 4,000. The cause is blockade of the cardioacceleratory fibers from T1 to T5 and a drop in venous return. It is usually preceded by bradycardia and hypotension. A true allergic reaction to a spinal anesthetic is typically limited to a hypersensitivity reaction that results in erythema and/or edema of the mucous membranes.
- For a bilateral block, the bevel of the needle should be facing cephalad. For a unilateral block, the bevel should be turned 45 degrees toward the side you wish to block.
- The segmental spread of an epidural is reliant primarily upon the volume of local anesthetic solution used. This is in contrast to spinal anesthesia, which relies more on the dose in milligrams of the local anesthetic, the baricity of the solution, and the patient's position to determine the spread of the anesthetic.
- Which position is associated with the highest rate for success when performing a combined spinal/epidural anesthetic? The odds for success are higher when the procedure is performed with the patient in a sitting, rather than lateral position. This is probably due to the higher hydrostatic pressure in the subarachnoid space in the sitting position, which makes the detection of CSF more likely.
- The main advantage of a two-level combined spinal/epidural anesthetic is that it allows the anesthesia provider the ability to test the function of the epidural catheter prior to placing the spinal anesthetic. Disadvantages of the two-level method include: difficulty distinguishing between the epidural test dose and CSF when performing the spinal anesthetic, CSF flow may be hindered by the compression of the dural sac by the epidural test dose, the epidural catheter can be cut by the spinal needle, and there is a risk of dural puncture with the epidural catheter.
- The dose of clonidine that can be added to the local anesthetic to augment a caudal block is 1 mcg/kg of bodyweight. This patient weighs 22 pounds (10 kilograms), so the appropriate dose is 10 micrograms of clonidine.
- Lidocaine and prilocaine are both suitable for Bier blocks because of the low risk of toxicity at the doses used at a minimum tourniquet time of 20 minutes. Because of its high risk for prolonged cardiotoxicity, bupivacaine in absolutely contraindicated for this procedure.
- Myopia or a previous history of scleral buckle surgery increases the risk of globe penetration by the needle as they both are associated with an increased anteroposterior length of the globe. A recessed eyeball also increases the risk of injury.
- General anesthesia is suitable for patients who are anticoagulated, cannot control their eye movement adequately, or suffer from too much anxiety or discomfort to tolerate any other forms of anesthesia. With topical anesthesia, there is little risk of ocular injury and the technique is perfect for an anticoagulated patient or a monocular patient who would be blind from the amaurosis induced by the retrobulbar block. Retrobulbar and peribulbar block would not be indicated for a patient who is anticoagulated.
- The median nerve provides motor innervation to the muscles of the thenar eminence, and the lumbrical muscles of the first and second fingers. The ulnar nerve supplies motor innervation to all of the small muscles of the hand with the exception of the lumbricals of the first and second finger and the muscles of the thenar eminence.
- When performing an intercostal block, the correct rib is identified by palpation and a skin wheal is raised with local anesthetic at the inferior border of the rib. A 22 to 25 gauge needle is walked off the inferior border of the rib and advanced about 0.5

centimeters under the rib. After negative aspiration for blood and air, 3-5 mL of local anesthetic is injected.



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